



nVIDIA

Certified Associate

AI Infrastructure and Operations



NVIDIA Certification Details



NVIDIA Certifications

	Infrastructure Professionals	Developers
Associate	AI Infrastructure and Operations (NCA-AIIO)	Generative AI and LLMs (NCA-GENL) Multimodal Generative AI (NCA-GENM)
Professional	AI Infrastructure (NCP-AII) AI Operations (NCP-AIO) AI Networking (NCP-AIN)	Generative AI and LLMs (NCP-GENL) Agentic AI (NCP-AAI) Accelerated Data Science (NCP-ADS) OpenUSD Development (NCP-OUUSD)

Exam Details: Associate – AI Infrastructure and Operations



- Duration:
 - One hour
- Price:
 - \$125
- Subject:
 - AI Infrastructure and operations
- Number of questions:
 - 50
- Prerequisites:
 - A basic understanding of data center infrastructure.

Topics Covered in the Exam

- Accelerated computing use cases
- AI, machine learning, and deep learning
- GPU architecture
- NVIDIA's software suite
- Infrastructure and operation considerations for adopting NVIDIA solutions



**NVIDIA-Certified Associate:
AI Infrastructure and Operations
Exam Study Guide**



Topics Covered in the Exam

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- AI, machine learning, and deep learning
- GPU architecture
- NVIDIA's software suite
- Infrastructure and operation considerations for adopting NVIDIA solutions



1. Fundamentals
2. Inside an AI Centric Data Center
3. NVIDIA Technology Stack
4. AI Workflows

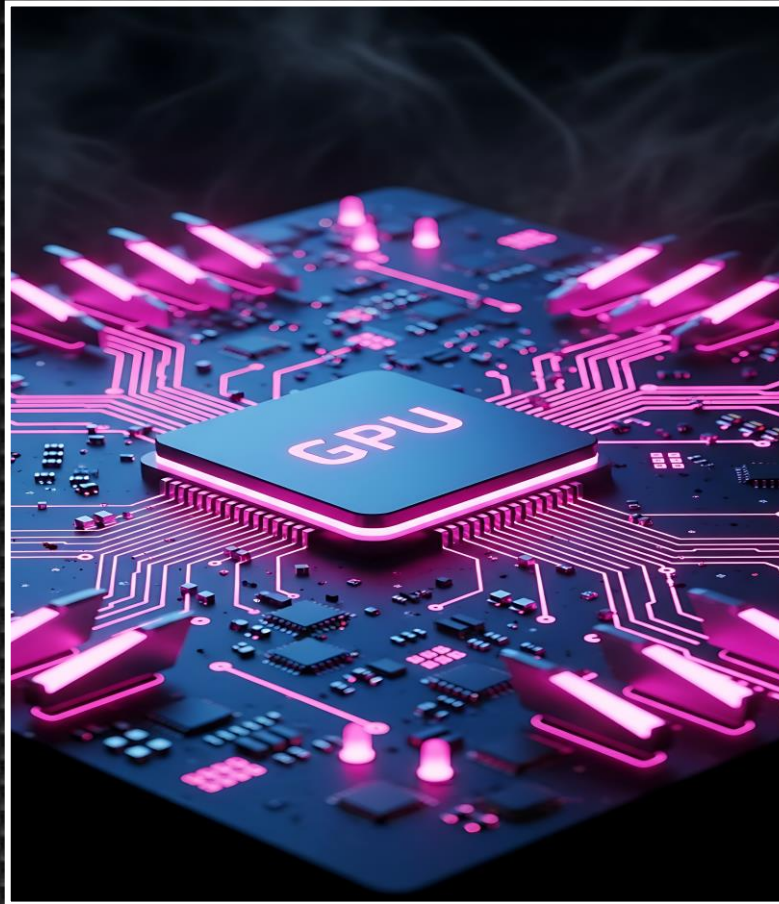
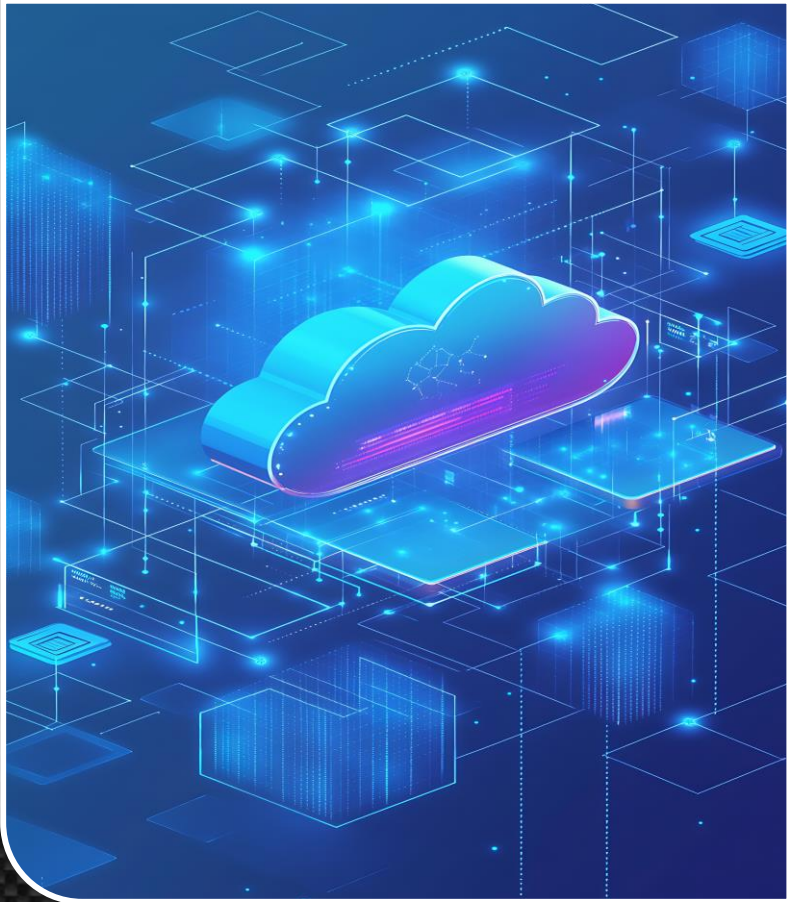
Module 1 - Fundamentals



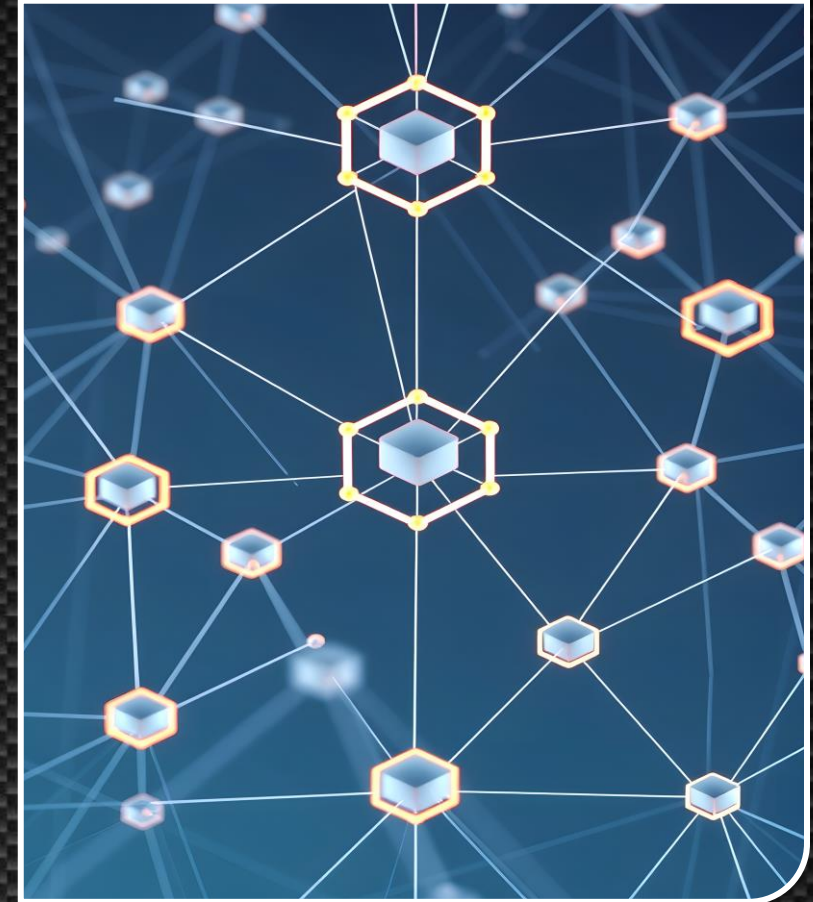
Drivers of AI Evolution



Explosion of Data



**Algorithm
Breakthrough**



**Computational
Power Growth**

AI use cases across industries

AUTOMOTIVE / AUTONOMOUS VEHICLES

- Real-time object detection & classification
- Autonomous decision-making
- Simulation for design/test



HEALTHCARE (MEDICAL IMAGING, GENOMICS)

- Automated medical image analysis
- Genomics pipelines, anomaly detection
- Clinical inference with low latency



VIDEO ANALYTICS / SURVEILLANCE

- Real-time video stream analysis
- Object detection & tracking
- Threat detection, Multi-camera stream processing



FINANCE AND BANKING

- Real-time fraud detection
- Transaction scoring at scale
- Ultra-low latency inference



RETAIL INDUSTRY

- Demand forecasting,
- Supply-chain optimization,
- Inventory management, personalized recommendations



MANUFACTURING

- Automated quality control (image recognition)
- Predictive simulations, supply-chain logistics optimization



AI, ML, DL, Gen AI

AI, ML, DL, Gen AI

AI

Machine simulating human intelligence and decision making



ML

Ability of machine to learn without explicitly being programmed



DL

Ability of machines to process data in a way that is inspired by the human brain (by using artificial neural network).



Gen AI

Ability of machines to create new and original content such as text, images, audio, or code based on learned data and patterns.



Analogy

AI

Machines that can play chess based on rules.



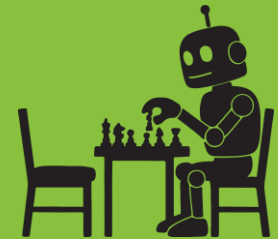
ML

Machines that learn to play chess from analyzing past chess games played by humans.



DL

Machines that can learn to play chess by playing against themselves.



Gen AI

Machines that can create a new game based on rules of chess and given prompts.



Transformer Model

- The Transformer model architecture
 - “Attention” enabled models to scale the understanding of relationships between words
 - Efficiently use parallel computing

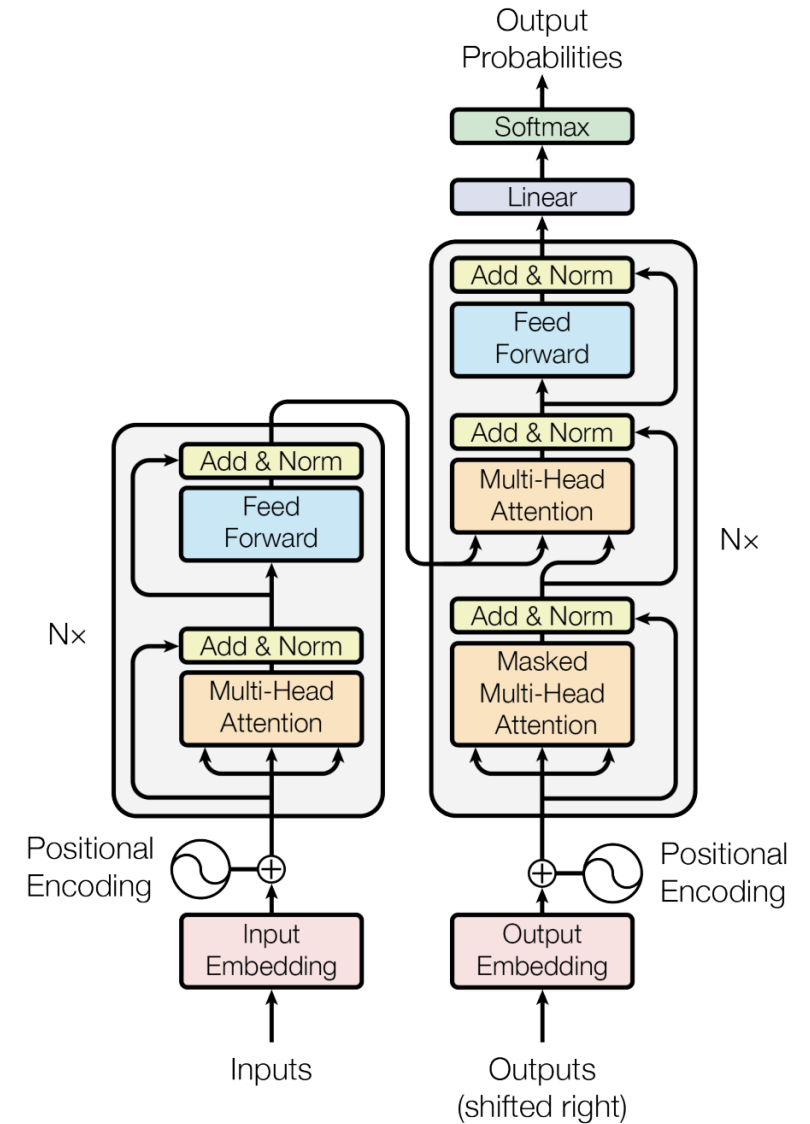


Figure 1: The Transformer - model architecture.

Transformer Model – Example

- Next word prediction with a transformer model

Sentence

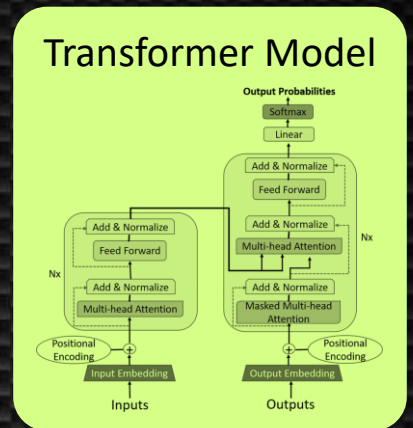
The quick brown fox jumps over the lazy _

~~person.~~

~~rabbit.~~

dog.

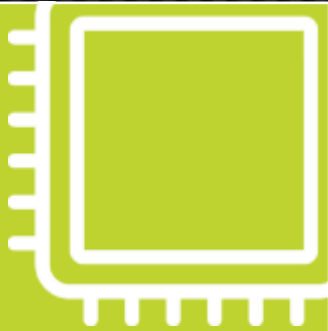
The quick brown fox jumps over the lazy dog.



Module 2 - Inside an AI centric Data Center



Inside an AI centric Data Center



Compute



Network



Storage

**Support Infrastructure
(Power, Cooling, Security etc.)**

Key Constraints in Deploying High-Density GPU Workloads



Limited electrical capacity
per rack

GPUs demand high and consistent
power supply



Heat generated by dense
GPU clusters

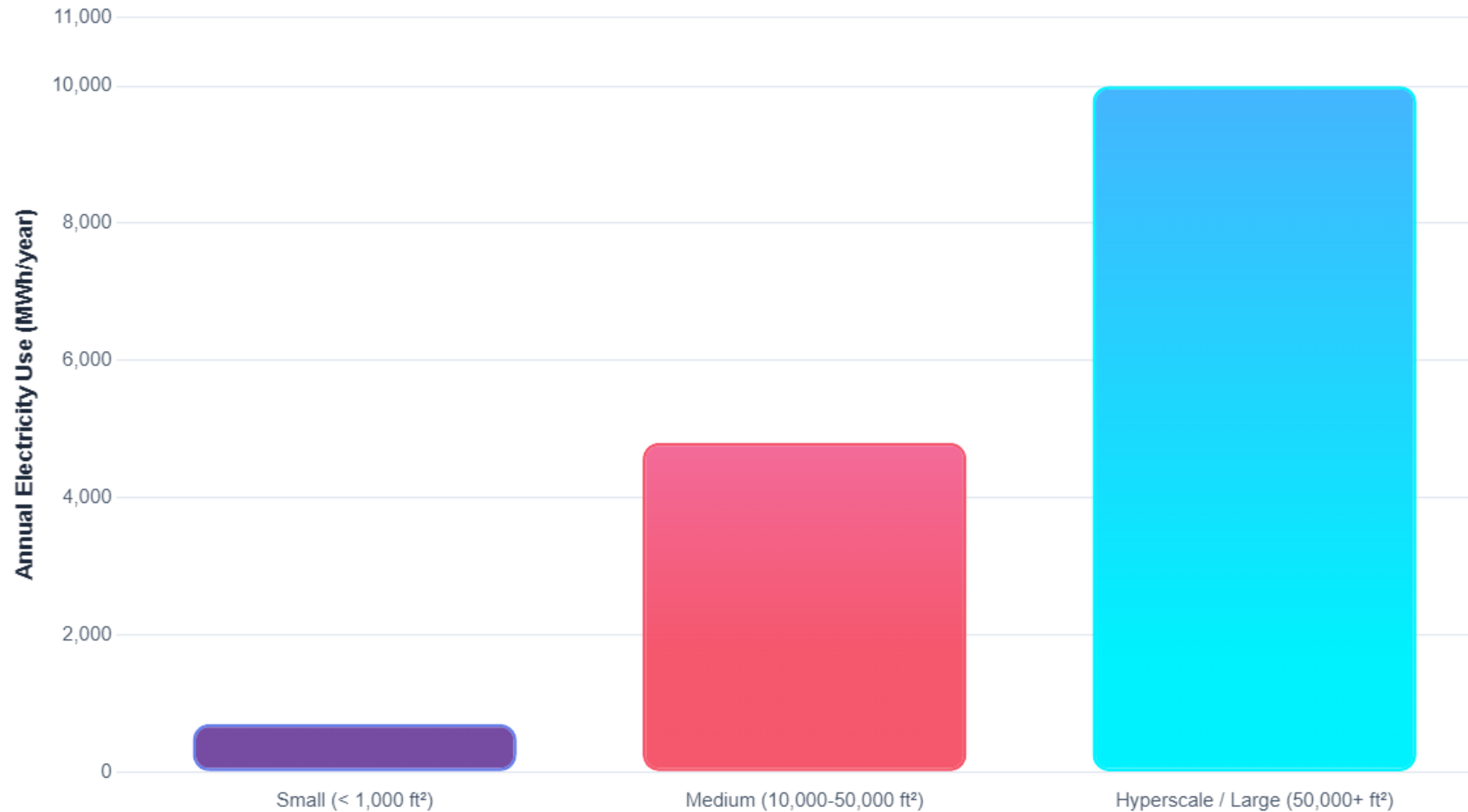
Cooling capacity of racks/rooms
becomes a bottleneck



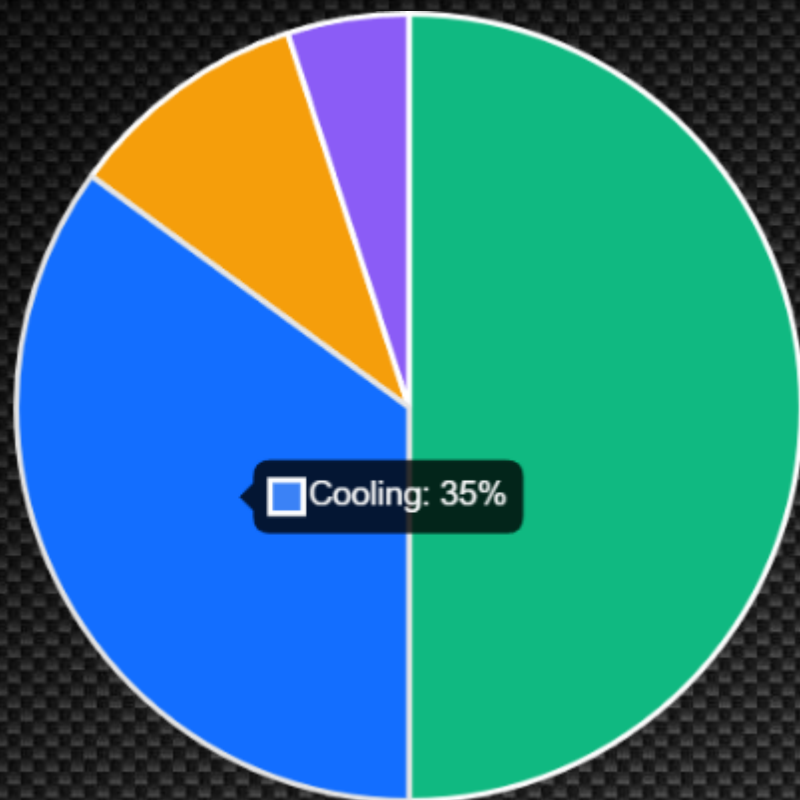
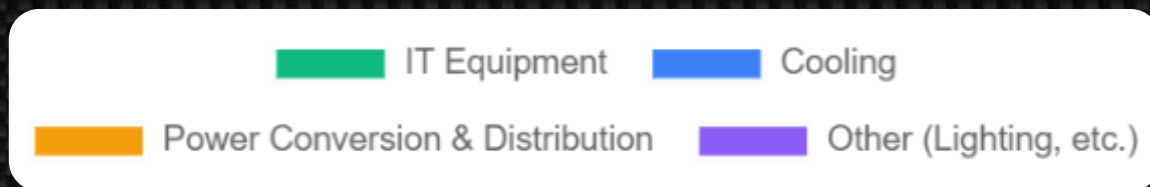
Rack density and available
floor space

Physical limits restrict how many
servers can be deployed

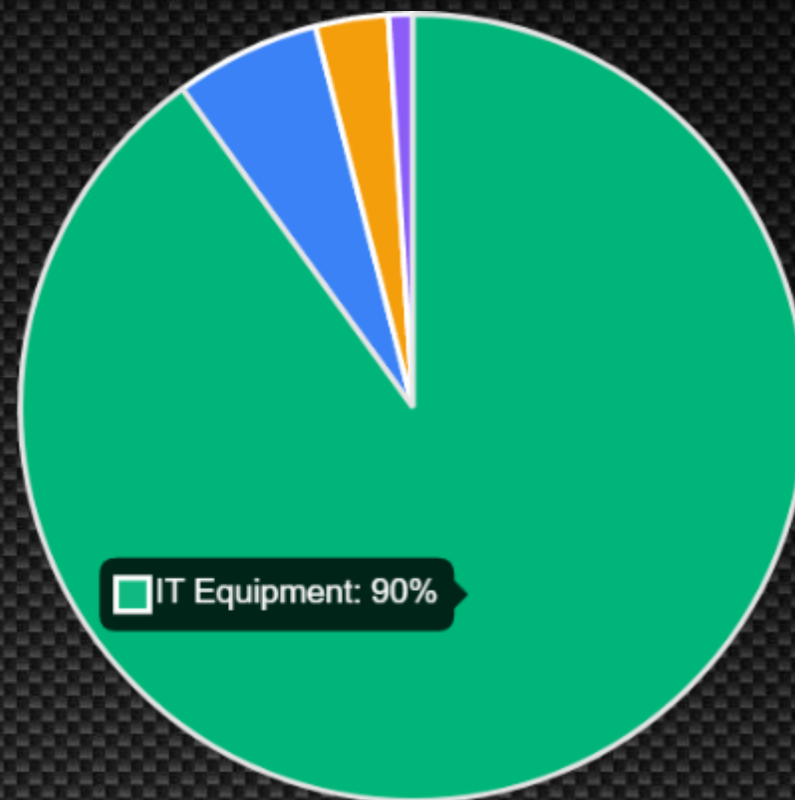
Average data centre electricity use by size



Power Consumption



Older Data Centers



Modern Data Centers

Power Usage Effectiveness (PUE)

- Definition

- A metric that compares the total energy consumed by a data center to the energy consumed by IT equipment (servers, storage, networking).

- Formula:

$$\text{PUE} = \frac{\text{Total Facility Energy}}{\text{IT Equipment Energy}}$$

- Why it matters:

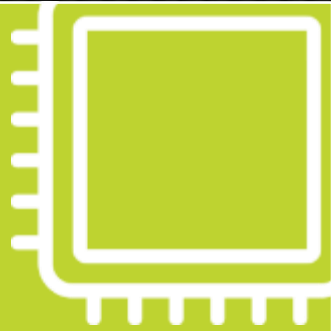
- Helps measure data center energy efficiency.
- Guides optimization in cooling, power distribution, and facility design.



- Interpretation:

- Lower PUE → Greener, more cost-effective operations.
- 1.2 → Highly efficient, close to ideal. (AWS/Google)
- 2.0 → Twice as much total energy used as IT load

Inside an AI centric Data Center



Compute



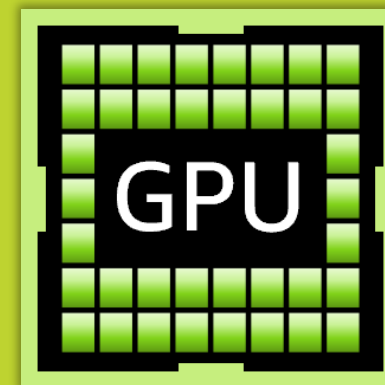
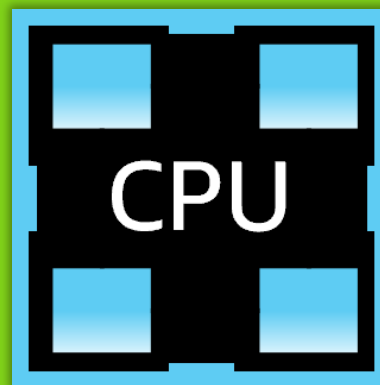
Network



Storage

Compute

Compute



Why we needed GPUs?

- GPUs were first created to handle complex graphics rendering in video games and applications, so images, animations, and 3D environments could be displayed smoothly and realistically.



Quake (1996)

started using 3D accelerator cards (e.g., 3Dfx Voodoo)



Unreal Tournament (1999)

Quake III Arena (1999)

NVIDIA released the GeForce 256 (world's first GPU)



Doom 3 (2004)

NVIDIA introduced support for DirectX 8 shaders, enabled modern graphics realism.

When did GPUs start being used for Machine Learning?



Idea

Researchers realized GPUs, designed for graphics, excel at parallel math — perfect for ML training.



Acceleration

In 2006, NVIDIA released CUDA, letting developers use GPUs for general computing, sparking ML adoption.



Breakthrough

In 2012, AlexNet trained on GPUs achieved a major leap in image recognition, proving GPU power for AI.

CPU vs. GPU



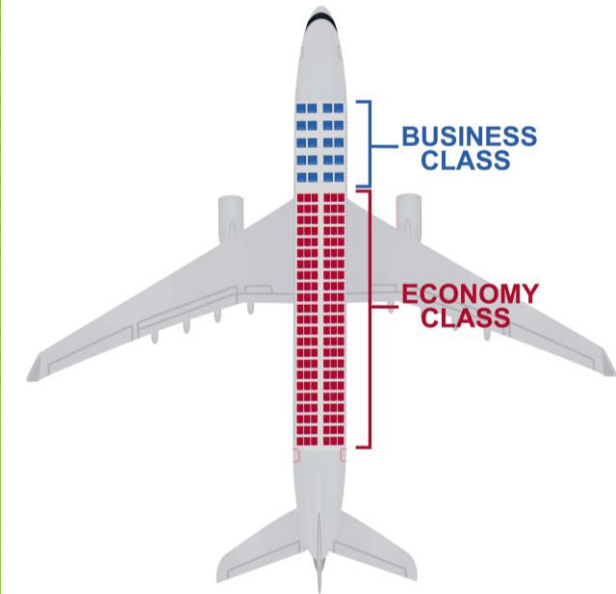
An Analogy – CPU vs. GPU

Private Jet

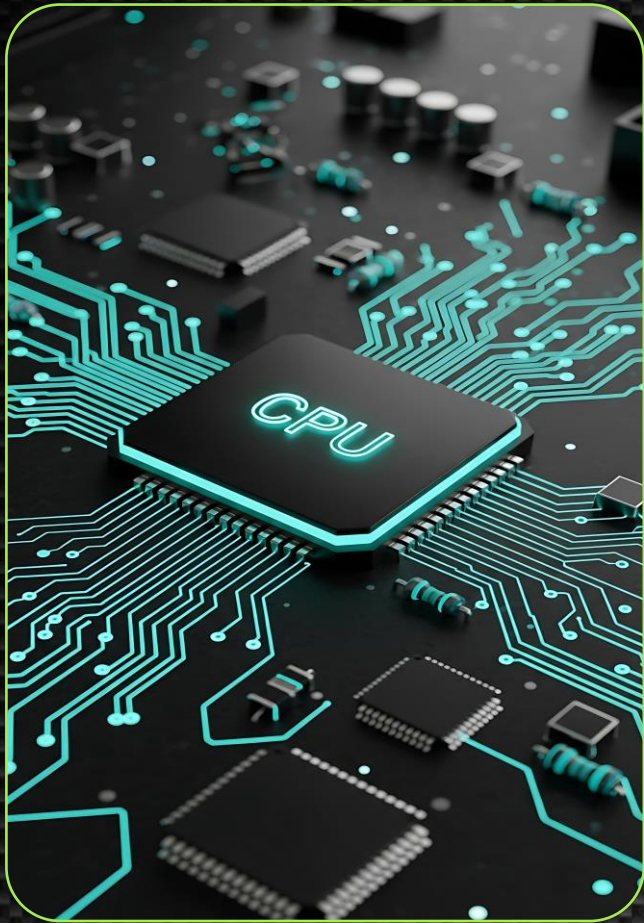


Few, spacious, luxurious	Seats	Many, simple, packed
Can fly anywhere, anytime	Flexibility	Fixed routes, less flexibility
Individual speed & custom trips	Focus	Mass transport of many passengers
Best for small groups	Efficiency	Best for large groups going same way
Personalized, diverse needs	Purpose	Specialized, parallel transport

Commercial Flight



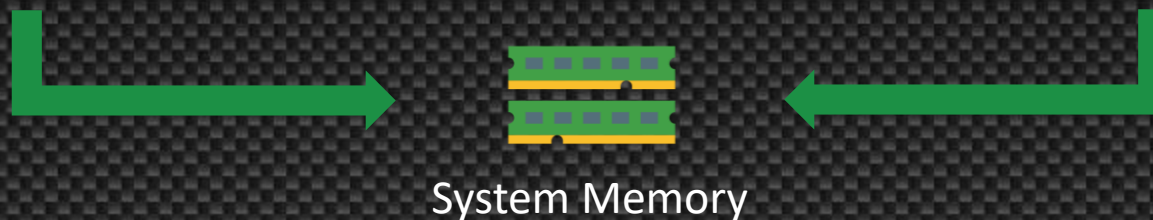
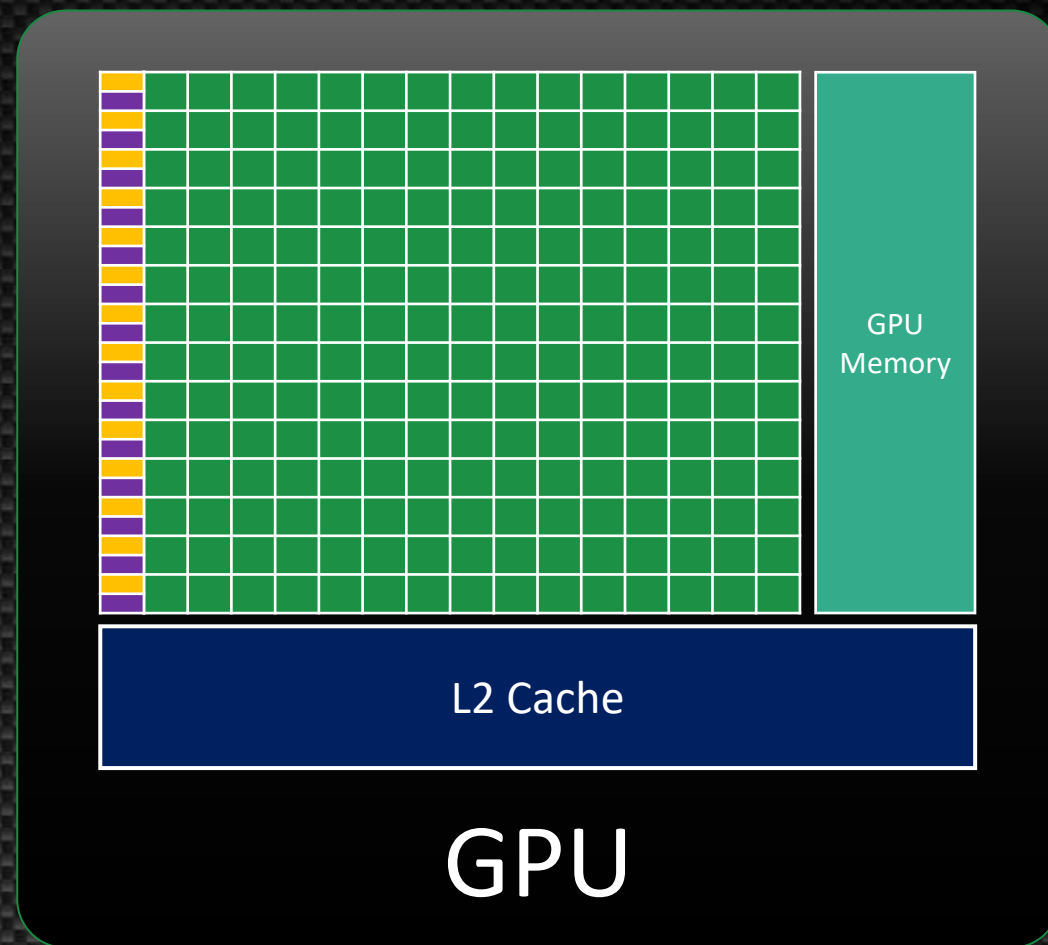
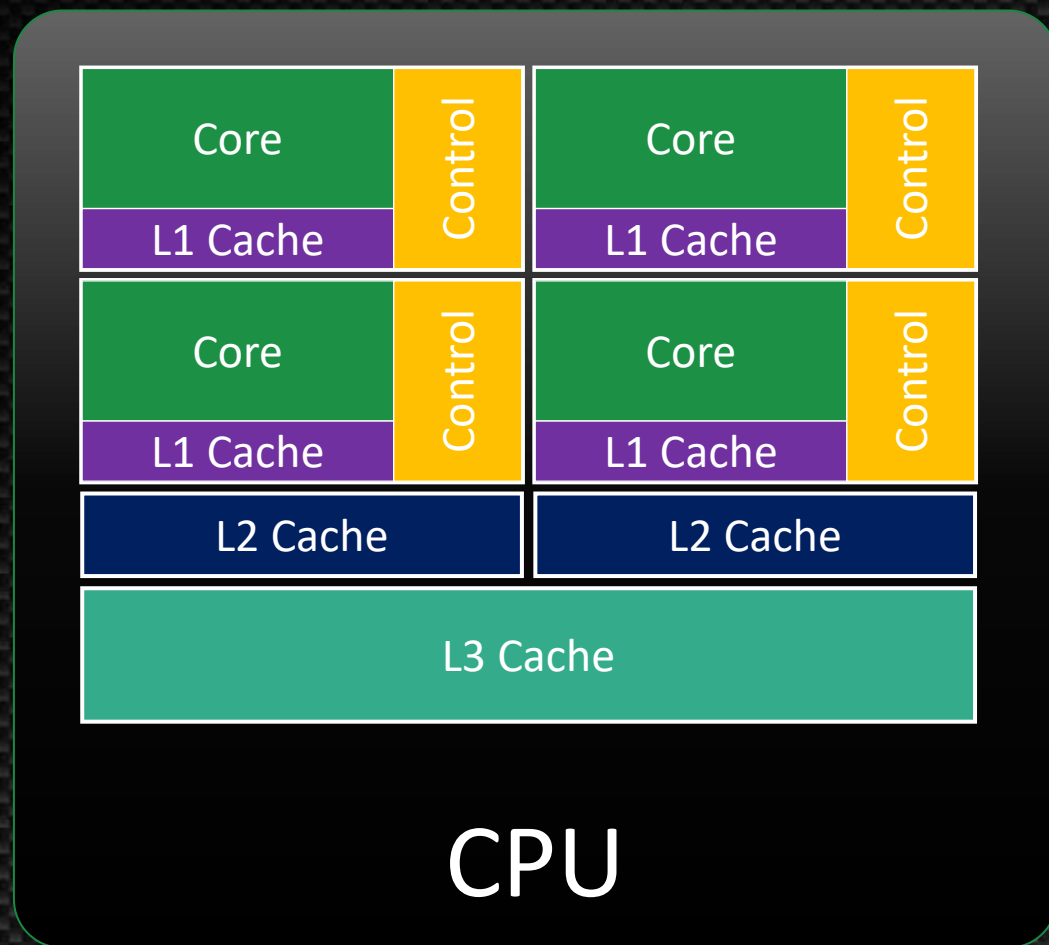
CPU vs. GPU



Few (4–16), powerful, complex	Cores	Many (hundreds–thousands), simpler
Very flexible, general-purpose	Flexibility	Less flexible, optimized for parallel tasks
Low latency, quick response	Focus	High throughput, bulk processing
Best for small, complex, varied tasks	Efficiency	Best for large, repetitive, parallel tasks
OS, general computing, apps	Use Cases	Graphics rendering, AI, simulations, mining

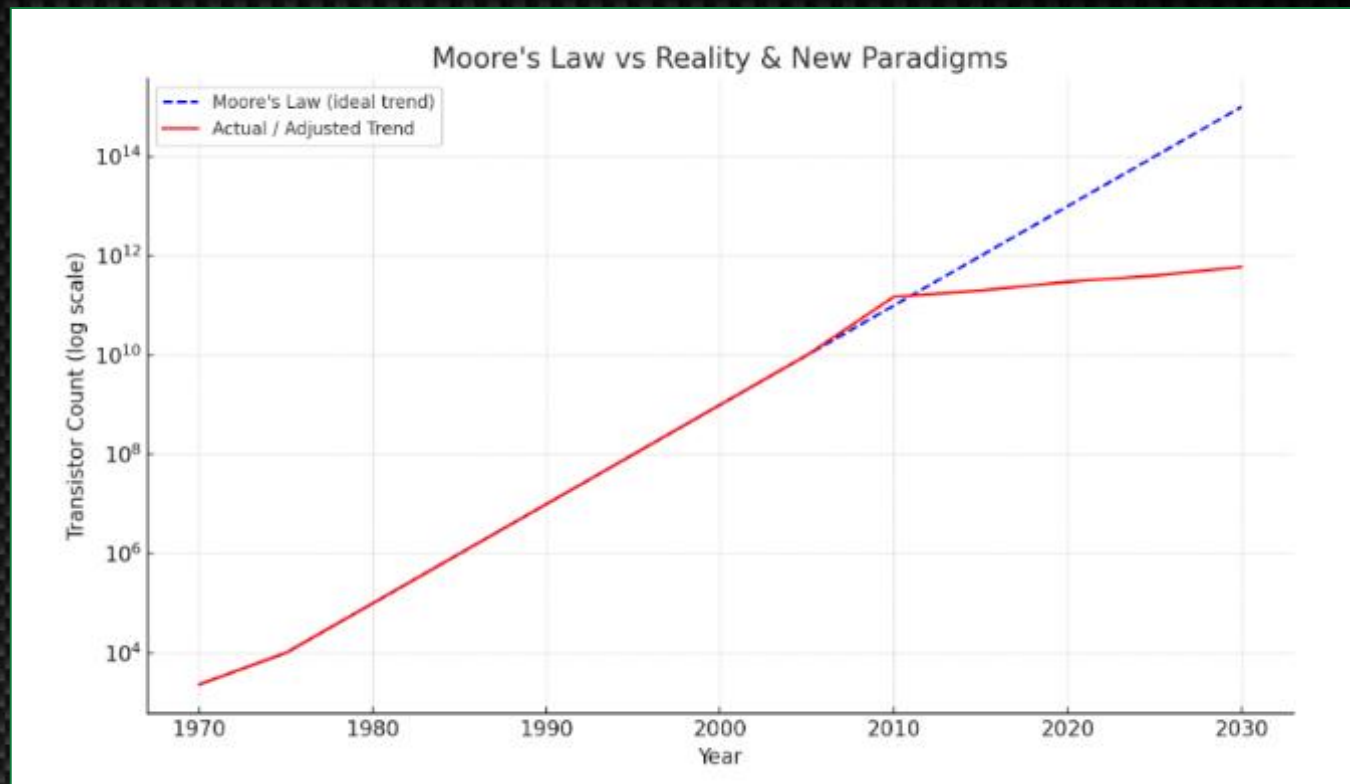


CPU vs. GPU – Architecture



Beyond Moore's law

- 1970s–2010s: Moore's Law Scaling
 - Doubling of transistors every ~2 years
 - Cheaper computing power per dollar
- 2010s–Present: Slowing of Moore's Law
 - Transistor scaling is harder & costlier (3nm, 2nm nodes)
 - Doubling now takes ~3–4 years, cost per transistor rising
- New Paradigms (Present & Future)
 - GPUs & Parallelism: Specialized acceleration for AI/graphics
 - Chiplets: Multiple smaller chips combined for scalability
 - 3D Stacking: Vertical layers to boost density
 - AI Accelerators: Custom silicon for massive speed-ups



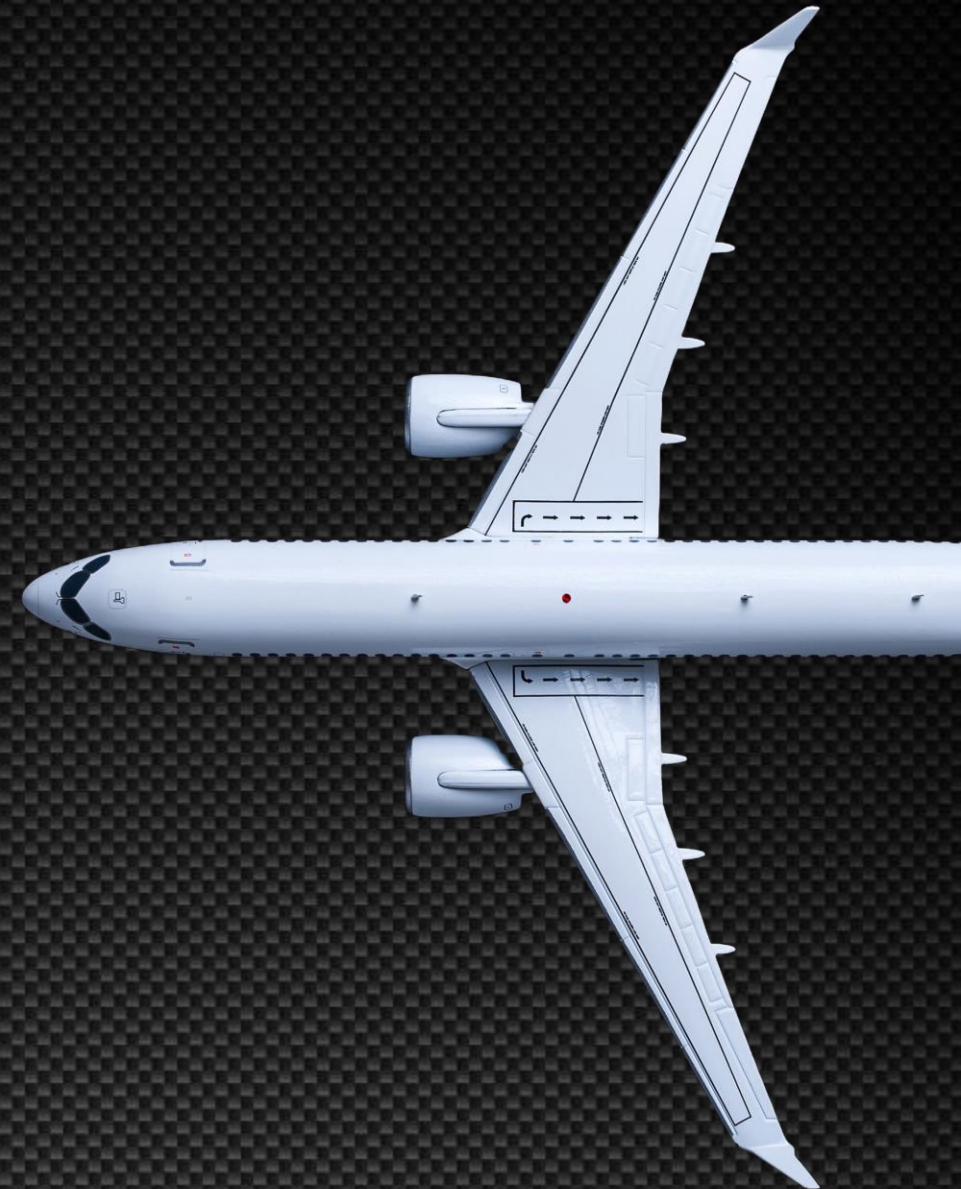
Gordon Moore

CPU, GPU and DPU

(Data Processing Unit)



Pilots fly the plane, but the crew makes it possible



CPUs and GPUs compute, but DPUs make it possible

- A DPU (Data Processing Unit) is a specialized processor designed to handle data-centric tasks such as:

Networking

- Packet processing
- Load balancing
- Overlay/underlay networking
- RDMA (Remote Direct Memory Access)

Storage

- Compression / decompression
- Encryption for data at rest
- RAID/erasure coding
- Data deduplication

Security

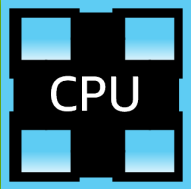
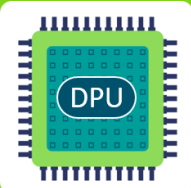
- Firewalling & packet inspection
- IPsec / TLS offload
- Multi-tenant isolation
- Zero Trust enforcement

- Think of it as the "control tower + ground services" of a data center — ensuring that data gets where it needs to go securely, efficiently, and without overloading the CPU or GPU.

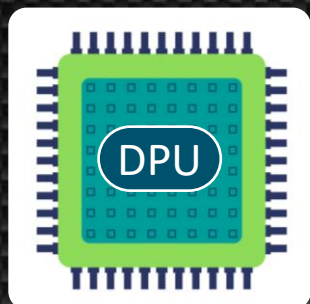
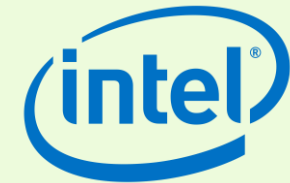
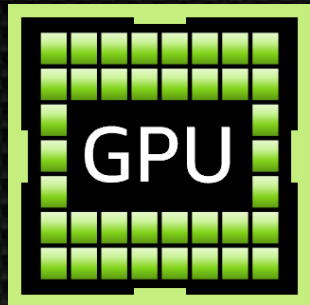
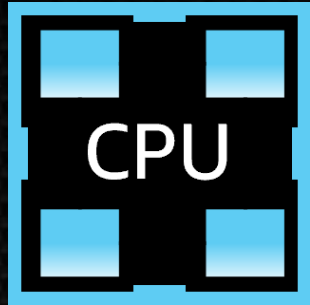
A DPU Unit



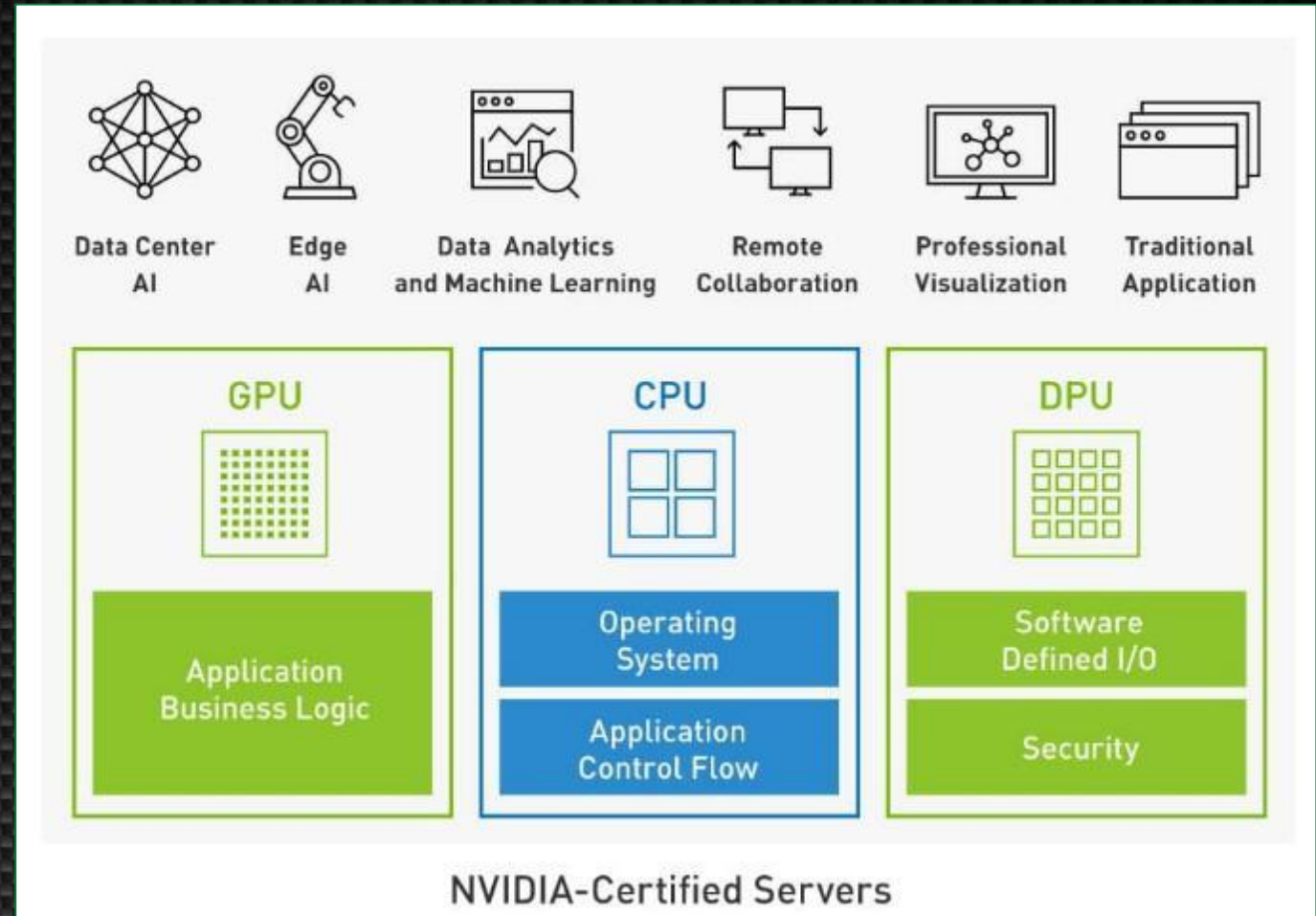
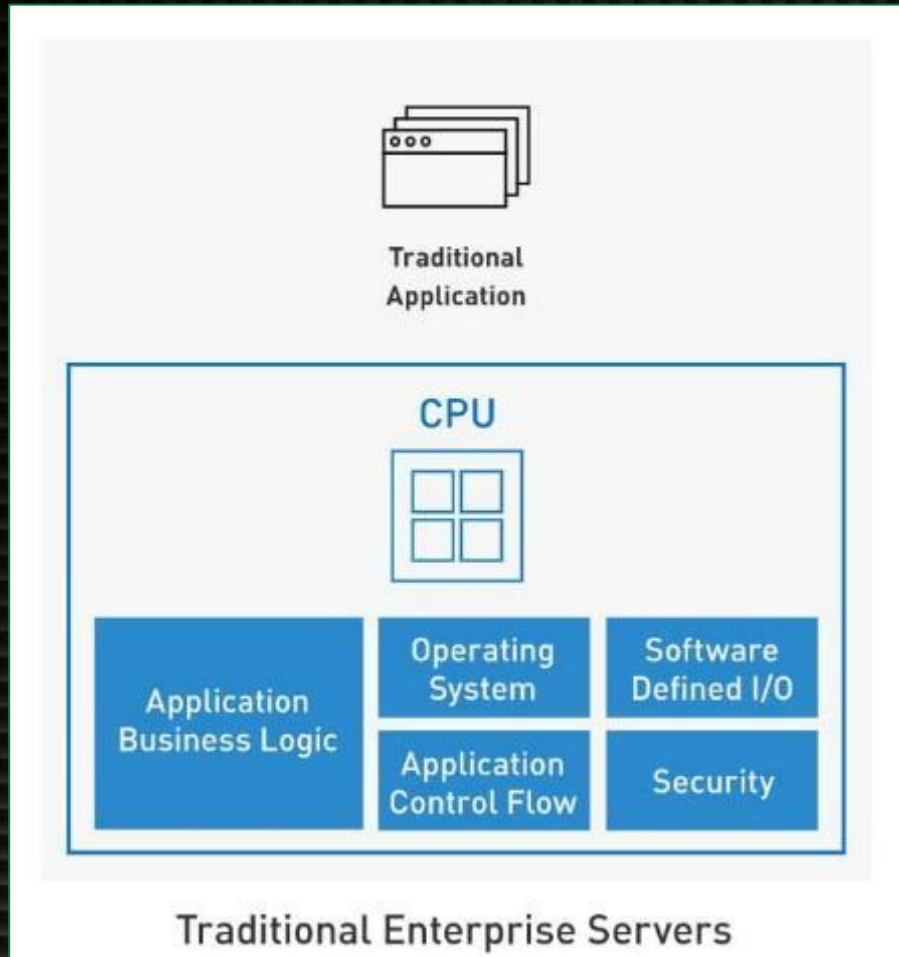
A DPU is designed to offload, accelerate, and isolate infrastructure workloads.

Processor	Main role	Best At	Weak at	Travel Analogy
	General-purpose compute	- Running applications - Decision-making - Sequential tasks - OS management	Scaling to thousands of parallel tasks	Private Jet – flexible, can go anywhere, but limited seats
	Parallel compute accelerator	- Graphics rendering - AI/ML training - Scientific simulations - Parallel math	- General-purpose logic - OS-level operations	Commercial Airliner – carries hundreds efficiently, but only on fixed routes
	Data movement & infrastructure offload	- Networking (packet processing) - Storage acceleration - Encryption - Security firewalls	- Running user apps, - Heavy math computations	Airport Ground Crew + Air Traffic Control – manages different aspects so planes can focus on flying

Market Leaders

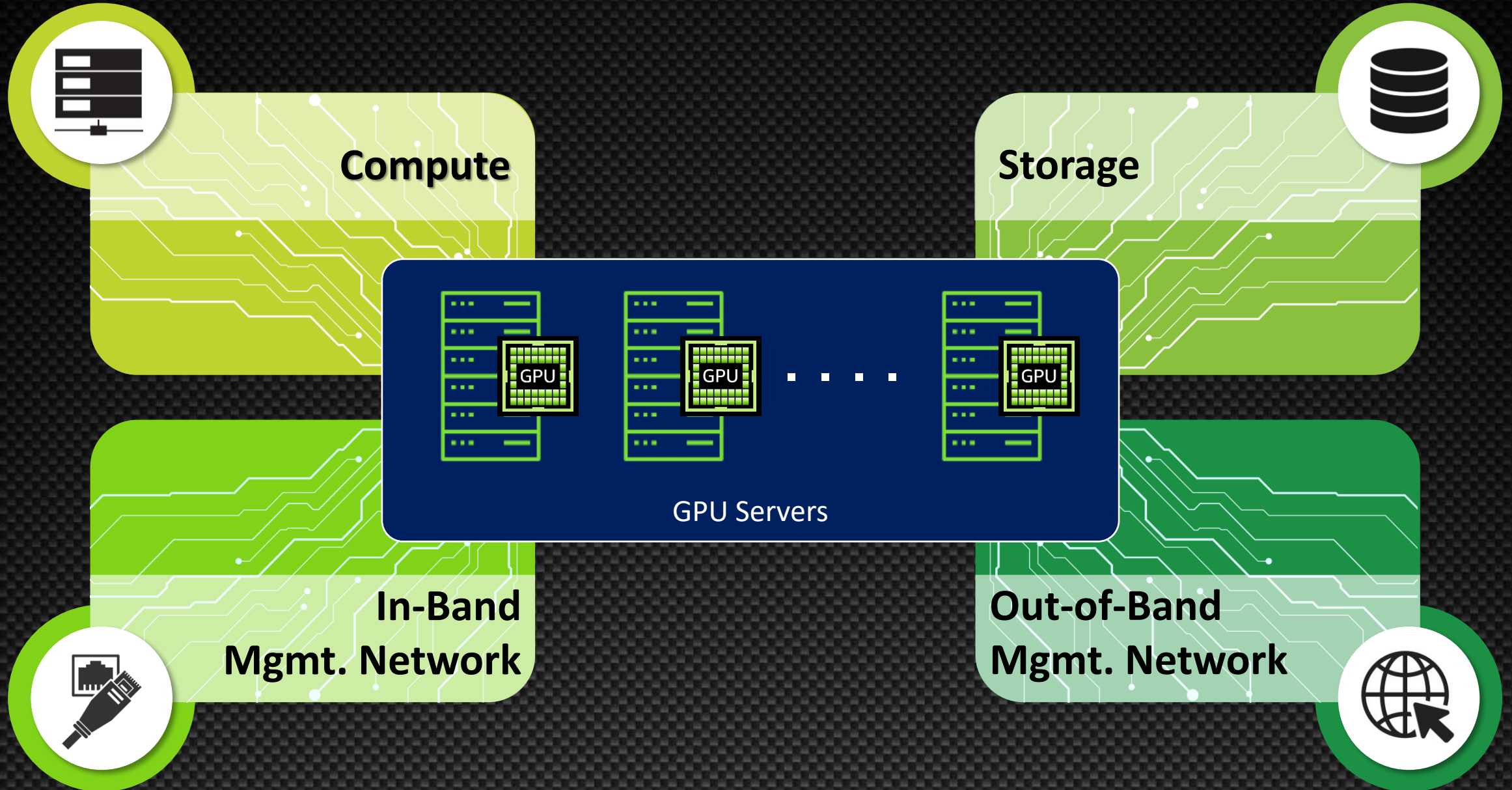


A modern server



Network

Network inside an AI centric Data Center



Why the Separation Matters?

Performance
Isolation

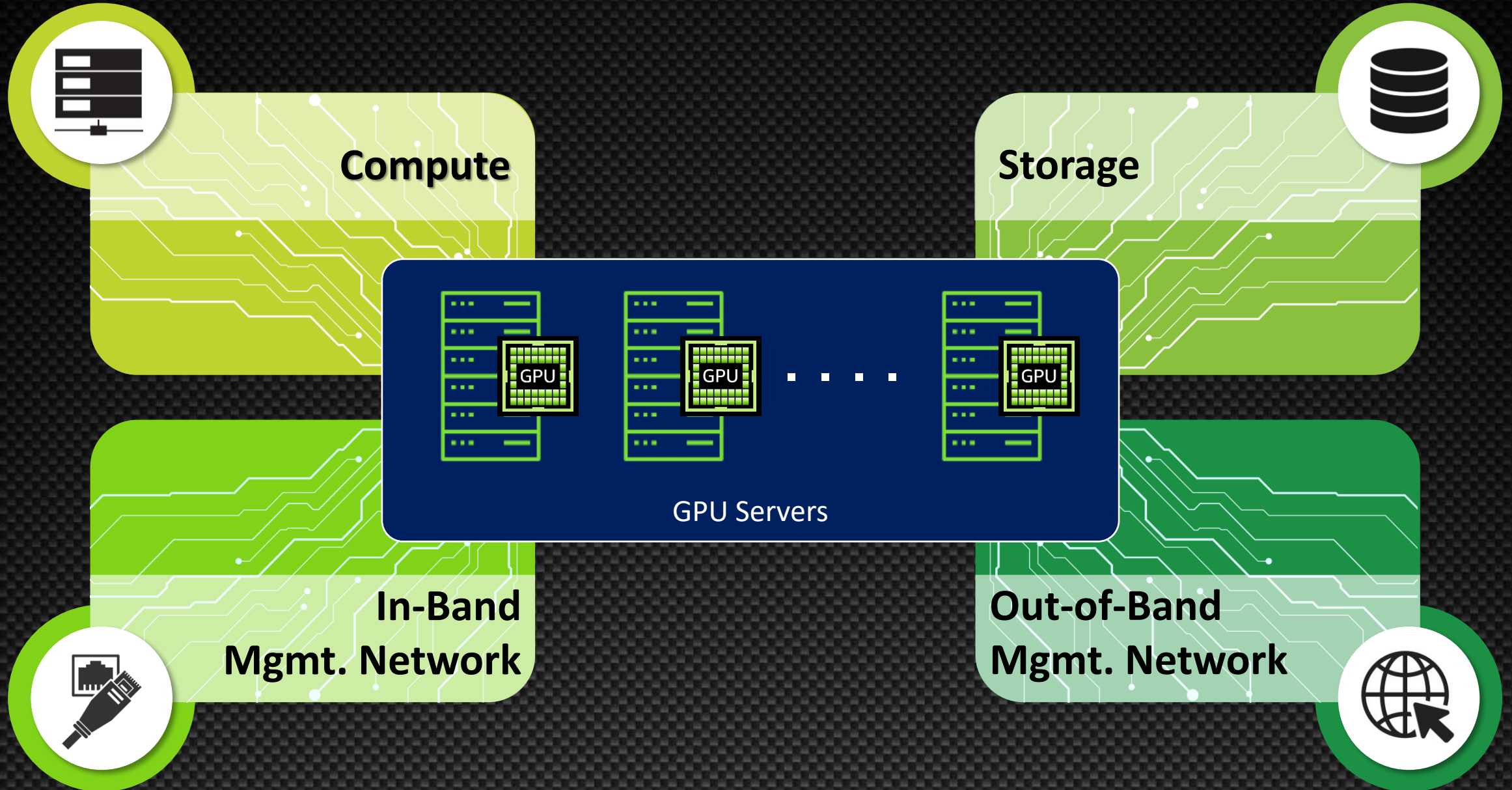
Latency
Sensitivity

Failure isolation
& robustness

Security

Scalability

Network inside an AI centric Data Center



NETWORK FABRIC

Purpose / Traffic Types

Physical / Logical Implementation

Key Design Features & Considerations



Compute

- GPU-to-GPU communication (within and across nodes)
- Backbone for training and inference

- InfiniBand / RoCE or NVLink fabrics
- High-bandwidth interconnects between compute nodes

- Extremely high throughput & ultra-low latency
- Must scale reliably with GPU/node count



Storage

- Connects compute nodes to storage appliances
- Supports file systems, checkpoints, and I/O traffic

- InfiniBand or Ethernet (RoCE), depending on design

- Multi-GB/s throughput per node
- Must remain isolated from compute traffic



In-Band Mgmt. Network

- Handles control-plane traffic, cluster management like SSH, DNS, Job scheduling
- Provides access to code repositories, external repos.

- Ethernet-based leaf-spine design
- Logical separation with VLAN/VXLAN/EVPN

- Requires reliability and moderate-high bandwidth
- Security and traffic isolation are critical



Out-of-Band Mgmt. Network

- Provides remote management function (power control, serial console) even if servers are off
- Used for recovery when in-band is unavailable

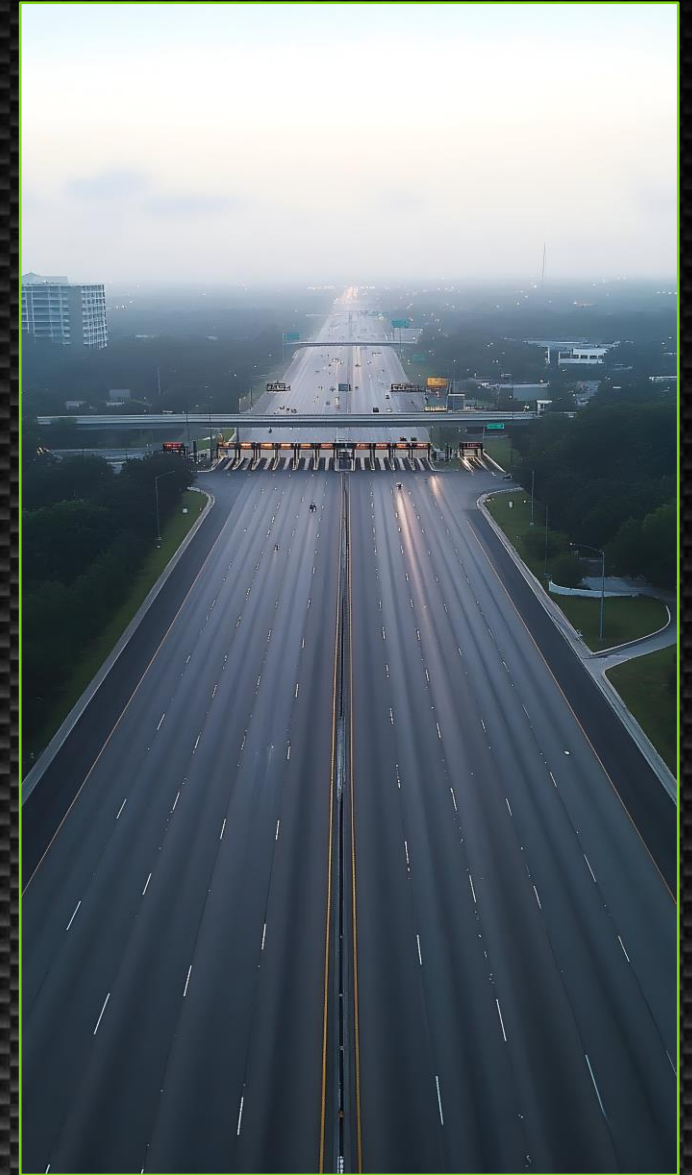
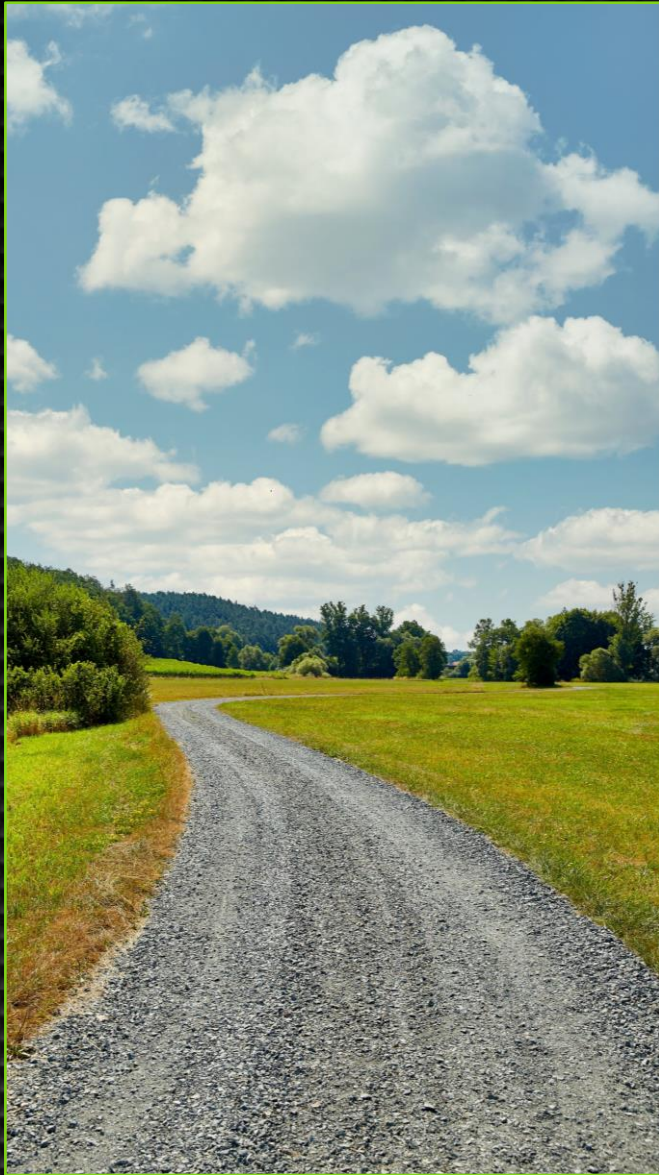
- Separate physical ports & low-speed switches

- Must be always available and redundant
- Strong access control and security are essential

Ethernet vs. InfiniBand



Evolution of Roads



Highway vs. Bullet Train Track



Shinkansen bullet trains at a depot in Fukuoka in 1975. Photograph: The Asahi Shimbun/Getty Images

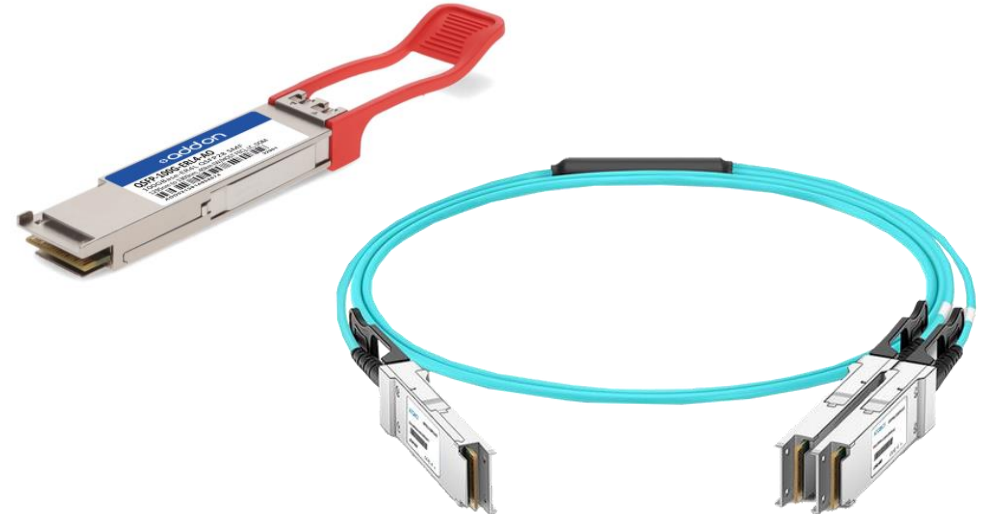
Two prominent networking technologies



Ethernet



InfiniBand

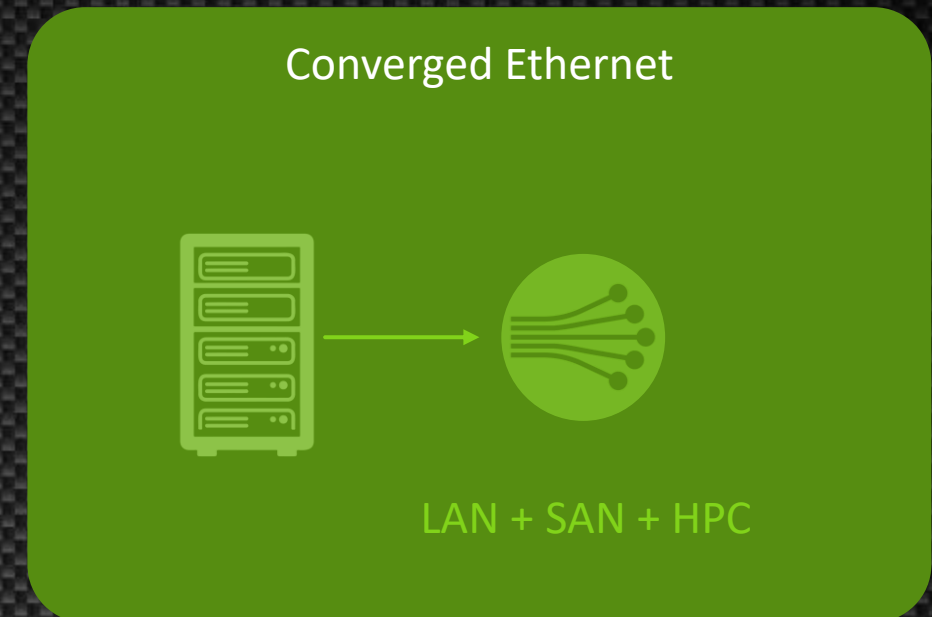
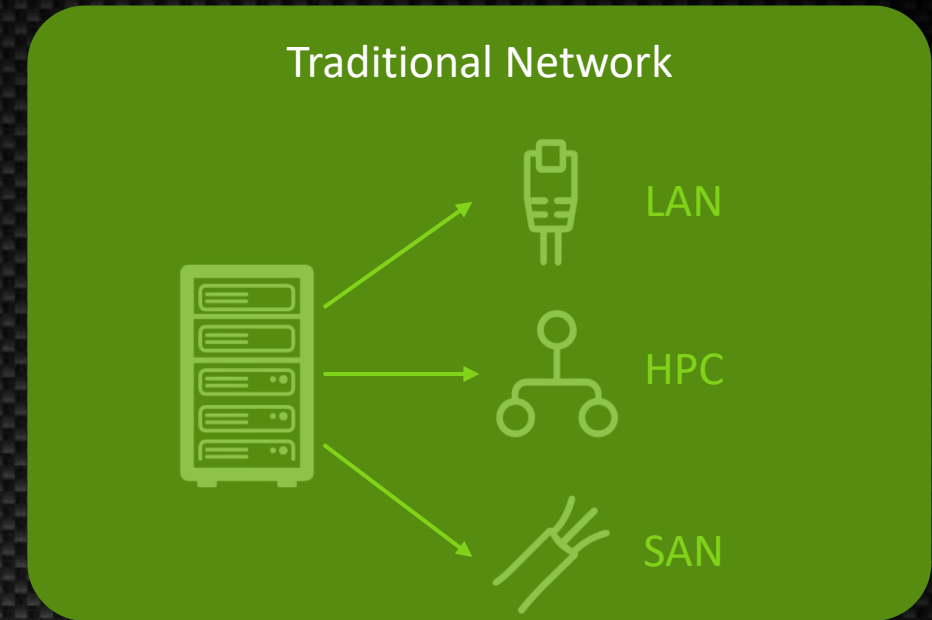


Ethernet vs. InfiniBand

Feature	Ethernet	InfiniBand
Analogy	Like a highway system: good for all vehicles (cars, trucks, buses), but traffic and stoplights slow things down	Like a dedicated bullet train line: only for high-speed trains, fewer stops, ultra-fast but limited to specific routes
History	Born in the 1970s for office networking, became the global standard	Born in the 2000s for supercomputing, remains a niche but essential technology for HPC and AI
Purpose	General-purpose networking (LANs, WANs, internet)	High-performance computing (HPC), AI clusters, data centers
Speed	1 Gbps → 400 Gbps	10 Gbps → 400 Gbps
Latency	Higher (~10–100 μ s typical)	Extremely low (~1–2 μ s)
Protocol stack	Uses TCP/IP stack (adds overhead)	Uses RDMA (Remote Direct Memory Access), bypasses CPU, very low overhead
Cost	Cheaper, commodity hardware widely available	More expensive, specialized hardware
Flexibility	Works everywhere: home, office, data centers, internet	Niche: mostly HPC, AI training clusters, storage interconnects
Ecosystem	Universally supported (all OS, devices, NICs)	Specialized drivers & software
Reliability	Good for general networking, but congestion adds delay	Lossless or near-lossless, designed for deterministic performance

Converged Ethernet (CE)

- Verb - converge
 - Be adjacent or come together.
 - "The lines converge at this point"
- Converged Ethernet
 - One Ethernet fabric
 - Carrying multiple types of traffic (LAN, SAN, HPC).
 - Higher bandwidth
 - 40/100/200/400 Gbps Ethernet links.
 - Cost efficiency
 - Fewer networks to manage, lower power usage.
 - Lower latency
 - Through RDMA / RoCE bypassing CPU



NVIDIA specific implementation

Parameter	NVIDIA InfiniBand 	NVIDIA Converged Ethernet (CE) 
Compute Nodes	NVIDIA DGX / HGX servers with HCAs (Host Channel Adapters)	NVIDIA DGX / HGX servers with NICs (ConnectX Dx)
Network Interface	ConnectX-6/7/8 HCAs	ConnectX-6 Dx / 7 NICs
Switches	Quantum / Quantum-2	Spectrum / Spectrum-X
Protocol Support	Native InfiniBand + RDMA verbs	Ethernet, RoCEv2 (RDMA over Converged Ethernet)
Most used in	HPC, AI supercomputers, Top-500 clusters (ultra-low latency)	AI training clusters in enterprises / cloud, Ethernet data centers
Max Speed (per port)	Up to 800 Gb/s (XDR InfiniBand)	Up to 800 GbE (Spectrum-4 Ethernet)
Management	OpenSM (subnet manager for IB fabrics)	Not applicable; TCP/IP stack, Ethernet switching, and Ethernet-specific management tools.

Storage



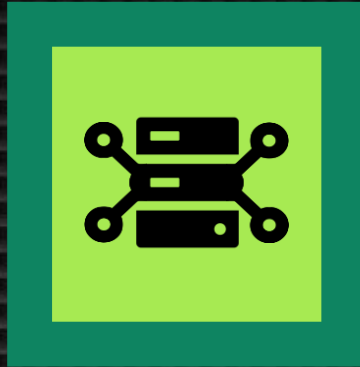
Storage

- AI workloads demand high throughput, low latency, and scalability.



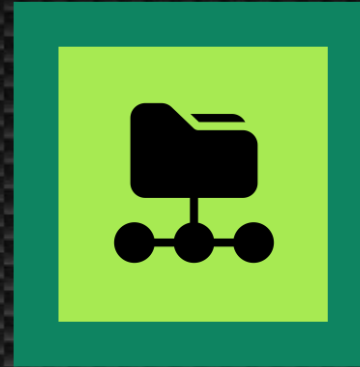
NVMe SSD
(Local Storage)

Used for fast I/O during model training and inference, limited capacity



Parallel File Systems
(Clustered Storage)

Used for shared, high-speed access across many GPUs/nodes in the cluster



Network File Systems
(Network Storage)

Used for distributing small datasets, configuration, and scripts across nodes



Object Storage
(Long-term storage)

Used for storing massive raw datasets, model checkpoints, and logs

- Tiered Storage (Hybrid Approach)

- Hot data (active training) → kept on **NVMe/parallel file system**.
- Warm/cold data (archival, old models, historical datasets) → stored in **object storage**.

Cloud vs. On-Prem



Cloud vs On-Prem GPU Infrastructure – Key Advantages



Cloud
Infrastructure

Aspect

On-Prem
Infrastructure



Lower cost barrier to entry

Key Advantage

Data security & sovereignty

Pay-as-you-go, no upfront cost

Cost Model

High upfront cost

Very flexible (scale up/down quickly)

Scalability

Limited by available hardware

Dependent on provider policies

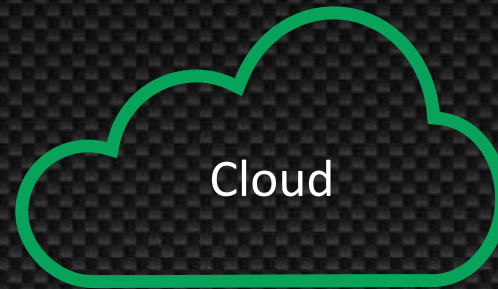
Compliance & Control

Full control over compliance & data

Key Advantages - Cloud vs On-Prem Infrastructure

Lower cost barrier to entry

- Pay-as-you-go model
- No upfront hardware purchase
- Flexible scaling up/down based on demand



Data security and sovereignty

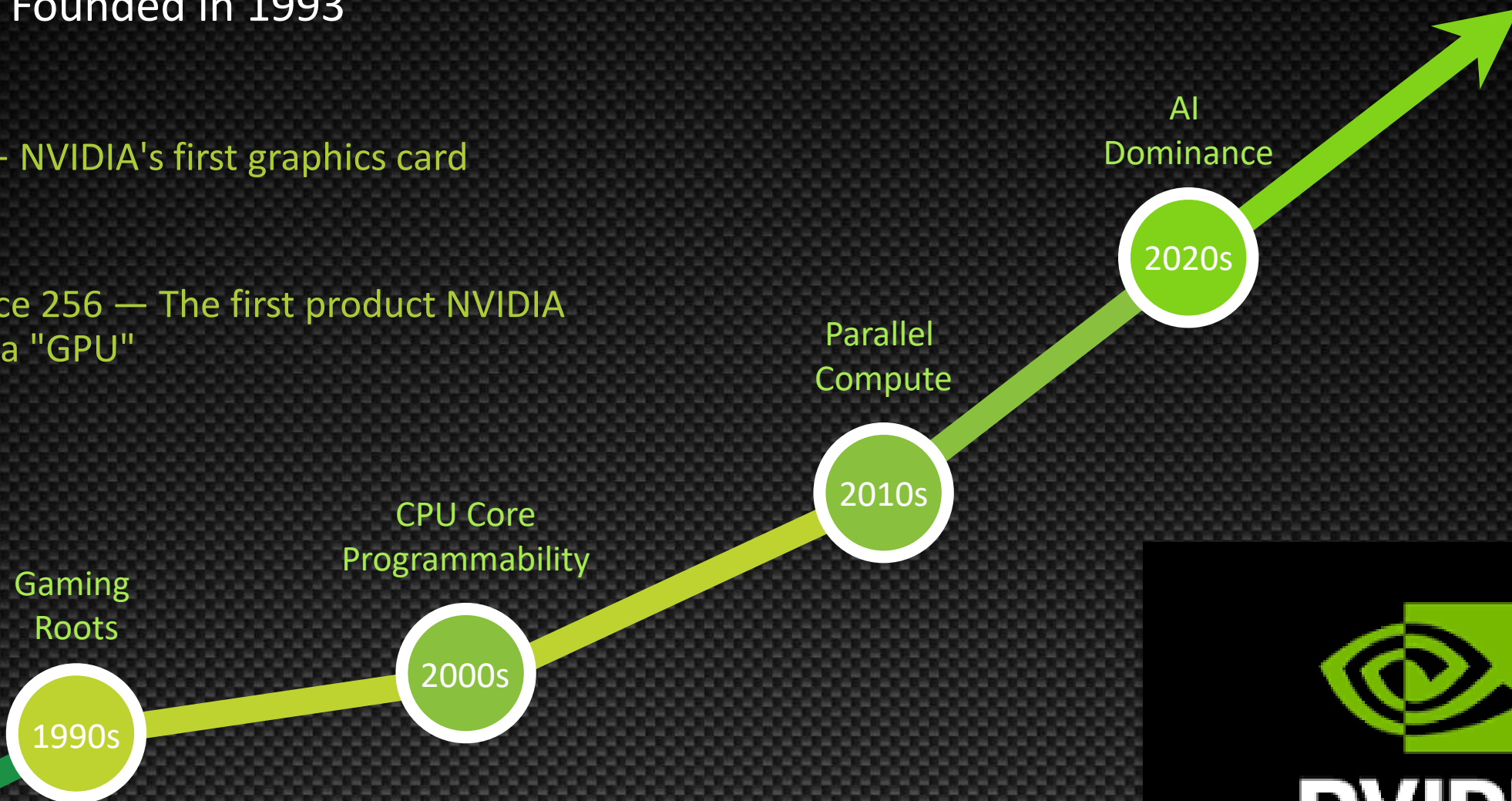
- Full control over sensitive data
- Easier compliance with local regulations
- No dependency on external cloud provider policies

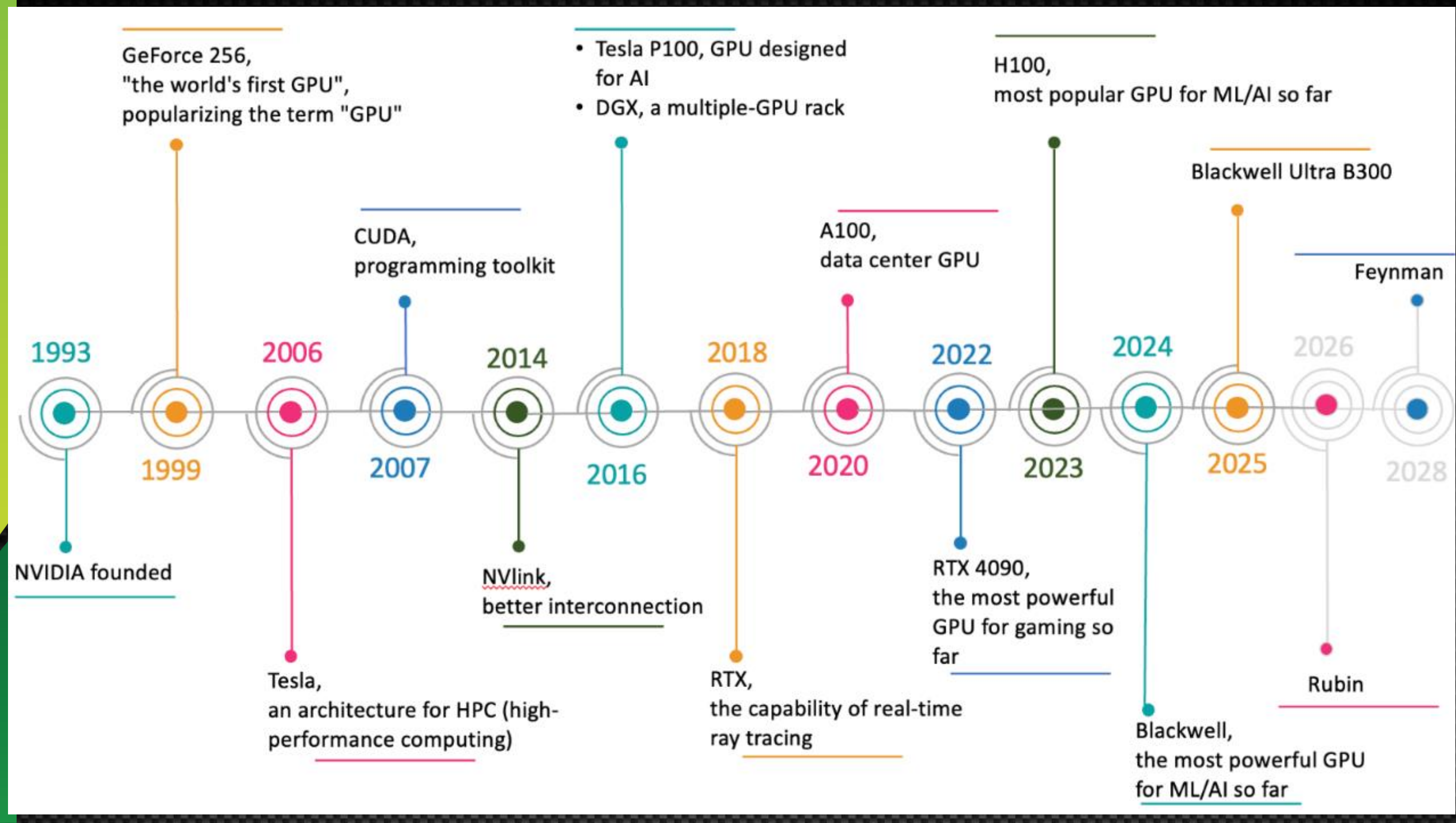
Module 3 - NVIDIA Technology Stack



NVIDIA: Powering AI GPU Innovation

- NVIDIA Founded in 1993
- 1995
 - NV1 — NVIDIA's first graphics card
- 1999
 - GeForce 256 — The first product NVIDIA called a "GPU"





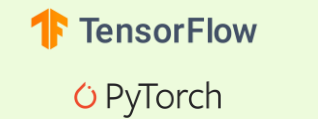
Layer 6: Applications & Vertical Solutions	Healthcare & Hospitals		Recommendation Systems		Inference Microservices	
	Clara		Merlin		NVIDIA NIMs	
Layer 5: Monitoring and Management	Monitoring & Management					
	nvidia-smi		DCGM		Base Command Manager	
Layer 4: Core Libraries	GPU Programming			GPU Communication		
	CUDA			NCCL		
Layer 3: OS, Driver and Virtualization	Operating System		GPU Driver		Virtualization	
	DGX OS		GPU Drivers		vGPU MIG (Multi-Instance GPU)	
Layer 2: Data Movement and I/O Acceleration	GPU Interconnect		GPU Direct		HPC Fabric	
	NVLink		RDMA, Storage		InfiniBand, OpenSM	
Layer 1: Physical Layer	Gaming and Workstation GPUs	Data Center AI server	AI Supercomputer	Networking Interface	DPUs	
	NVIDIA RTX Series	DGX Platform	DGX SuperPOD	ConnectX InfiniBand HCAs / NICs	BlueField DPUs / SuperNICs	

Integration

Containerization



ML Frameworks



Workload Mgmt.



Monitoring



Certified Vendors and Partners

Layer 1: Physical Layer



Layer 6: Applications & Vertical Solutions	Healthcare & Hospitals	Recommendation Systems	Inference Microservices		
	Clara	Merlin	NVIDIA NIMs		
Layer 5: Monitoring and Management	Monitoring & Management				
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	NVIDIA RTX Series	DGX Platform	DGX SuperPOD	ConnectX InfiniBand HCAs / NICs	BlueField DPUs / SuperNICs



NVIDIA RTX Series

- RTX name refers to NVIDIA's GPU architecture that includes: RT Cores, Tensor Cores, CUDA Cores
- It is a shared architecture used across gaming (GeForce RTX 40xx), professional workstations (RTX A5000, A6000), and data center GPUs (L4, L40) each tuned for its market
- RTX GPUs include hardware and software features for professional visualization, remote rendering, and virtualization

PRIMARY USE CASES

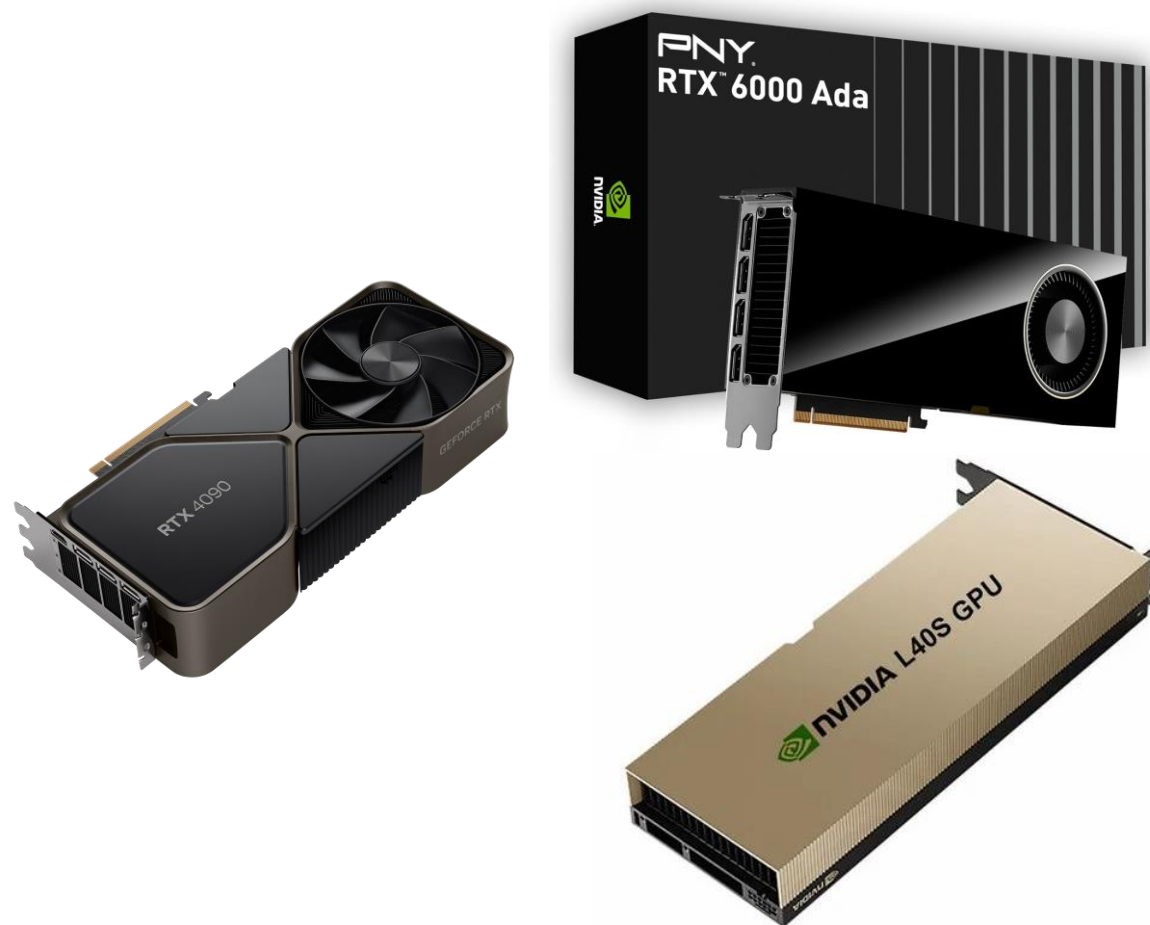
- Real-time collaboration and rendering in 3D design and visualization workflows
- Virtual workstation deployments for graphics-intensive teams via NVIDIA Virtual Workstations (vWS) across multiple users virtually.

SUITABLE FOR

- Design studios, architects, and engineering teams needing real-time visualization
- Enterprises deploying NVIDIA Certified Servers with RTX GPUs for remote collaboration

NOT SUITABLE FOR

- Large-scale AI model training or HPC workloads
- Edge or embedded systems with strict power/space constraints



DGX A100 Platform

- Data center AI system designed for training, inference, and analytics
- Integrates 8x NVIDIA GPUs with high-speed interconnects
- Optimized for large-scale workloads and multi-instance GPU (MIG) flexibility
- End-to-end solution for enterprise AI infrastructure



PRIMARY USE CASES

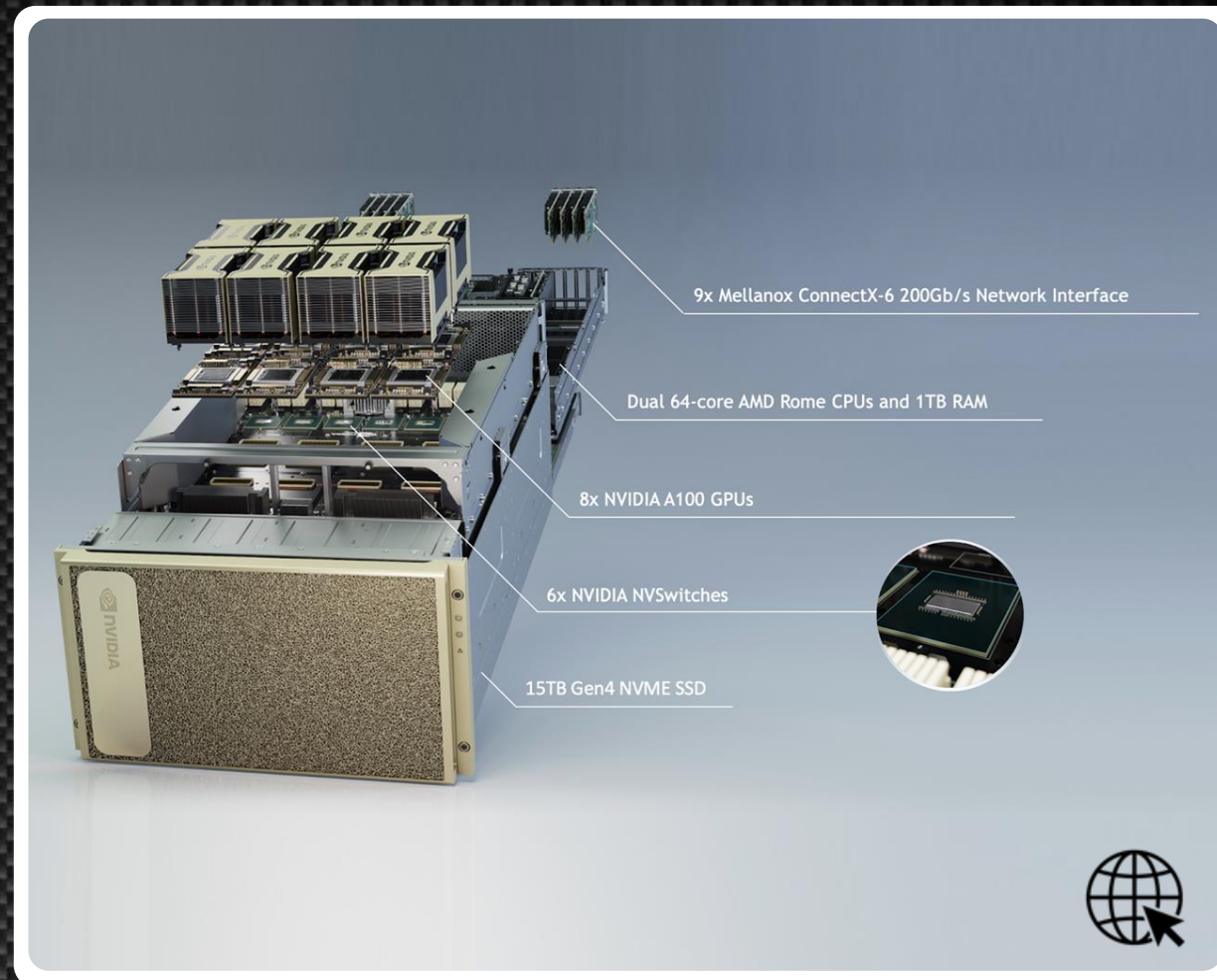
- Large-scale AI training (NLP, computer vision, speech)
- AI inference at scale with multi-tenant workloads
- High-performance data analytics and scientific work
- Federated learning and enterprise AI deployments
- Cloud-native and containerized AI workflows

SUITABLE FOR

- Research institutions and enterprise AI teams
- Training massive deep learning models
- Multi-user, multi-workload AI environments
- HPC workloads with high throughput demands

NOT SUITABLE FOR

- Consumer or small business use cases
- Edge or embedded deployments
- Low-power or cost-sensitive environments



DGX SuperPOD

- Scalable AI supercomputing infrastructure built from DGX systems
- Delivers exascale-class performance with interconnected DGX nodes
- Optimized for massive AI training, HPC, and enterprise-scale workloads
- Reference architecture for turnkey AI data centers

PRIMARY USE CASES

- Training foundation models and large language models
- HPC workloads (climate modeling, genomics, simulations)
- Large-scale AI research and innovation labs
- Multi-tenant AI cloud services within enterprises
- Federated learning across global infrastructures

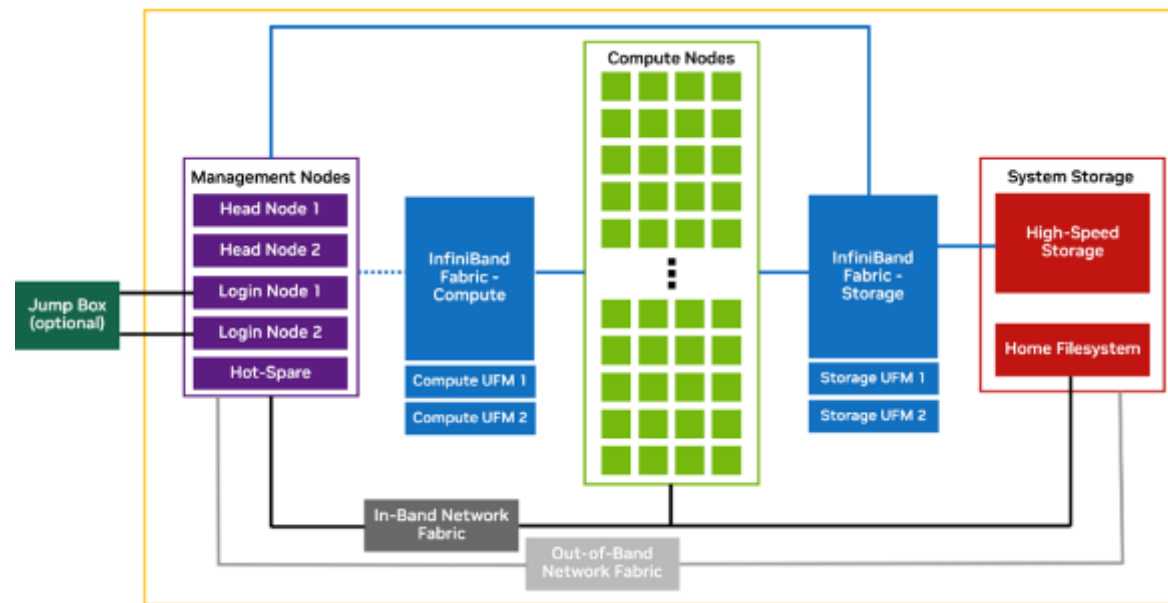
SUITABLE FOR

- Enterprises building AI centers of excellence
- National labs and research organizations
- Global-scale AI model training and deployment
- Multi-user, multi-cluster HPC and AI environments

NOT SUITABLE FOR

- Small to mid-size businesses
- Edge or embedded AI applications
- Cost-sensitive deployments

Logical depiction of the DGX SuperPOD



ConnectX InfiniBand HCAs / NICs

- High-performance interconnect adapters for InfiniBand
- Designed for ultra-low-latency networking and high throughput
- Scalable across AI, HPC, and enterprise data center workloads
- Supports advanced networking features for GPU-accelerated systems



PRIMARY USE CASES

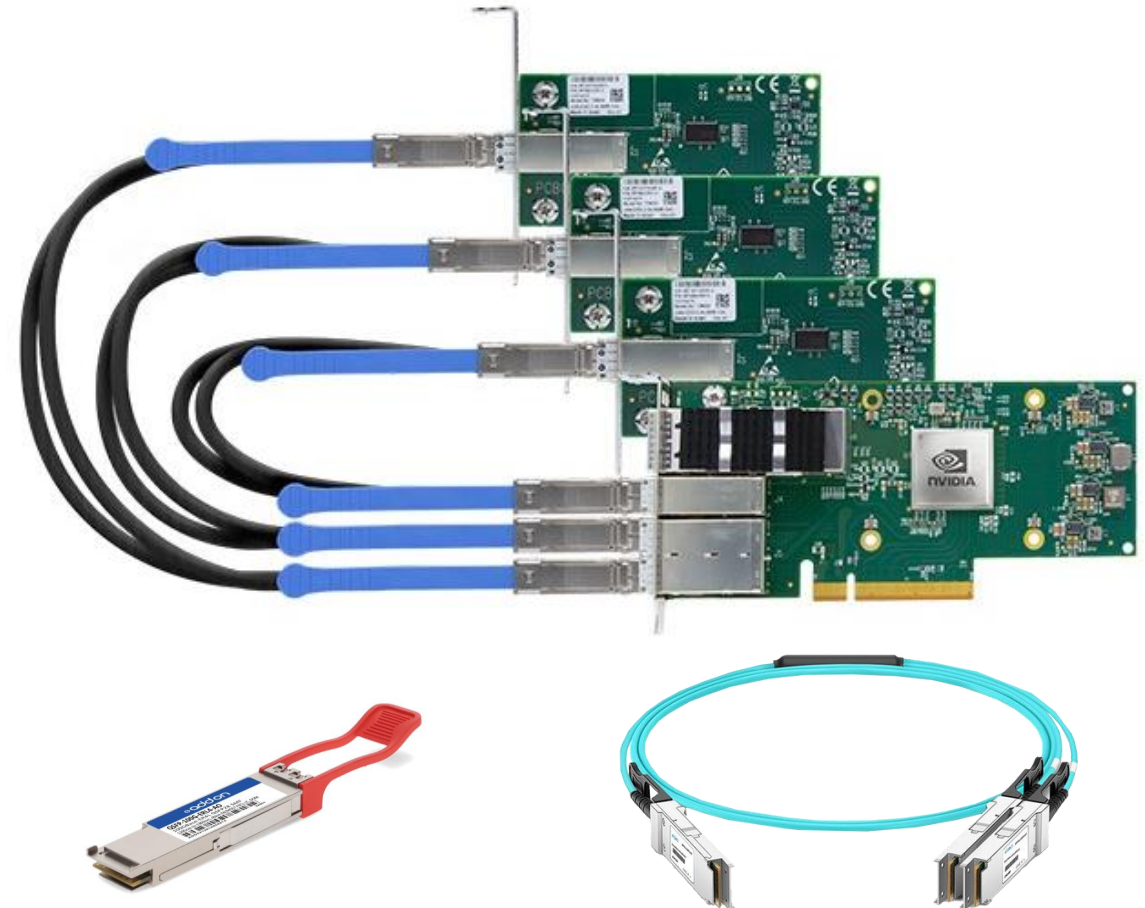
- AI training and inference clusters with GPU acceleration
- HPC workloads (scientific research, simulations)
- Data centers requiring fast, reliable networking
- Cloud-native workloads and hyperscale infrastructures
- Storage acceleration (RDMA, NVMe-oF, GPUDirect Storage)

SUITABLE FOR

- Large-scale AI and HPC clusters
- High-bandwidth, low-latency networking
- Data centers deploying RDMA and storage acceleration
- Multi-node training and computing environments

NOT SUITABLE FOR

- Consumer or small office networking
- Edge devices and embedded systems
- Workloads that do not require high-speed interconnects



BlueField DPUs / SuperNICs

- A smart NIC combining networking, compute, and acceleration engines to offload networking, storage, and security from host CPUs/GPUs. It acts like a mini-computer inside the network card, with its own CPU cores, memory, and high-speed network interfaces optimized for AI, HPC, and GPU-accelerated workloads.
- DOCA (Data Center Infrastructure on a Chip) is a software framework for programming and managing BlueField DPUs.

PRIMARY USE CASES

- Infrastructure offload (networking, storage, security)
- Zero-trust security enforcement in data centers
- Cloud-native and containerized workload acceleration
- AI and HPC clusters needing GPU-optimized networking
- Multi-tenant and hyperscale data center environments

SUITABLE FOR

- Zero-trust security and workload isolation
- Optimized networking/storage for AI and HPC
- Multi-tenant cloud environments
- Offloading infrastructure workloads from CPUs

NOT SUITABLE FOR

- Workloads not requiring network, storage, or security offload, multi-tenant isolation and high-throughput transfer.



NVIDIA Spectrum-4



NVIDIA BlueField-3

Hyperscale HPC/AI



B3240 - FHHL
2P x QSFP112, 400G, >75w

HPC/AI



B3140H - HHHL
1P x QSFP112, 400G, <75w

Cloud Computing



B3210/B3220 - FHHL
2P x QSFP112, 100/200G, >75w
B3210L/B3220L - FHHL
2P x QSFP112, 100/200G, <75w

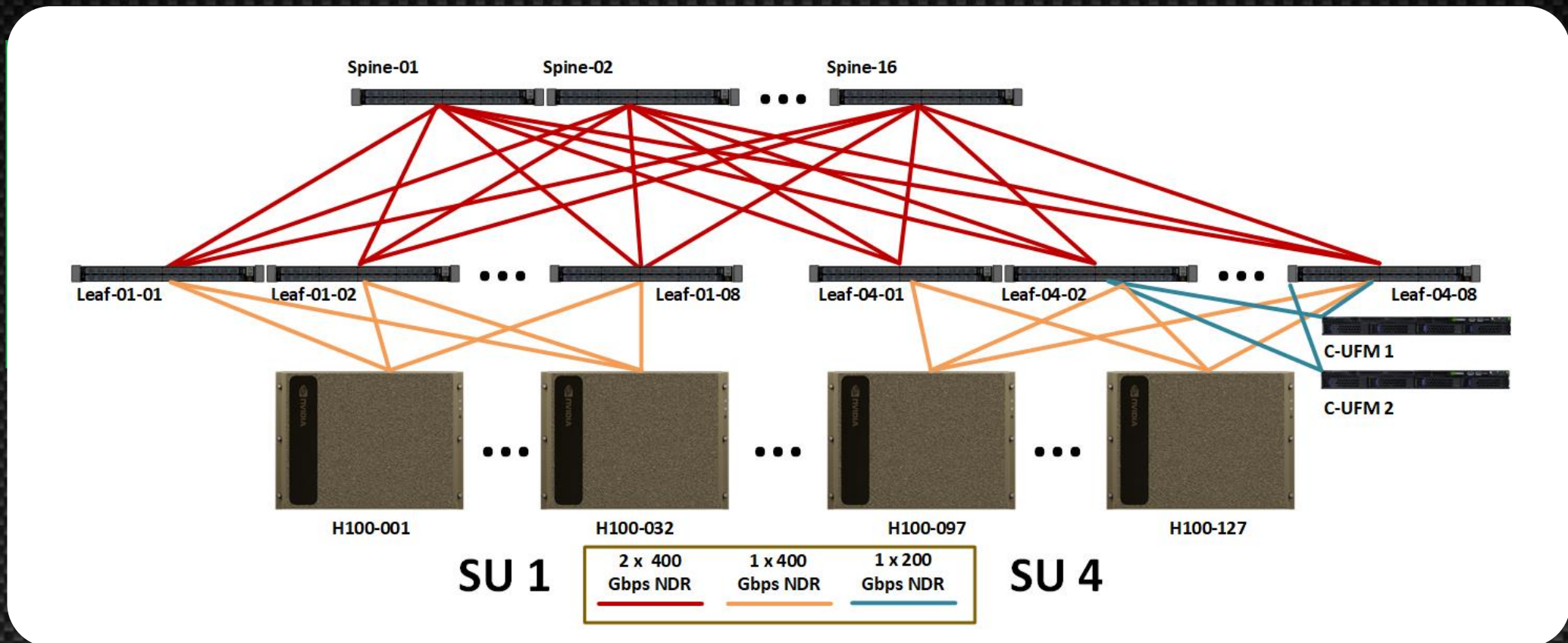
Storage



B3220SH - FHHL
2P x QSFP112, 200G, >75w

NVIDIA Reference Architectures

- Framework for system implementation
- Faster, reliable, cost-efficient design
- NVIDIA DGX & AI platforms



GPU Cores



Human Brain

FRONTAL LOBE
MOVEMENT
BEHAVIOR
PROBLEM SOLVING
SPEECH

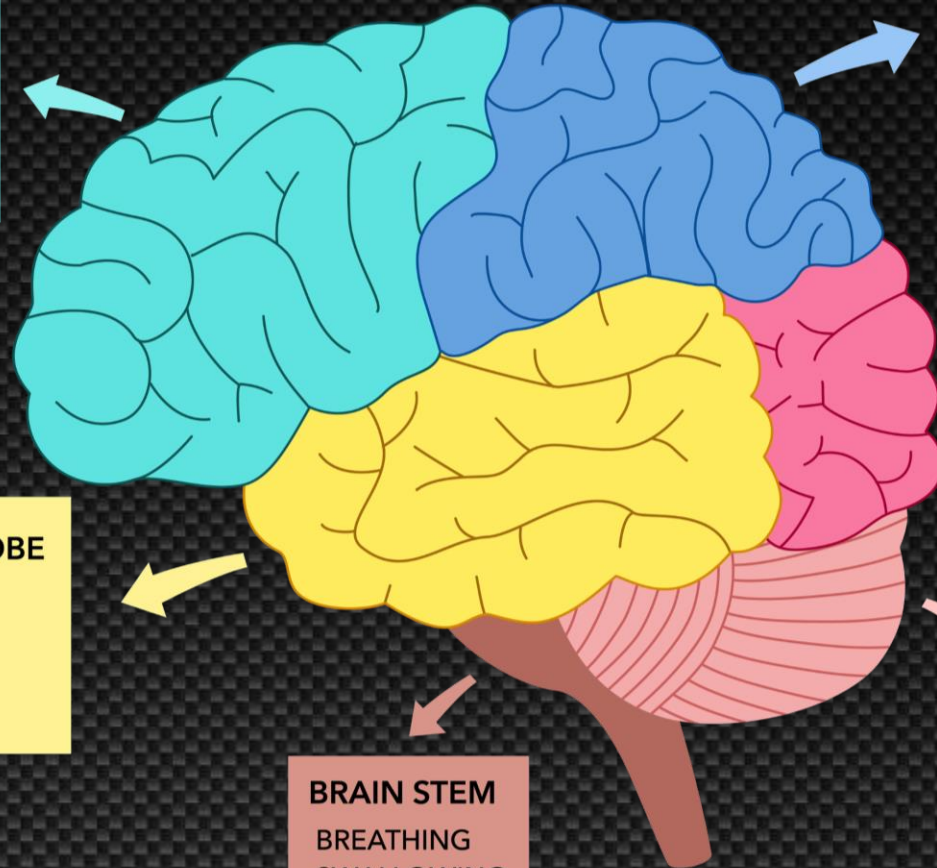
PARIETAL LOBE
INTELLIGENCE
LANGUAGE
SENSATION
REASONING

OCCIPITAL LOBE
SIGHT
VISUAL RECEPTION

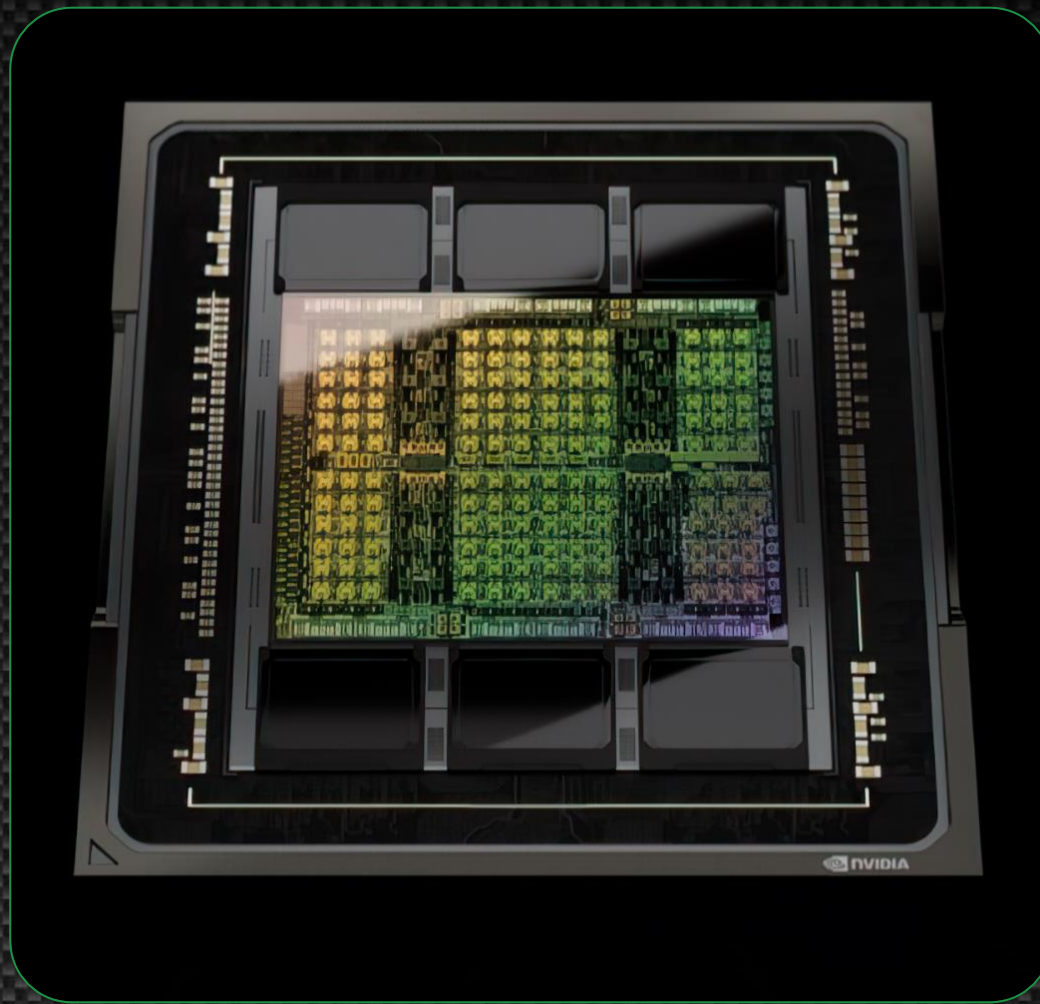
TEMPORAL LOBE
LANGUAGE
MEMORY
HEARING
EMOTION

BRAIN STEM
BREATHING
SWALLOWING

CEREBELLUM
BALANCE
MUSCLE CONTROL



GPU CORES

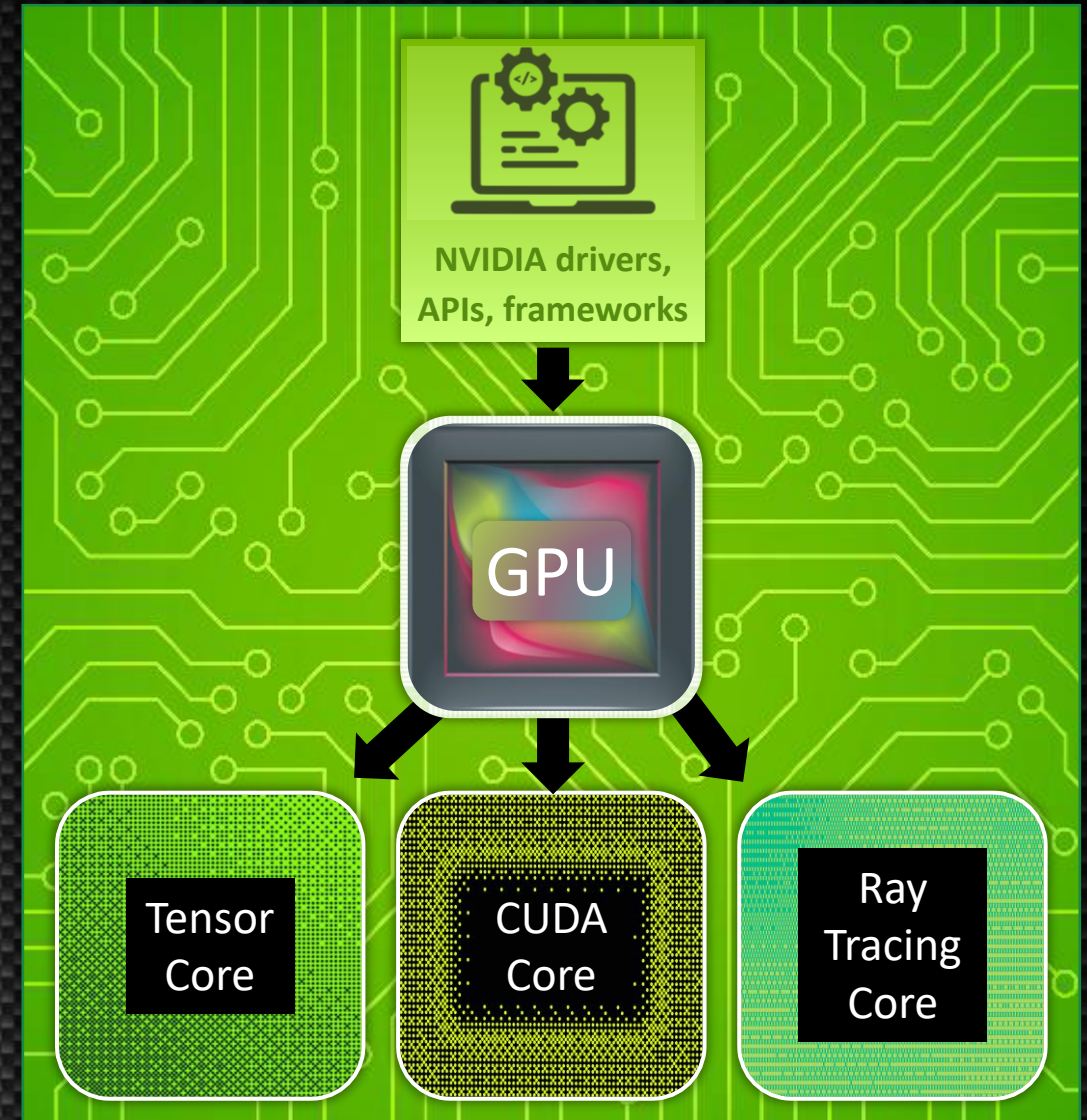


An analogy to understand GPU cores

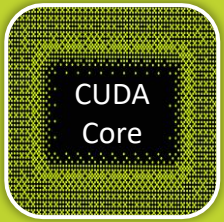
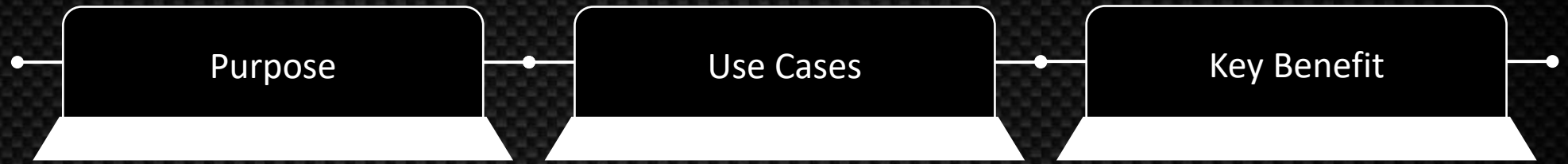
In a School



In a GPU



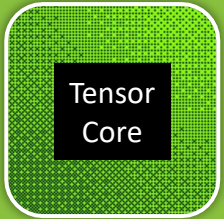
GPU Cores



General computation and
graphics rendering

Game graphics,
simulations

Flexibility in general parallel
tasks



AI and deep learning
acceleration

AI training/inference

Up to 10X AI workload
speedup



Real-time ray tracing and
light simulation

Realistic lighting in games

Realistic shadows,
reflections

Cyberpunk 2077

- Base Rendering (e.g. shapes, textures, physics, shading)
 - Sent to CUDA Cores. They do the heavy lifting of drawing the city, characters, vehicles.
- AI Enhancement (e.g. DLSS 3 upscaling, frame generation)
 - Sent to Tensor Cores. Upscale and even generate extra frames.
- Ray-Tracing (e.g. Lighting & Shadows)
 - Sent to RT Cores. Calculate how neon lights bounce off wet streets and glass windows.



NVIDIA GPUs

GPU Family	CUDA Cores	Tensor Cores	RT Cores
GeForce GTX	Yes	No	No
GeForce RTX	Yes	Yes	Yes
Data Center (V100, A100, H100)	Yes	Yes	No
Jetson (Edge AI)	Yes	Yes	No

Model	CUDA Cores	Tensor Cores	RT Cores
RTX 4090	16,384	512	128
A100	6,912	432	0

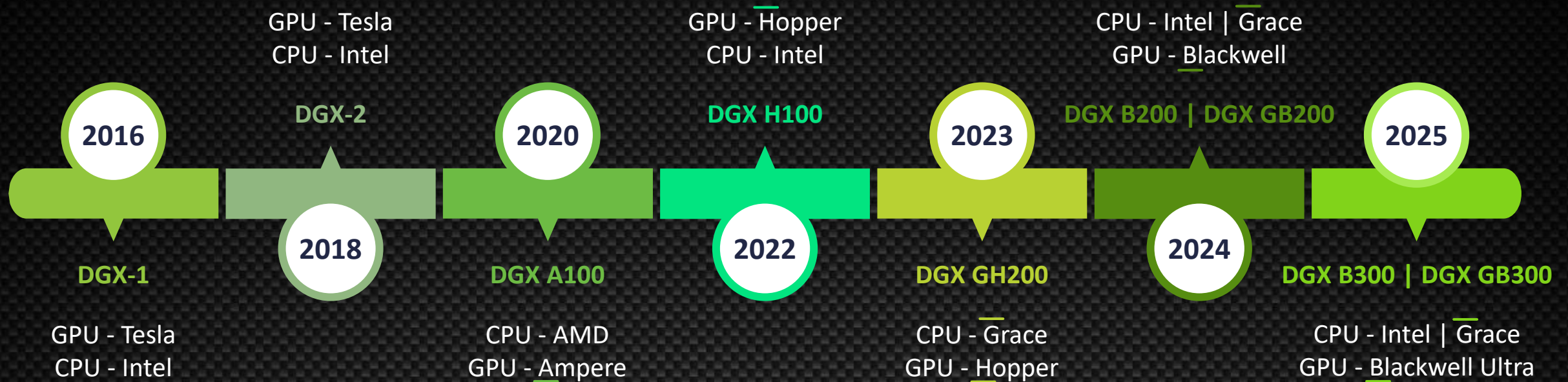
NVIDIA DGX System

DGX stands for Deep Learning System (DGX) —

- It's NVIDIA's hardware platform designed specifically for AI, ML, DL workloads.



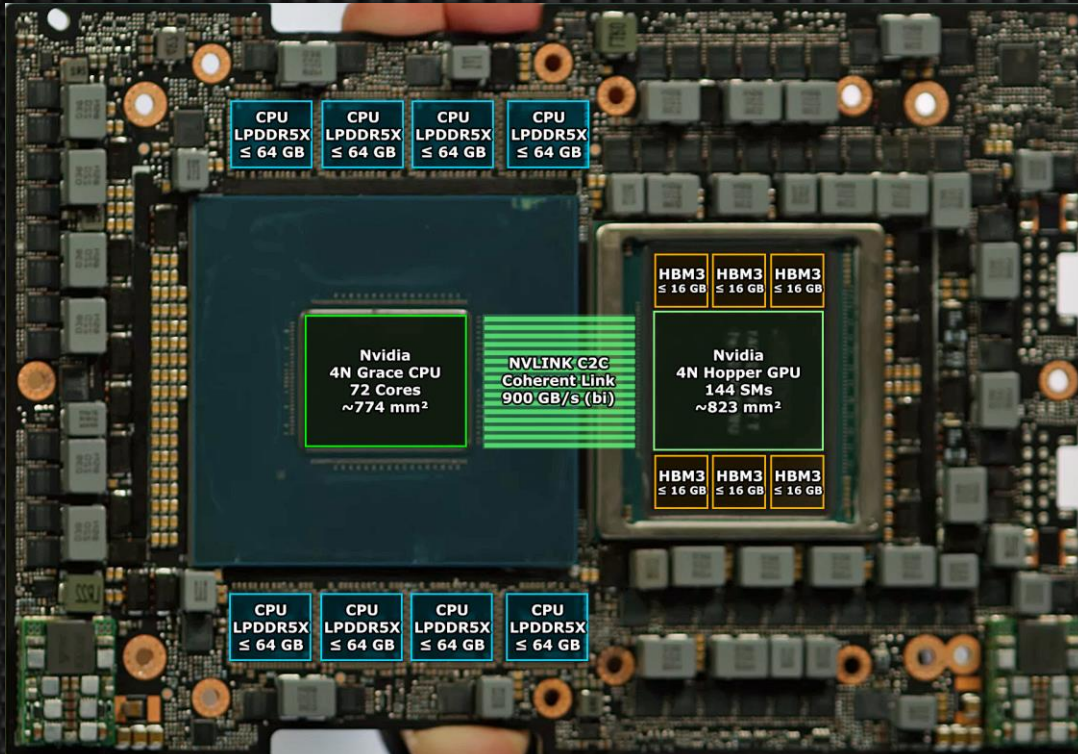
DGX Family – Timeline



Workstation	2017 - DGX Station (V100) 2021 - DGX Station (A100) 2025 - DGX Station (GB300)
Desktop	2025 - DGX Spark (GB10)

NVIDIA Grace

- NVIDIA Grace is a 72-core Arm-based CPU designed by NVIDIA to work seamlessly with NVIDIA GPUs (Hopper, Blackwell), forming Grace Hopper and Grace Blackwell Superchips.



Nvidia Grace Hopper (mixture of physical and product specific specs)

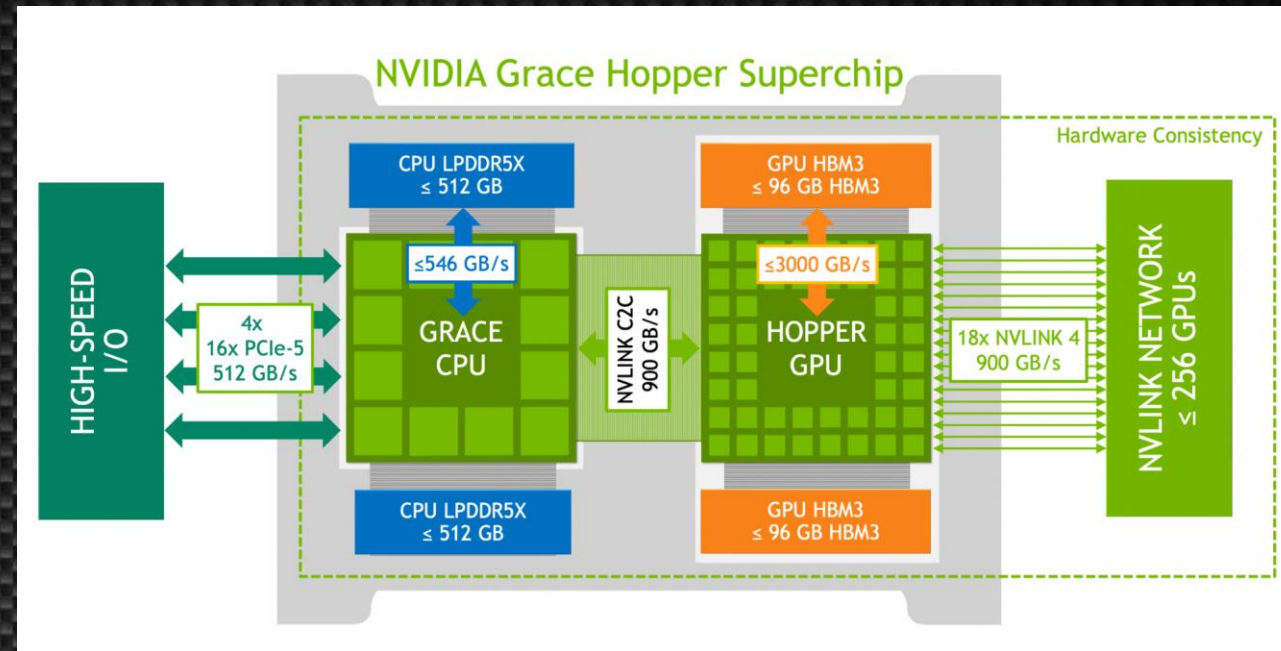
4N Grace CPU with 72 CPU Cores (Neoverse V2), ≥117 MB L3 Cache, up to 512 GB LPDDR5X memory @ 546 GB/s, 64x PCIe 5 lanes
4N Hopper GPU with 144 SMs (18432 FP32 Lanes), 60 MB L2 Cache, up to 96 GB HBM3 memory @ ~3000 GB/s

Grace Hopper image from Tom's Hardware - <https://www.tomshardware.com/news/nvidia-ceo-jensen-huang-grace-delay>

Compiled by Locuza, May 2023



NVIDIA Grace Hopper And NVIDIA Grace Superchip



NVIDIA DGX Platform

Software Stack

 OS	 Monitoring & Mgmt.	 Model Training	 Cluster Management
NVIDIA DGX OS (based on customized Ubuntu)	GPU Driver (includes nvidia-smi), DCGM	CUDA Toolkit, cuDNN, NCCL	Base Command Manager NVIDIA System Management

Hardware Components

 CPU	 GPU	 Interconnect	 Storage	 Network
Dual AMD or Intel Xeon CPUs or NVIDIA Grace	8× NVIDIA A100 or H100 Tensor Core GPUs	NVLink, NVSwitch, PCIe Gen5	Multiple Local NVMe SSDs	Dual or quad InfiniBand / Ethernet

DGX Family

 Workstation	 Server	 Cluster (SuperPOD)
To develop and prototype AI models locally.	Rack-scale AI system to run production- scale workloads.	Cluster of DGX systems at supercomputing scale.

Deployment Options

 On-Prem	 Public Cloud	 DGX Cloud Partners
Self/Partner-led deployment using NVIDIA Reference Architectures	AWS (DGX Cloud), Microsoft Azure, Google Cloud, Oracle Cloud etc.	CoreWeave, Lambda Labs, Crusoe, Yotta, SoftBank, GMI Cloud

Compare DGX A100 vs. DGX H100/H200

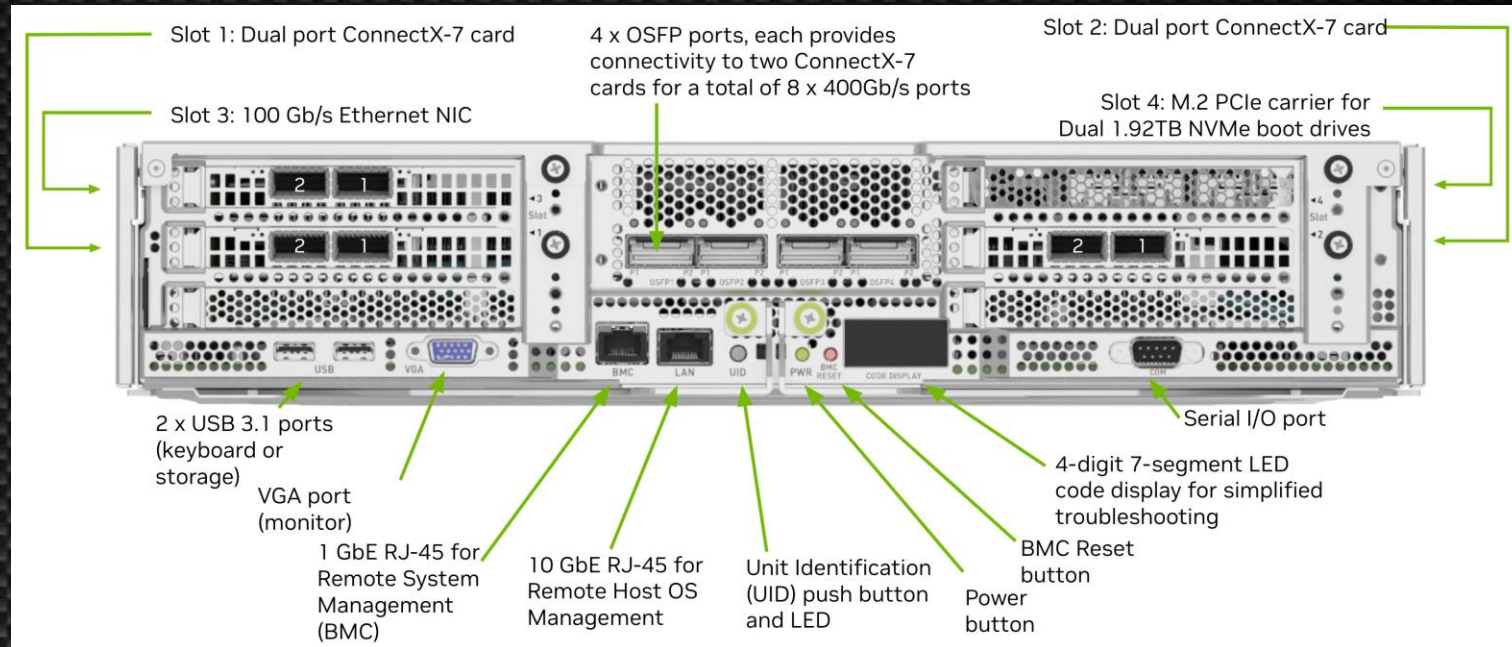
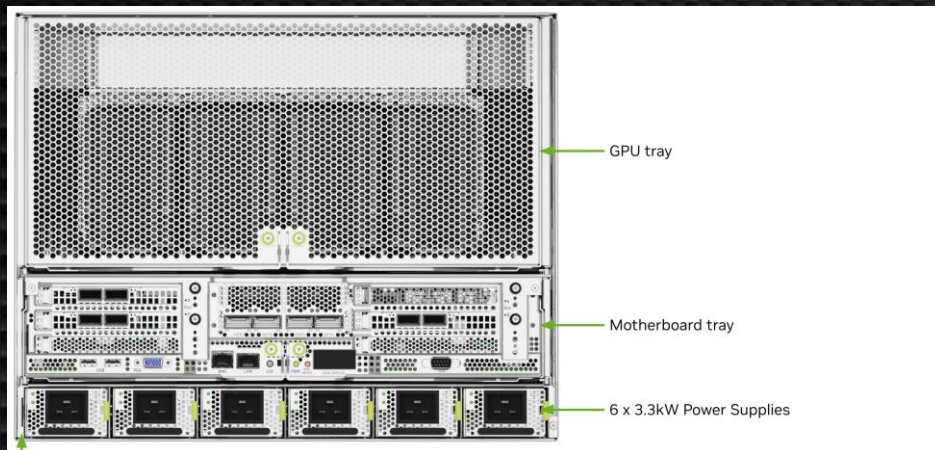
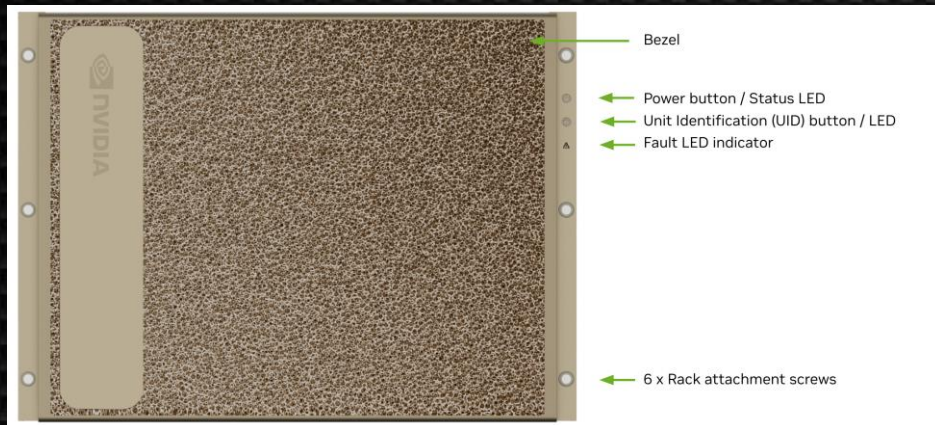


Feature	DGX A100	DGX H100 / H200
GPU	8× NVIDIA A100 Tensor Core GPUs	8× NVIDIA H100 Tensor Core GPUs
GPU Memory	80 GB per GPU (640 GB total)	H100 - 80 GB per GPU (640 GB total) H200 - 141 GB per GPU (1,128 GB total)
CPU	Dual AMD	Dual Intel Xeon
System Memory	1 TB	2 TB
Networking	8× Single-Port Mellanox ConnectX-6 VPI 200Gb/s HDR InfiniBand	10× Single-Port NVIDIA ConnectX-7 VPI 400Gb/s InfiniBand/Ethernet

DGX H100

- User Guide

- <https://docs.nvidia.com/dgx/dgxh100-user-guide/introduction-to-dgxh100.html>



Baseboard Management Controller (BMC) for out-of-band system management

The only 1 GbE RJ45 interface in the system.

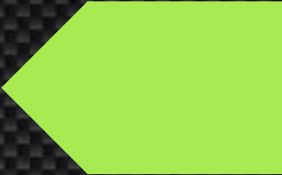
Supports Redfish, IPMI, SNMP, KVM, and Web user interface

Compare DGX A100 vs. DGX H100/H200



Feature	DGX A100	DGX H100 / H200
GPU	8× NVIDIA A100 Tensor Core GPUs	8× NVIDIA H100 Tensor Core GPUs
GPU Memory	80 GB per GPU (640 GB total)	H100 - 80 GB per GPU (640 GB total) H200 - 141 GB per GPU (1,128 GB total)
CPU	Dual AMD	Dual Intel Xeon
System Memory	1 TB	2 TB
Networking	8× Single-Port Mellanox ConnectX-6 VPI 200Gb/s HDR InfiniBand	10× Single-Port NVIDIA ConnectX-7 VPI 400Gb/s InfiniBand/Ethernet
In-Band Mgmt.	2× 1 GbE RJ45	2× 100 GbE
Out-of-Band Mgmt.	1× 1 GbE RJ45 (BMC)	1× 1 GbE RJ45 (BMC)
NVSwitch	6×	4×
Power Usage	6.5 kW max	10.2 kW max
Storage	15 TB Gen4 NVMe SSD	30.72 TB Gen4 NVMe SSD
Software Stack	NVIDIA DGX OS 5, CUDA 12.2	NVIDIA DGX OS 6, CUDA 12.4
Target Use Cases	AI training, inference, data analytics, HPC	Large-scale AI training, inference, data analytics, HPC

Layer 6: Applications & Vertical Solutions	Healthcare & Hospitals	Recommendation Systems	Inference Microservices		
	Clara	Merlin	NVIDIA NIMs		
Layer 5: Monitoring and Management	Monitoring & Management				
	nvidia-smi	DCGM	Base Command Manager		
Layer 4: Core Libraries	GPU Programming		GPU Communication		
	CUDA		NCCL		
Layer 3: OS, Driver and Virtualization	Operating System	GPU Driver	Virtualization		
	DGX OS	GPU Drivers	vGPU MIG (Multi-Instance GPU)		
Layer 2: Data Movement and I/O Acceleration	GPU Interconnect	GPU Direct	HPC Fabric		
	NVLink	RDMA, Storage	InfiniBand, OpenSM		
Layer 1: Physical Layer	Gaming and Workstation GPUs	Data Center AI server	AI Supercomputer	Networking Interface	DPU
	NVIDIA RTX Series	DGX Platform	DGX SuperPOD	ConnectX InfiniBand HCAs / NICs	BlueField DPUs / SuperNICs

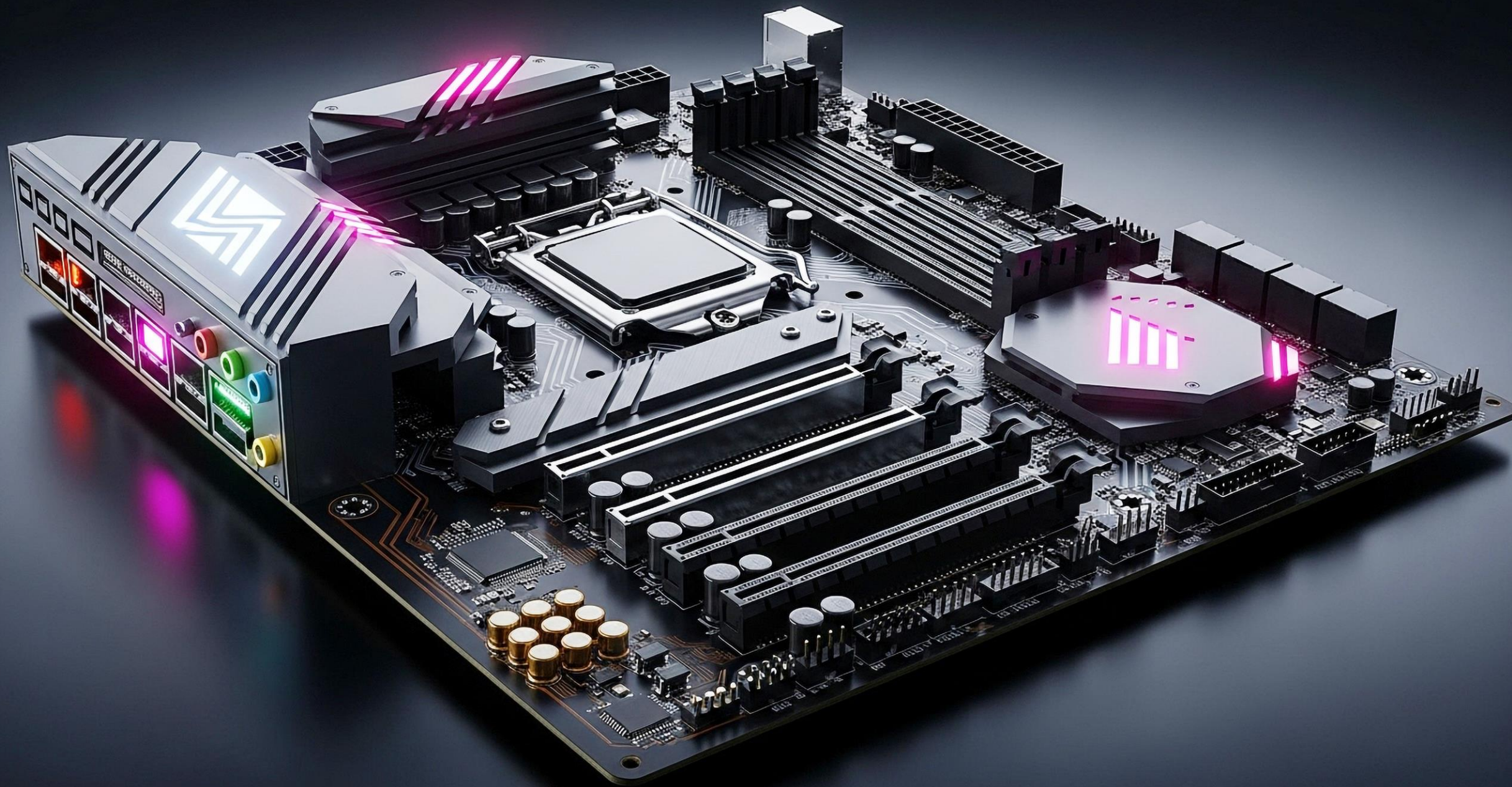


Layer 2: Data Movement and I/O Acceleration

GPU Interconnect	GPU Direct	HPC Fabric
NVLink	RDMA, Storage	InfiniBand, OpenSM

Skybridge at Petronas Tower











NVLink

- NVLink creates a high-speed, direct connection between multiple GPUs or between a CPU and GPU. Unlike the slower standard PCI Express (PCIe) bus, NVLink is a dedicated, wire-based link designed for high-performance data transfers.
- NVLink requires both hardware and software to function. The hardware components create the physical pathways, while the software protocols and drivers manage the data transfer and memory pooling.

- **NVLink bridges:** For consumer and professional graphics cards
- **Integrated NVLinks:** The NVLink technology is integrated directly into the silicon
- **NVSwitch:** For large-scale AI and supercomputing clusters, NVIDIA uses dedicated NVSwitch chips

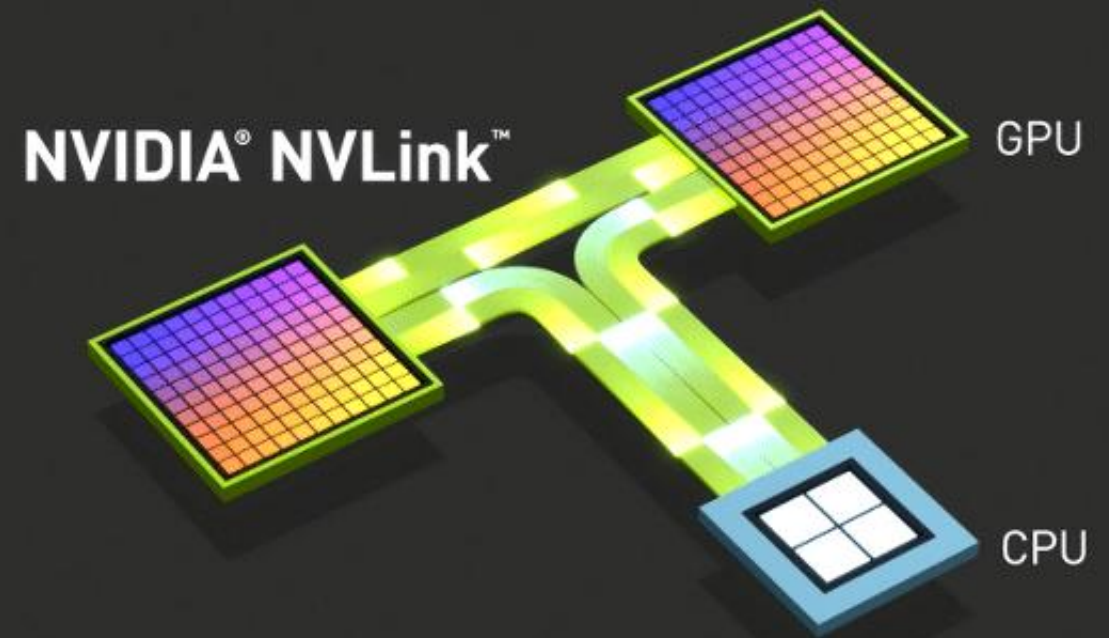
SUITABLE FOR

- AI researchers and enterprises training large models
- HPC clusters needing fast GPU interconnects
- Multi-GPU servers and supercomputers
- Workloads limited by PCIe bandwidth

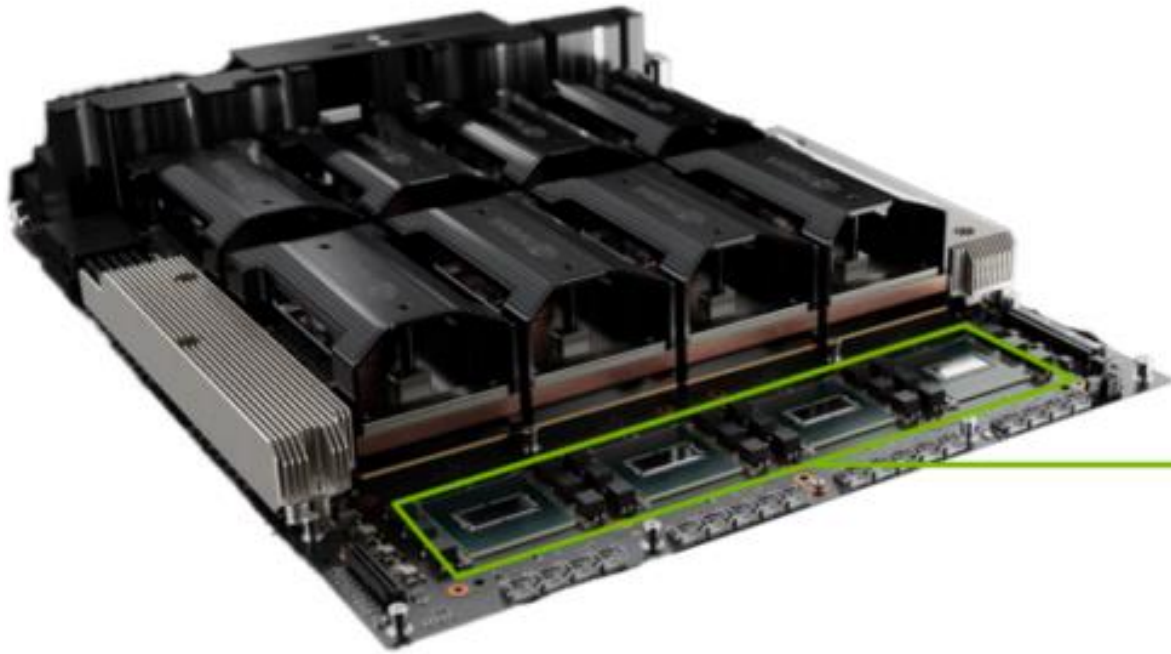
NOT SUITABLE FOR

- Single-GPU or low-power systems
- Consumer or small-scale deployments
- Workloads without heavy GPU-to-GPU communication

A Private express highway between GPUs
(and sometimes CPUs)

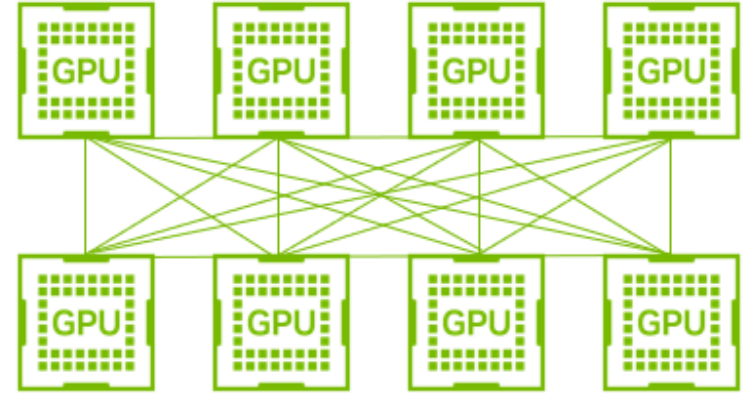


NVSwitch

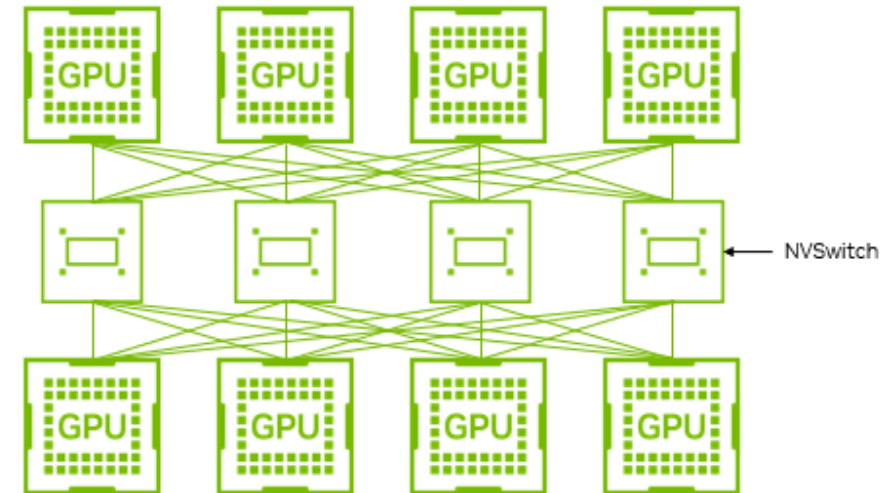


NVSwitch

Multi-GPU Configuration without NVSwitch



Multi-GPU Configuration with NVSwitch



Layer 2: Data Movement and I/O Acceleration

GPU Interconnect	GPU Direct	HPC Fabric
NVLink	RDMA, Storage	InfiniBand, OpenSM

InfiniBand

- High-performance interconnect standard for low latency, high bandwidth communication among servers, and storage.
- Open standard, though today NVIDIA is the primary commercial supplier of InfiniBand hardware.
- Offered via adapters, switches, cables, and silicon from Mellanox (acquired by NVIDIA in 2020).
- With Mellanox, NVIDIA gained end-to-end networking (InfiniBand + Ethernet) capabilities.

PRIMARY USE CASES

- Distributed GPU clusters (AI training, HPC) requiring fast inter-GPU communication
- High-performance compute and simulation workloads
- Storage networks using RDMA (Remote Direct Memory Access)

SUITABLE FOR

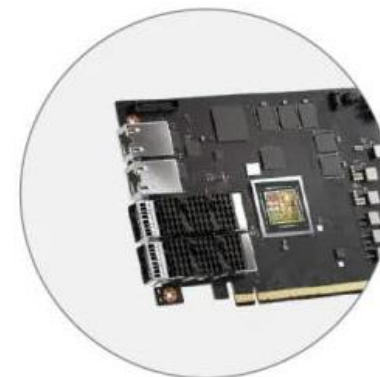
- Large-scale AI / HPC clusters
- Environments where inter-node communication is performance-critical
- Scenarios needing minimal latency and high throughput

NOT SUITABLE FOR

- Small-scale setups where Ethernet is sufficient
- Use cases with minimal data exchange between nodes
- Environments where cost trumps performance



ADAPTER



DPU



SWITCH



CABLE

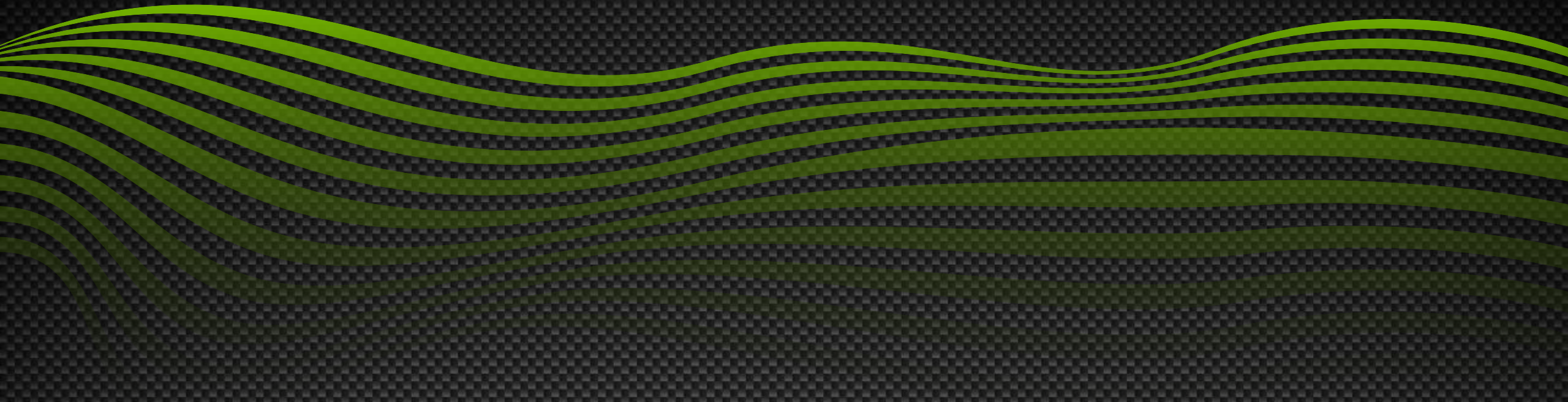
NVIDIA specific implementation

Parameter	NVIDIA InfiniBand 	NVIDIA Converged Ethernet (CE) 
Compute Nodes	NVIDIA DGX / HGX servers with HCAs (Host Channel Adapters)	NVIDIA DGX / HGX servers with NICs (ConnectX Dx)
Network Interface	ConnectX-6/7/8 HCAs	ConnectX-6 Dx / 7 NICs
Switches	Quantum / Quantum-2	Spectrum / Spectrum-X
Max Speed (per port)	Up to 800 Gb/s (XDR InfiniBand)	Up to 800 GbE (Spectrum-4 Ethernet)
Management	OpenSM (subnet manager for IB fabrics)	Not applicable; TCP/IP stack, Ethernet switching, and Ethernet-specific management tools.

Layer 2: Data Movement and I/O Acceleration

GPU Interconnect	GPU Direct	HPC Fabric
NVLink	RDMA, Storage	InfiniBand, OpenSM

Direct Memory Access (DMA)



DMA (pre cursor to RDMA)

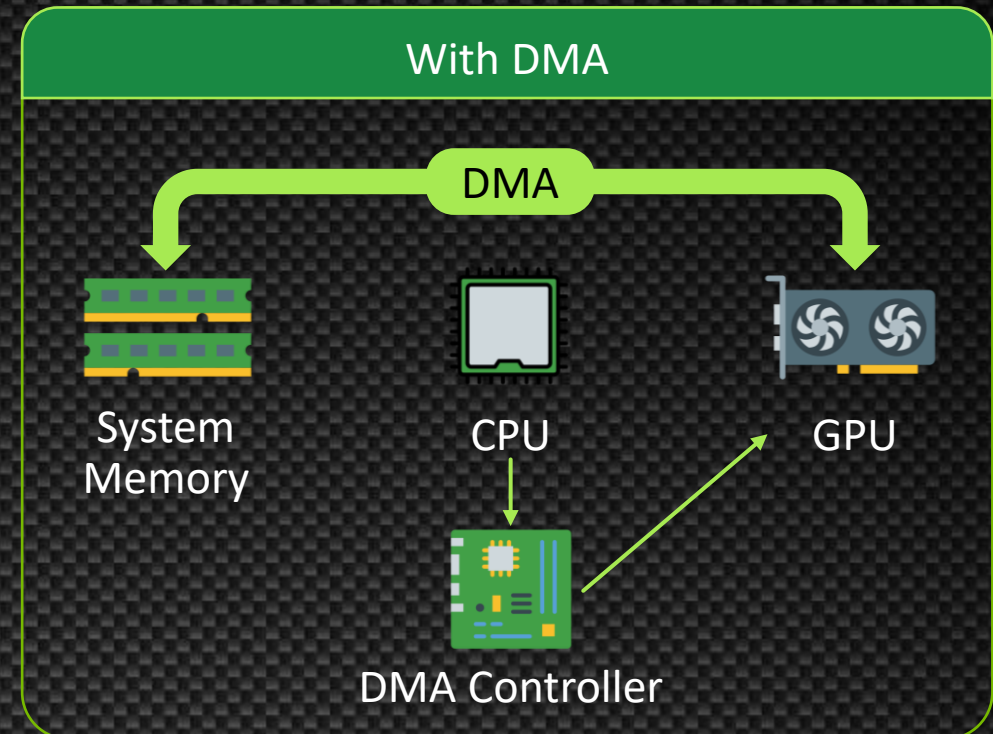
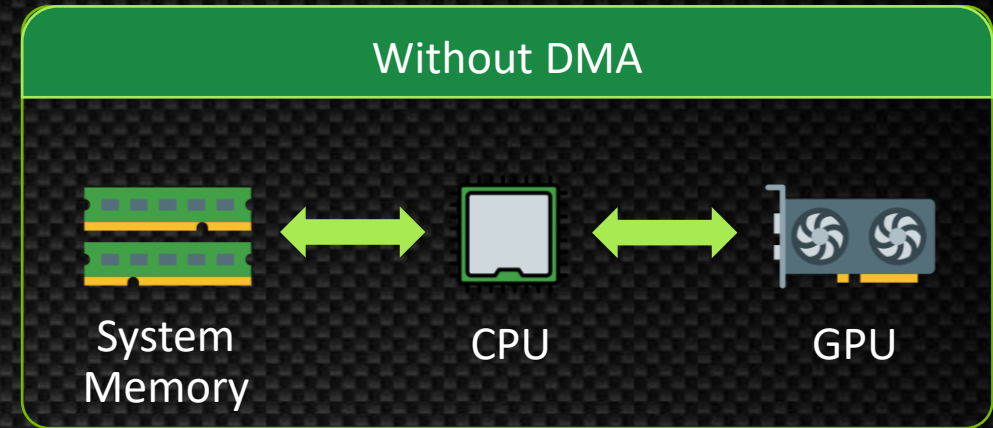
- Direct Memory Access (DMA)
 - A hardware feature that lets supported devices transfer data to/from memory without the CPU acting as middleman.
- Why It Matters for AI:

Without DMA:

- CPU must copy every byte from device → memory
- CPU utilization: 90-100% for data transfer
- GPU sits idle waiting for data
- Training slows to a crawl

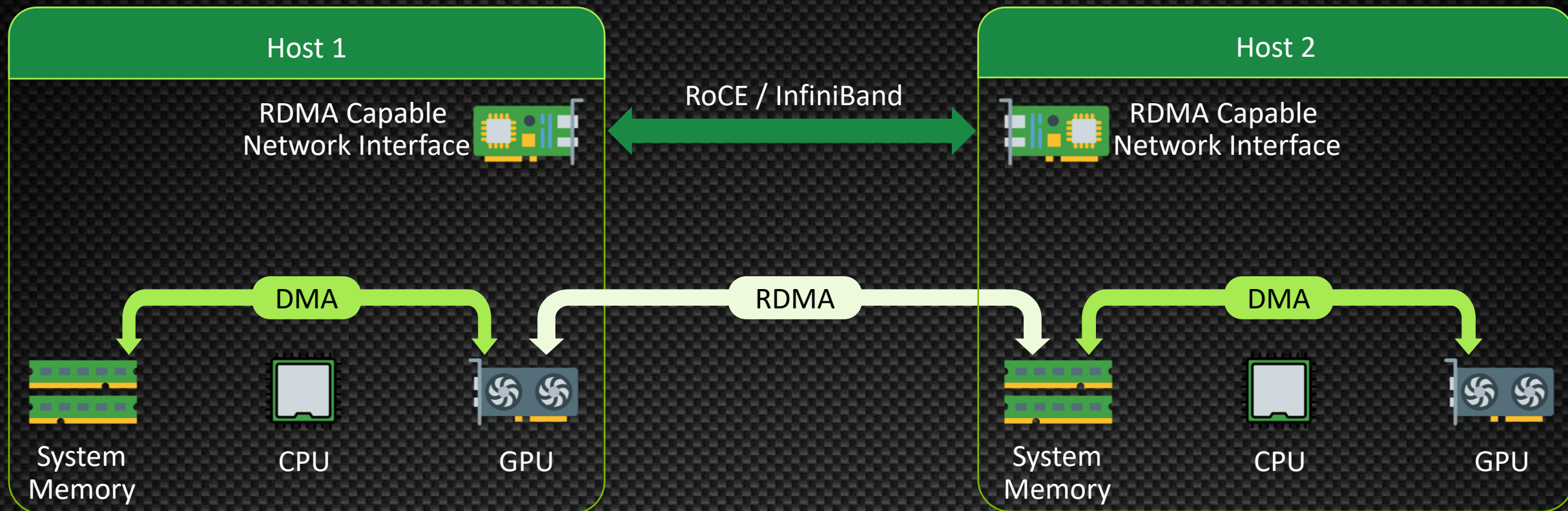
With DMA:

- Device controller handles transfers directly
- CPU utilization: 5-10% (just setup + notification)
- CPU/GPU free for actual computation
- 10-100x faster data movement



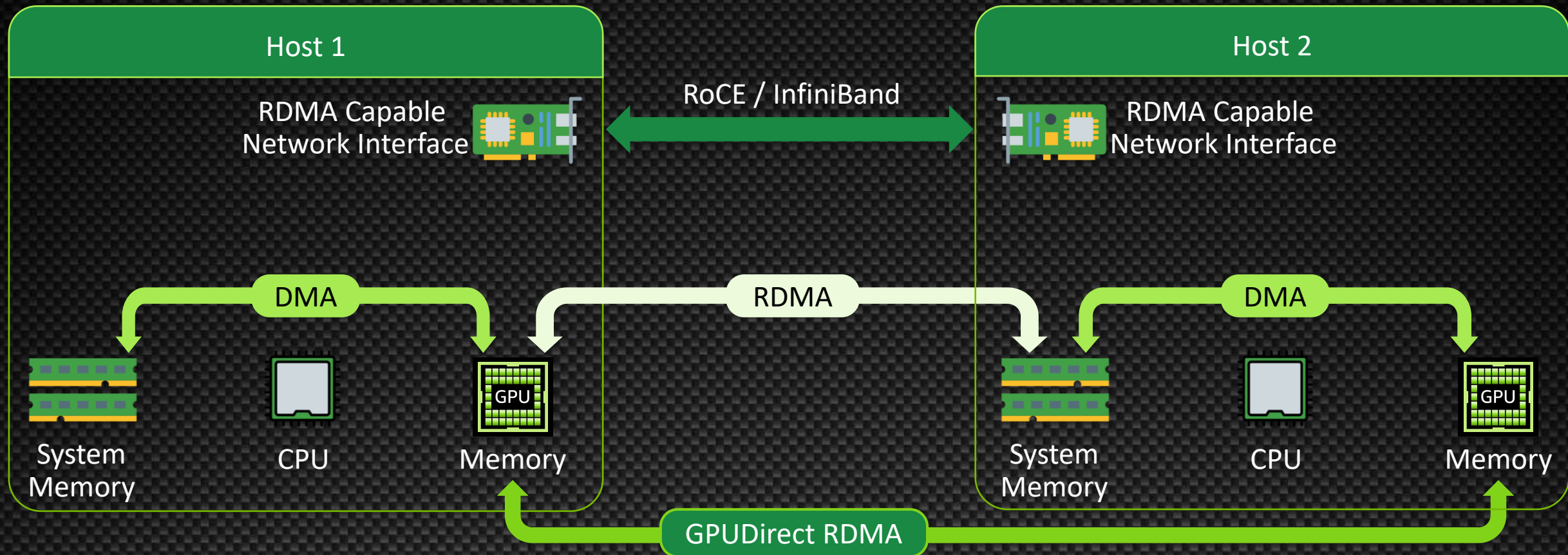
Remote DMA (RDMA)

- With RDMA, devices can bypass the CPU and not only access the local host's memory but also access memory on another host.



GPUDirect RDMA

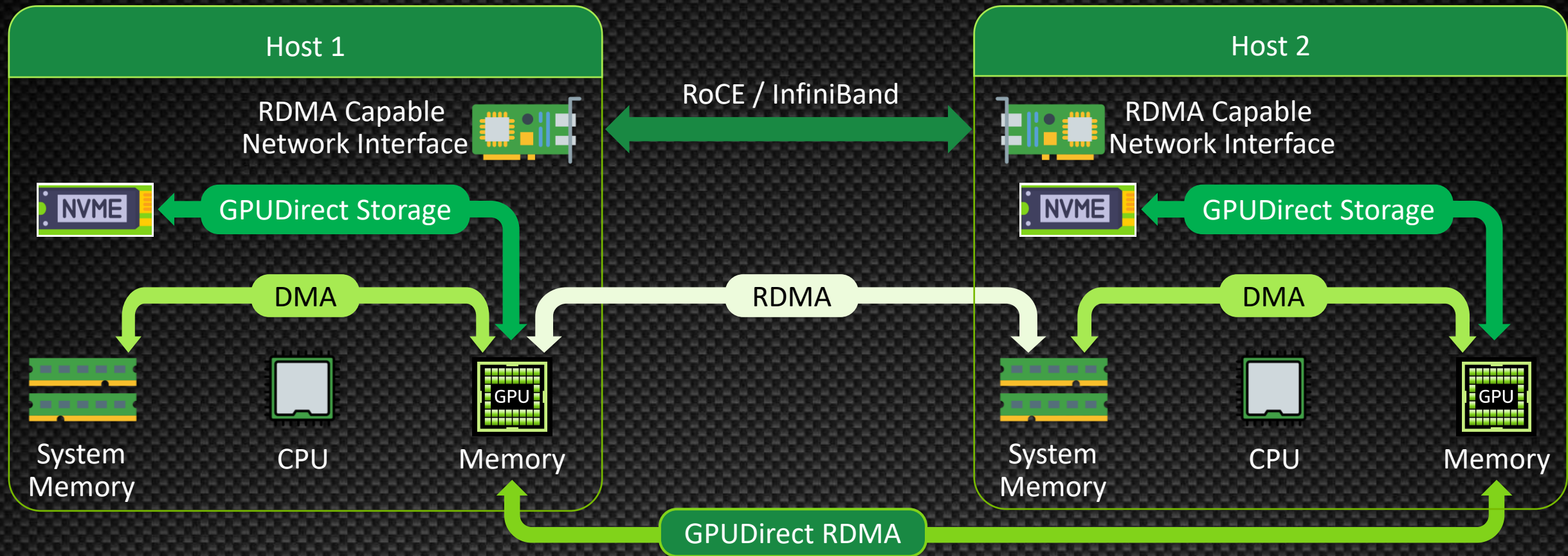
- NVIDIA extension of RDMA designed for GPUs.
- Allows direct access to GPU memory bypassing CPU and OS.



It requires: NVIDIA GPUs, NVIDIA drivers, RDMA-capable NIC

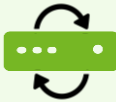
GPUDirect Storage

- Direct GPU-to-Storage transfers bypassing CPU and system memory
- Reduces I/O bottlenecks in training



It requires: NVIDIA GPUs, NVIDIA drivers, RDMA-capable NIC

Quick Comparison

Feature	GPUDirect RDMA	GPUDirect Storage
Scope	Works across hosts (GPU ↔ remote GPU/NIC via RDMA) 	Works within a host (GPU ↔ local or networked storage) 
Main Use Case	Low-latency GPU-to-GPU or GPU-to-NIC data transfer for HPC, AI clusters	High-throughput GPU data loading from NVMe/RAID/parallel storage
Data Path	GPU memory ↔ RDMA NIC (bypasses OS, system memory & CPU)	Storage device ↔ GPU memory (bypasses OS, system memory & CPU)
Key Benefit	Ultra-low latency networking between GPUs across nodes	High-bandwidth I/O for feeding large datasets into GPUs
Hardware Needs	NVIDIA GPU + RDMA-capable NIC (e.g., Mellanox ConnectX)	NVIDIA GPU + NVMe or parallel storage (e.g. IBM Spectrum Scale, WekaFS)

Layer 6: Applications & Vertical Solutions	Healthcare & Hospitals	Recommendation Systems	Inference Microservices		
	Clara	Merlin	NVIDIA NIMs		
Layer 5: Monitoring and Management	Monitoring & Management				
	nvidia-smi	DCGM	Base Command Manager		
Layer 4: Core Libraries	GPU Programming		GPU Communication		
	CUDA		NCCL		
Layer 3: OS, Driver and Virtualization	Operating System	GPU Driver	Virtualization		
	DGX OS	GPU Drivers	vGPU MIG (Multi-Instance GPU)		
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	NVLink	RDMA, Storage	InfiniBand, OpenSM		
Layer 1: Physical Layer	Gaming and Workstation GPUs	Data Center AI server	AI Supercomputer	Networking Interface	DPUs
	NVIDIA RTX Series	DGX Platform	DGX SuperPOD	ConnectX InfiniBand HCAs / NICs	BlueField DPUs / SuperNICs



Layer 3: OS, Driver and Virtualization

Operating System	GPU Driver	Virtualization
DGX OS	GPU Drivers	vGPU MIG (Multi-Instance GPU)

DGX OS

- NVIDIA's customized Ubuntu 22.04-based operating system for DGX Systems.
- NVIDIA-Optimized Kernel for AI, ML, and analytics workloads on DGX systems.
- Latest version DGX OS 7.2.1 released on August 19, 2025
- Comprehensive Software Stack: Includes NVIDIA GPU drivers, CUDA Toolkit, cuDNN, NCCL, DCGM, Docker Engine, and more.



Layer 3: OS, Driver and Virtualization

Operating System	GPU Driver	Virtualization
DGX OS	GPU Drivers	vGPU MIG (Multi-Instance GPU)

GPU Drivers

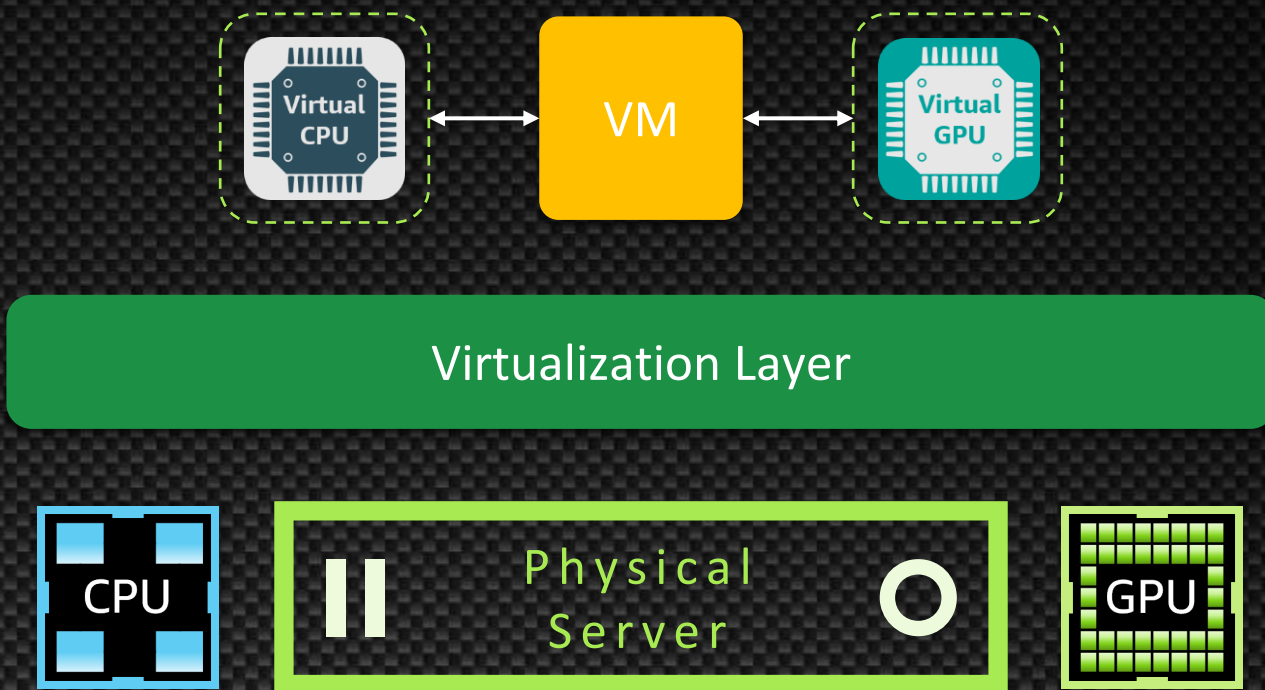
- Software that connects NVIDIA GPUs with the operating system and applications.
- Purpose:
 - Enables GPU acceleration for AI, HPC, and graphics
 - Manages GPU resources, memory, and performance
 - Latest (2025):r580 – supports CUDA 13.0 and DGX OS 7
- Key Features:
 - Optimized for compute and graphics workloads
 - Supports containers, virtualization, and multi-GPU setups
- Can be downloaded from the official NVIDIA website by selecting the GPU model and OS version



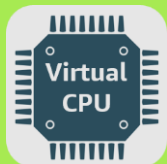
Layer 3: OS, Driver and Virtualization

Operating System	GPU Driver	Virtualization
DGX OS	GPU Drivers	vGPU MIG (Multi-Instance GPU)

GPU Virtualization

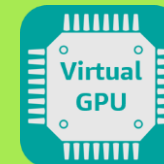


Quick comparison



CPU
Virtualization

Aspect



GPU
Virtualization

Very mature, standardized

Maturity

Still evolving, vendor-specific

Easy to allocate cores/threads

Granularity

Harder – GPUs don't split neatly, need vGPU/MIG

Strong isolation by hypervisors

Isolation

Isolation is harder; needs special drivers

Near-native with minimal overhead

Performance

Near-native in passthrough, variable in sharing

Time-slicing or core allocation

Sharing Model

Hardware slicing, driver mediation, or API interception

Why virtualize GPUs?

COST REDUCTION BENEFITS



- Share one GPU across multiple users or VMs.
- Reduce hardware expenses significantly.

IMPROVED RESOURCE UTILIZATION



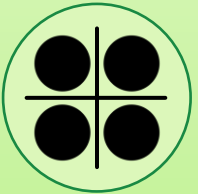
- Idle GPU resources allocated to other tasks.
- Increase overall efficiency in data centers.

FLEXIBLE WORKLOAD SCALING



- Dynamically adjust GPU resources per demand.
- Scale workloads without physical hardware changes.

SECURE PERFORMANCE ISOLATION



- Each virtual GPU instance is fully isolated.
- Prevents one workload affecting another.

• Two common methods

vGPU

MIG

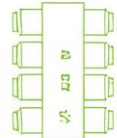
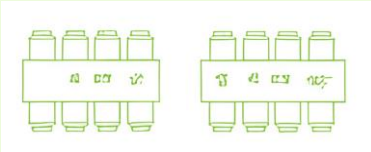
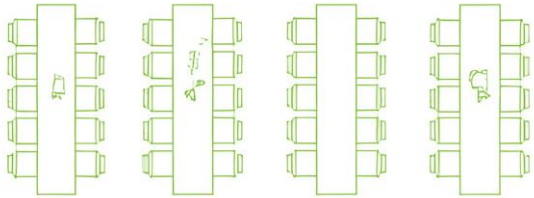
Analogy – A co-working space



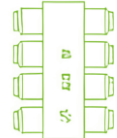
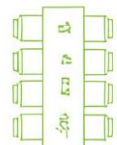
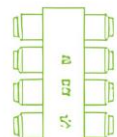
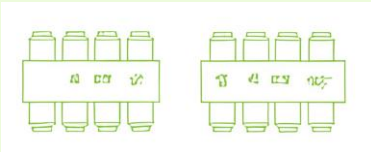
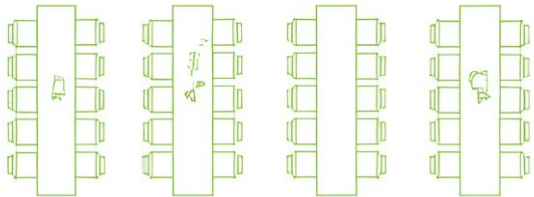
- WeWork
- Mindspace
- LABS
- Landmark
- Second Home
- Work.Life
- Huckletree
- Uncommon
- Runway East
- Fora
- Boutique Workplace
- Spacemade
- Clockwise
- Knotel
- Regus

A shared co-working space

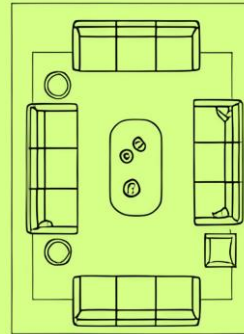
Company A



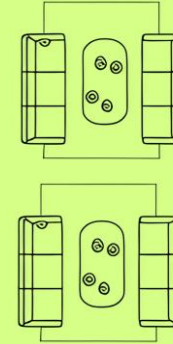
Company B



Common Area



Lobby



Elevator

Elevator

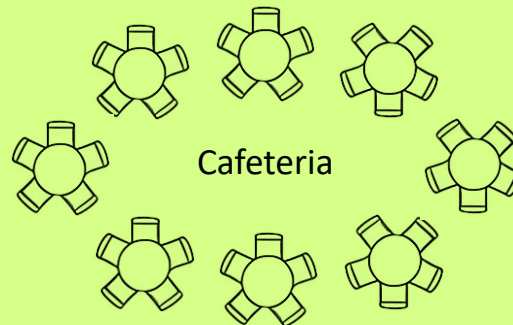
Elevator

Elevator

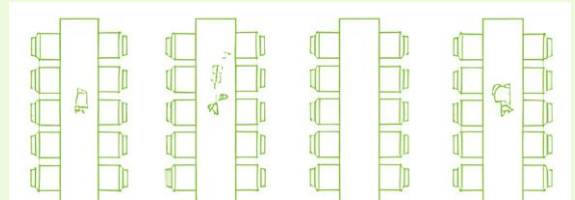
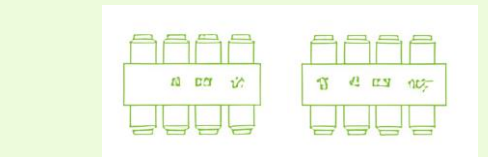
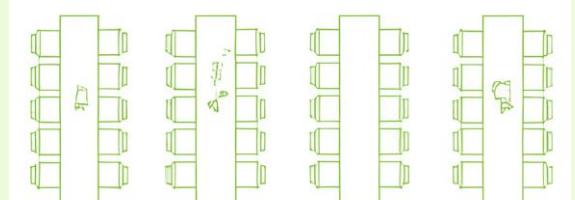
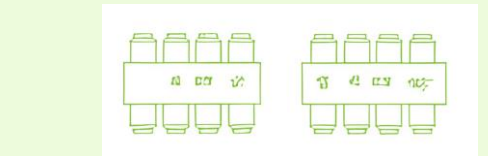
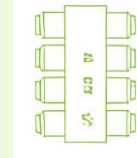
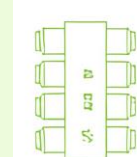
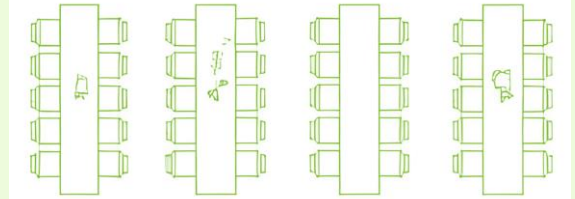
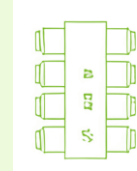
Elevator

Elevator

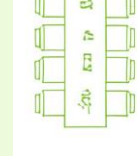
Cafeteria



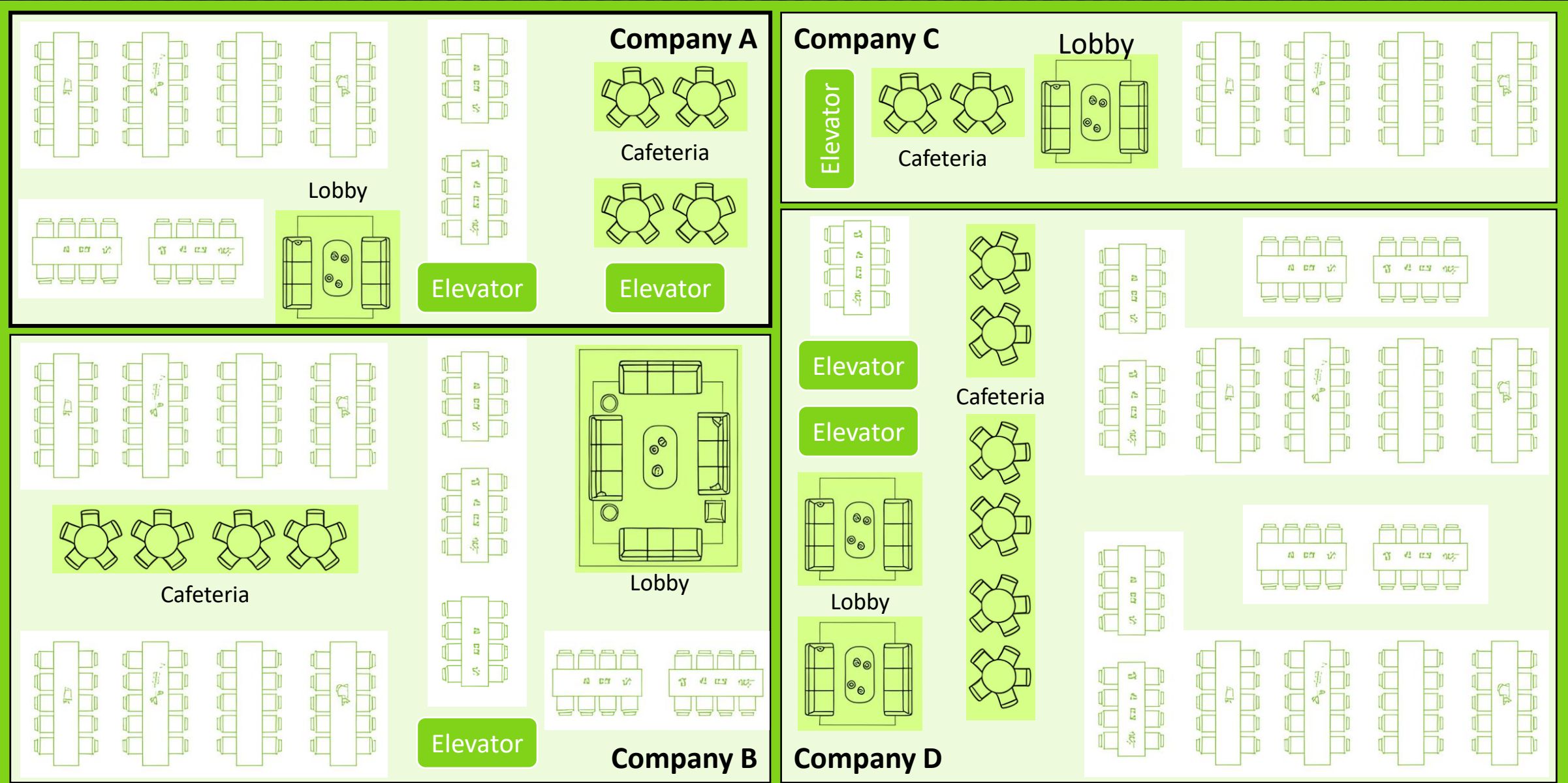
Company C



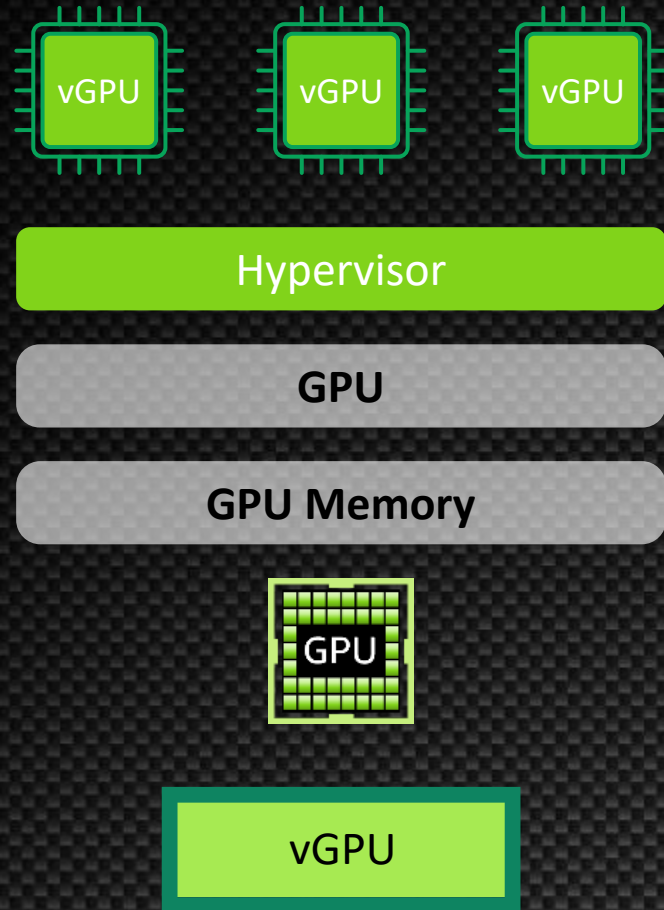
Company D



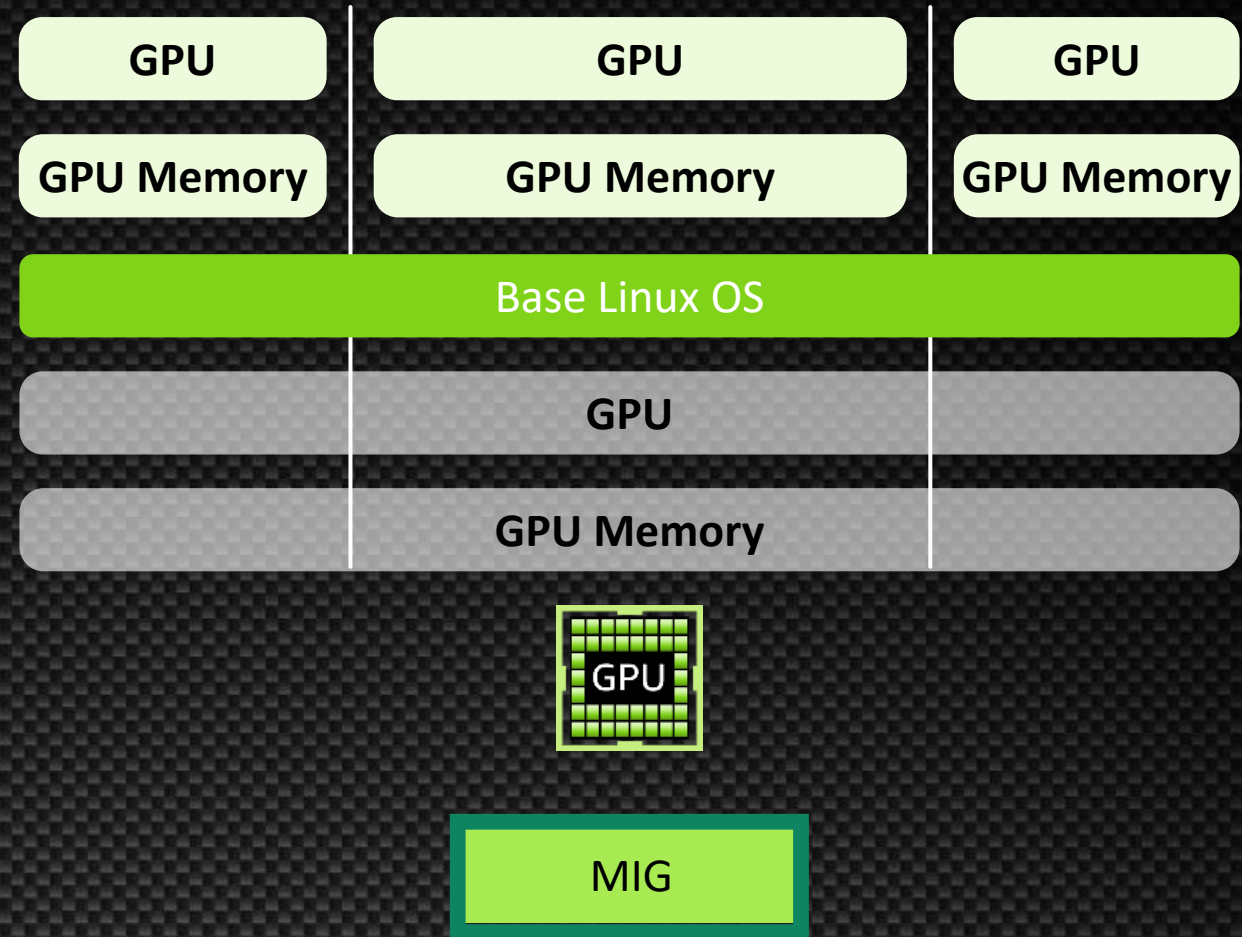
A little more isolation



Simple explanation

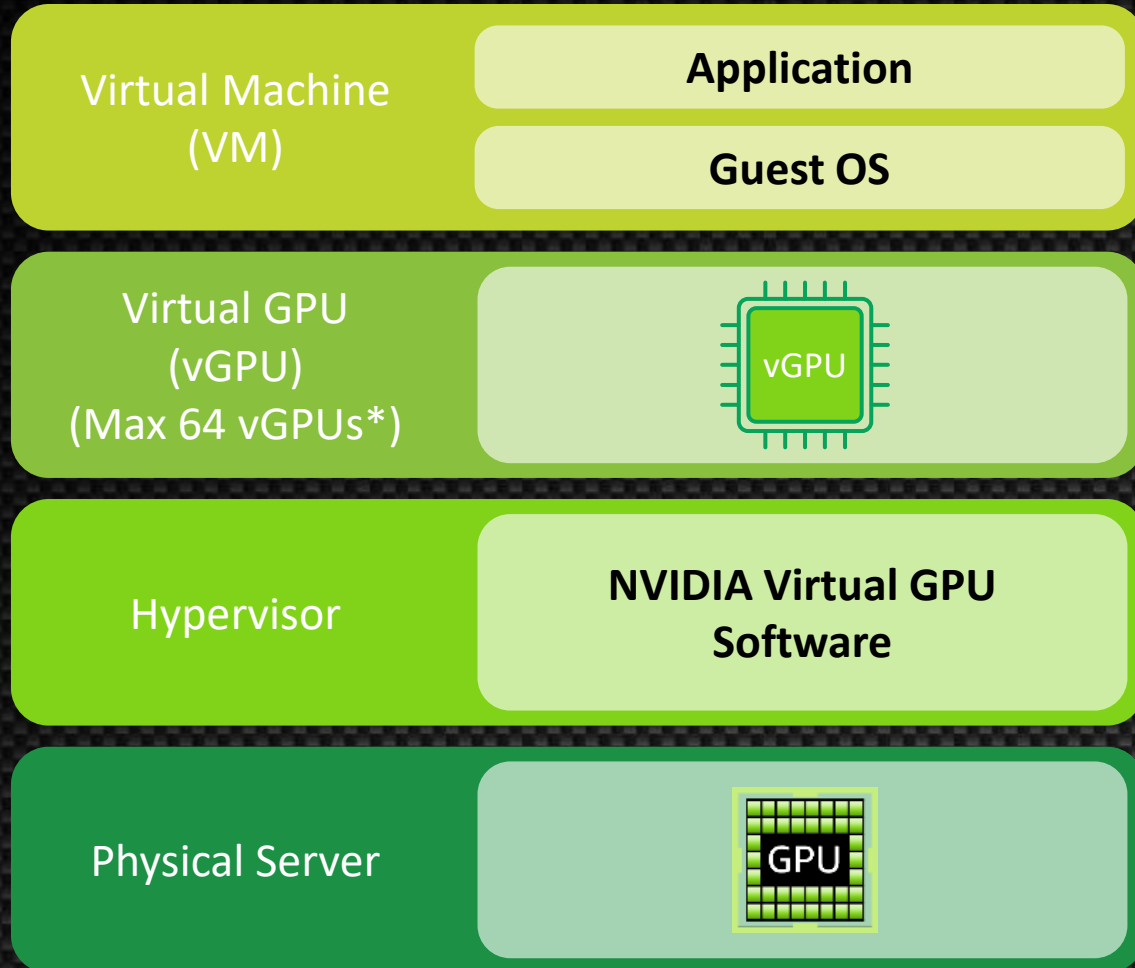


Software level isolation

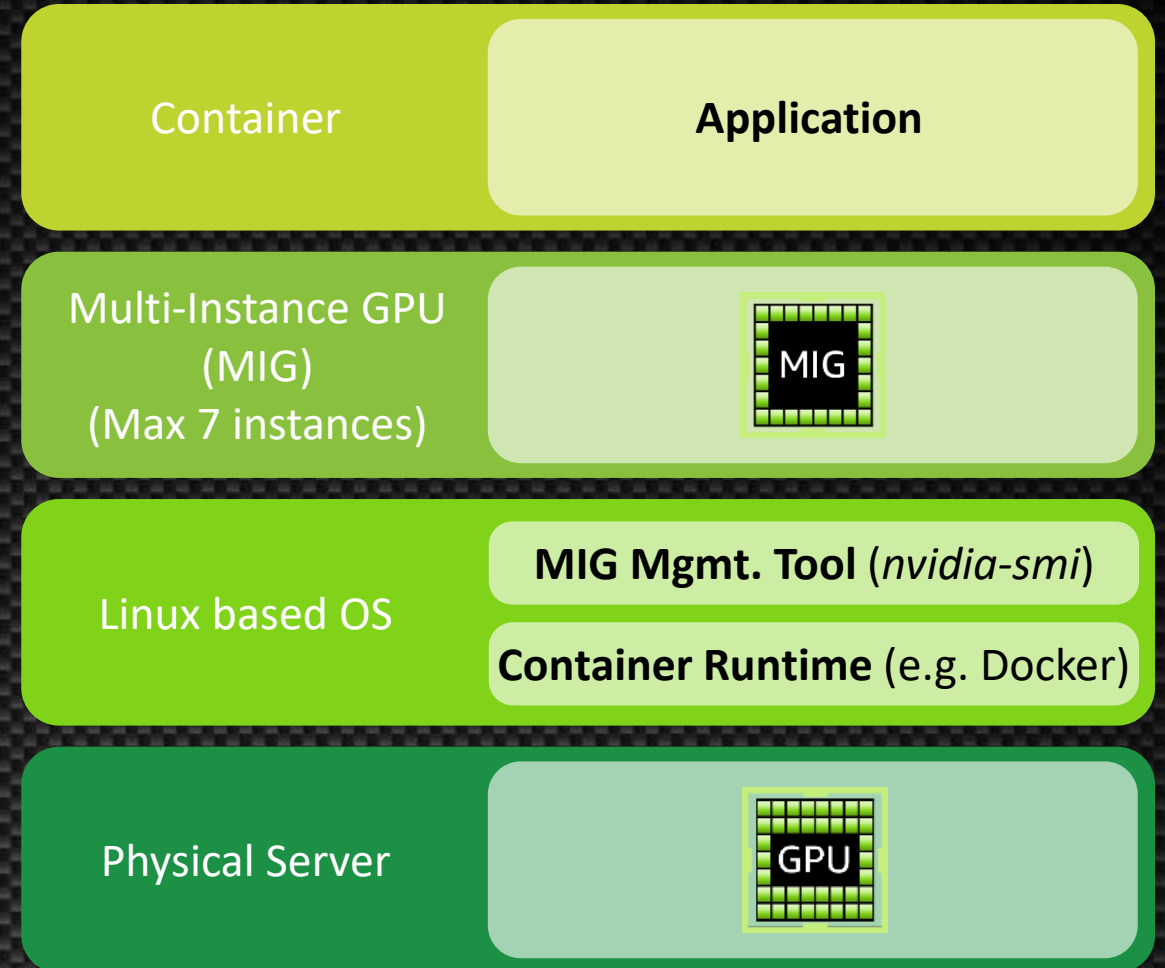


Hardware level isolation

vGPU vs. MIG



vGPU



MIG



vGPU vs. MIG



Feature	NVIDIA vGPU	NVIDIA MIG
Isolation	Software-based, hypervisor-controlled	Hardware-enforced, strong isolation
Max Partitions (Per GPU – not per server)	Up to 64 (depending on the GPU model)	7
Supported GPUs	Many NVIDIA GPUs (Quadro, RTX, data center)	Newer data center GPUs (A100, H100, L40S, etc.)
Use Case	VDI, VM-based AI/ML, GPU sharing in VMs	HPC, AI/ML, containers
Host Layer	Hypervisor (VMware, Citrix, RHV)	Bare-metal Linux (no hypervisor needed)
Software	vGPU Manager + Guest VM NVIDIA driver	Standard NVIDIA driver + MIG tools
Performance Predictability	Shared; may vary with load	Consistent per instance

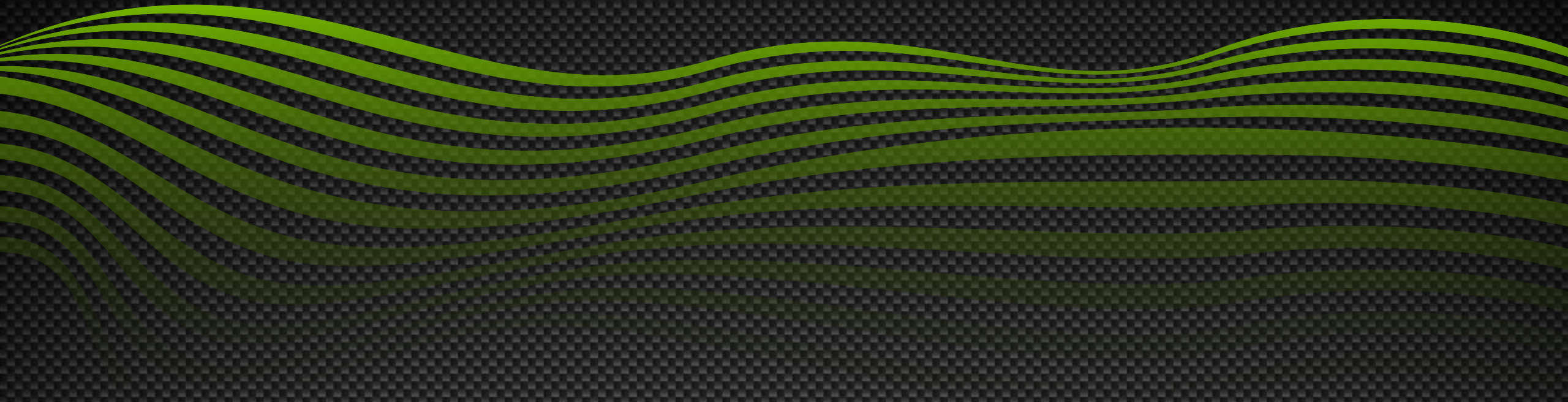
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Layer 4: Core Libraries

GPU Programming	GPU Communication
CUDA	NCCL

Compute Unified Device Architecture (CUDA)



An Analogy for CUDA



x 100



x 100

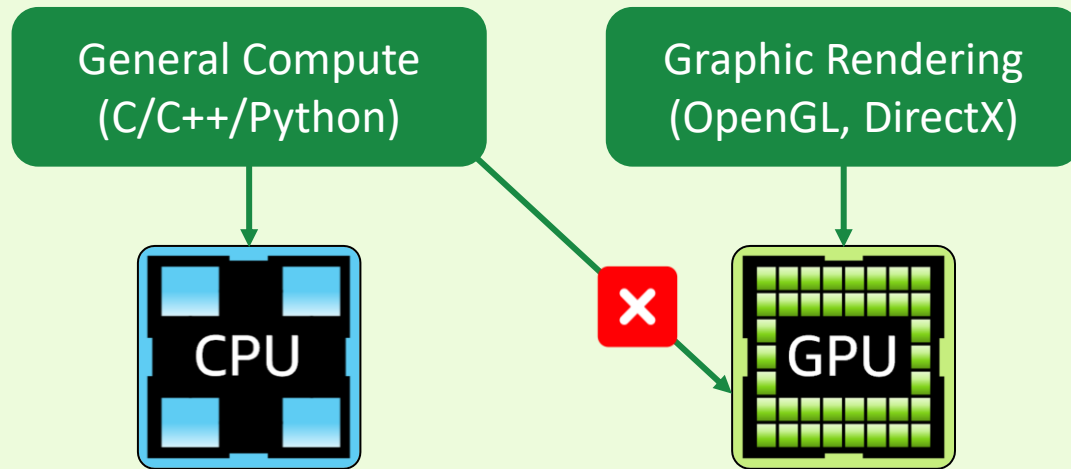


x 100



The CUDA difference

Before CUDA

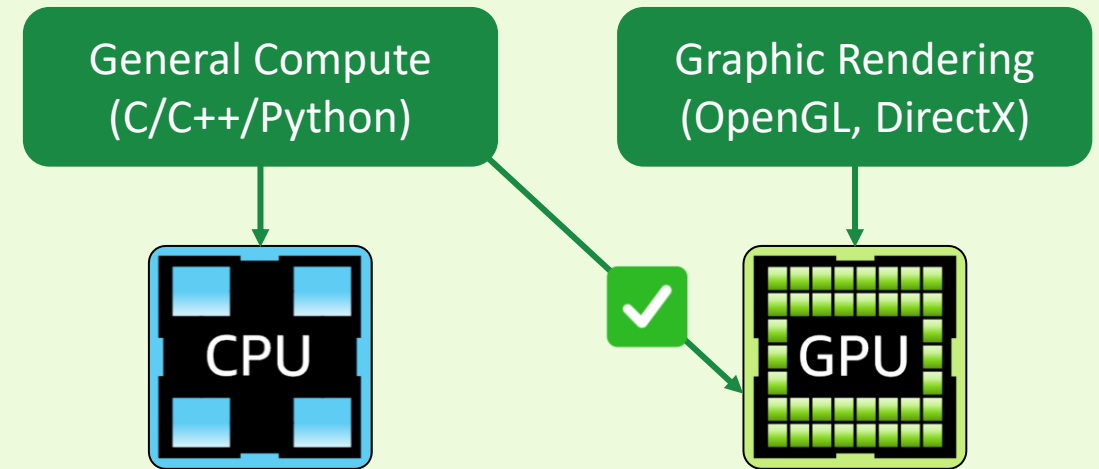


- GPU locked behind graphics APIs
- Required "tricking" for non-graphics work
- CPU handled most general computing
- GPU cores mostly idle for non-graphics tasks
- Complex, impractical development

GPU Supports

- | | |
|----------------------------------|--------------------------|
| ✓ Graphics rendering (games, 3D) | ✗ Image/video processing |
| ✗ Machine learning & AI | ✗ Financial modelling |

After CUDA



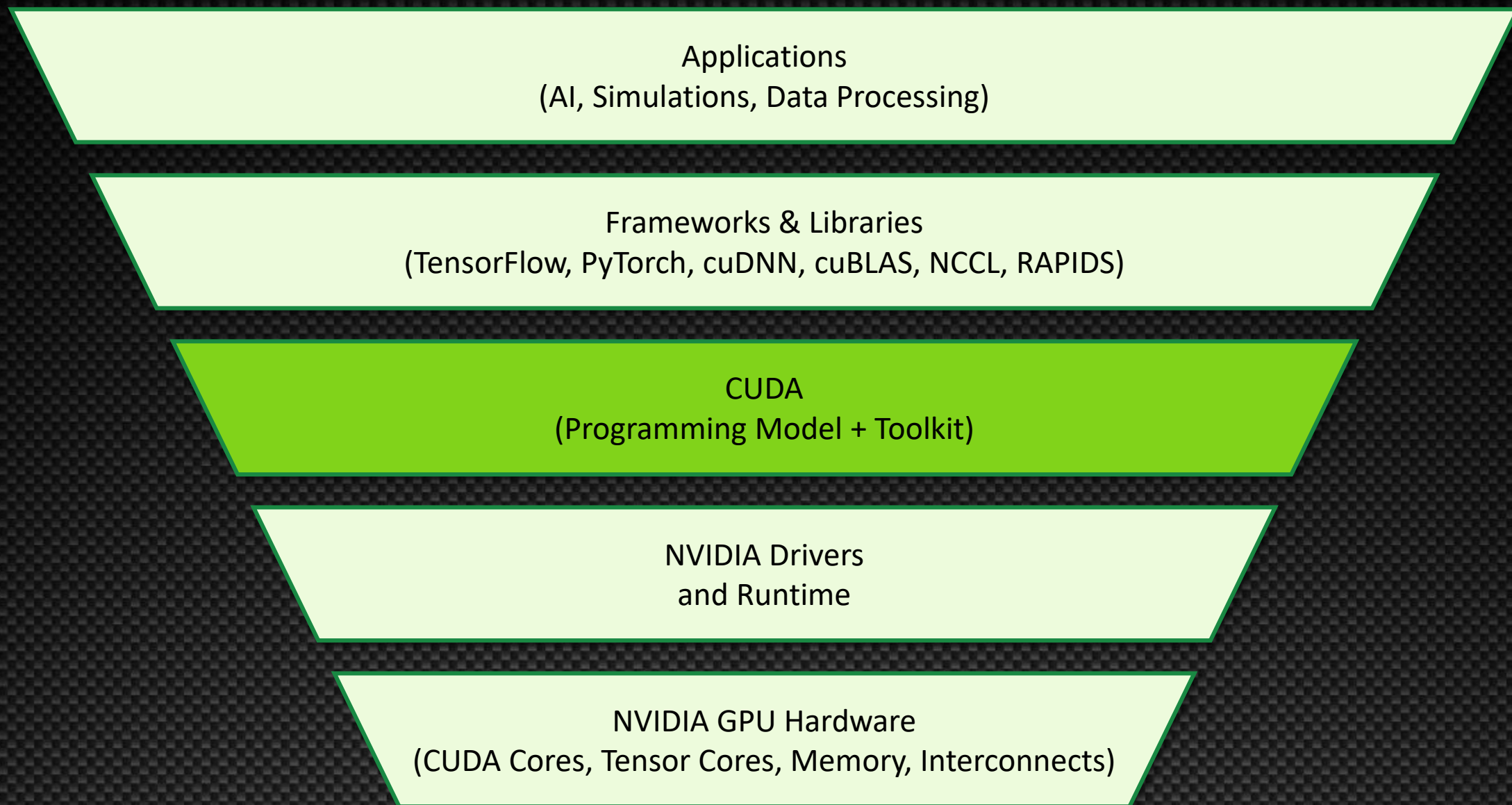
- Direct GPU programming model
- Native C/C++/Python support
- GPU as general-purpose accelerator
- Thousands of cores utilized efficiently
- Straightforward parallel programming

CUDA enables parallel programming: the CPU organizes, GPU executes massively in parallel.

GPU Supports

- | | |
|----------------------------------|--------------------------|
| ✓ Graphics rendering (games, 3D) | ✓ Image/video processing |
| ✓ Machine learning & AI | ✓ Financial modelling |

CUDA: Bridging Software and NVIDIA Hardware



CUDA = A Bridge Between Normal Code and GPU Power

- Why "Parallel Computing Platform"?
 - Parallel = Many things happening at the same time
- Imagine painting a fence:
 - Sequential (CPU): One person paints left to right
 - Parallel (GPU with CUDA): 1000 people each paint one section simultaneously
 - CUDA gives you the tools to organize and coordinate those 1000 workers effectively.
- Why "Programming Model"?
 - A "model" is just a way of thinking about and organizing your code:
 - Traditional programming model: Do step 1, then step 2, then step 3...
 - CUDA parallel model:
 - Break problem into small identical tasks
 - Launch thousands of threads (workers) to do them simultaneously, Collect results when everyone finishes

Your Code (C/C++/Python)

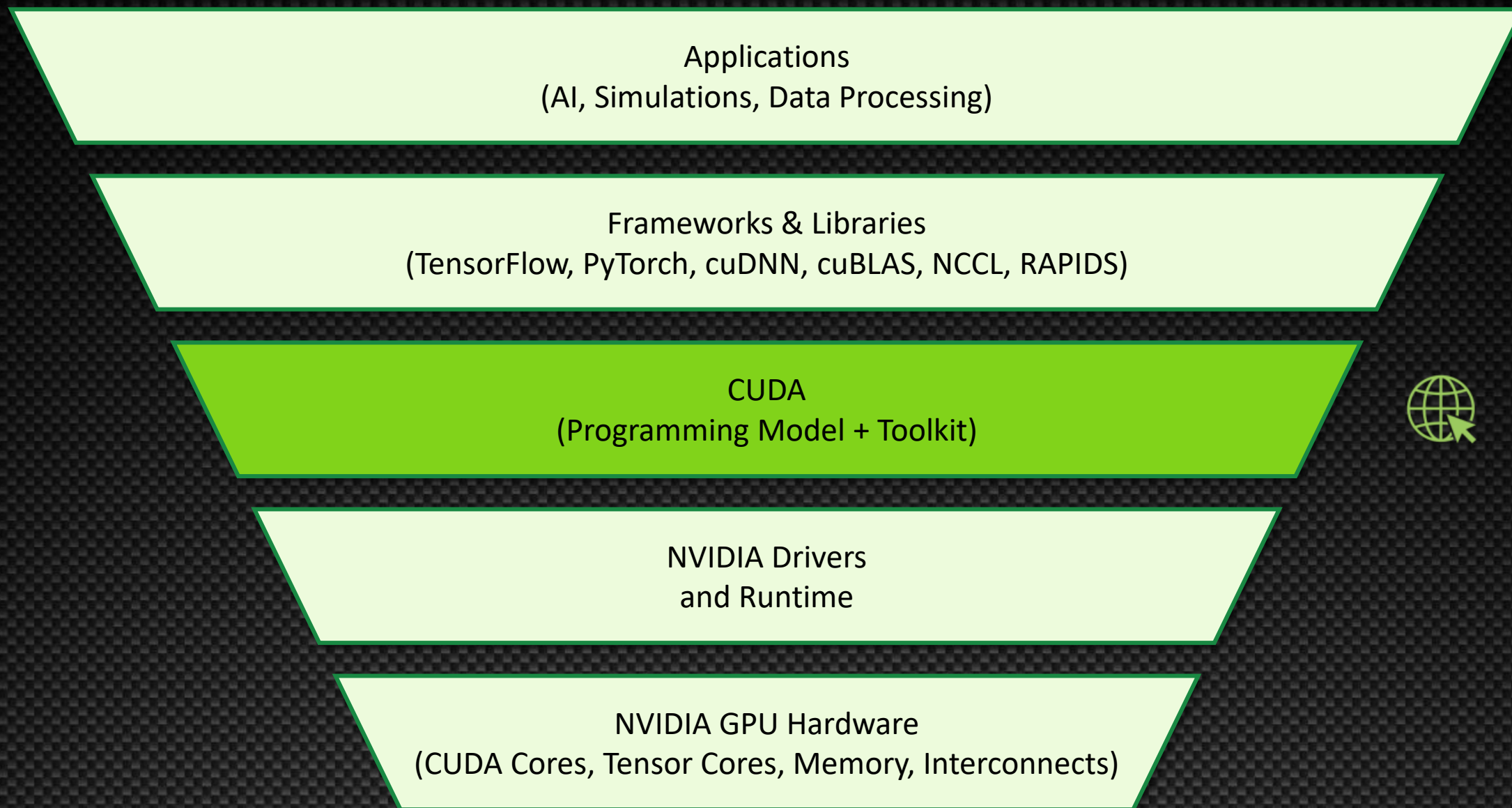


CUDA translates

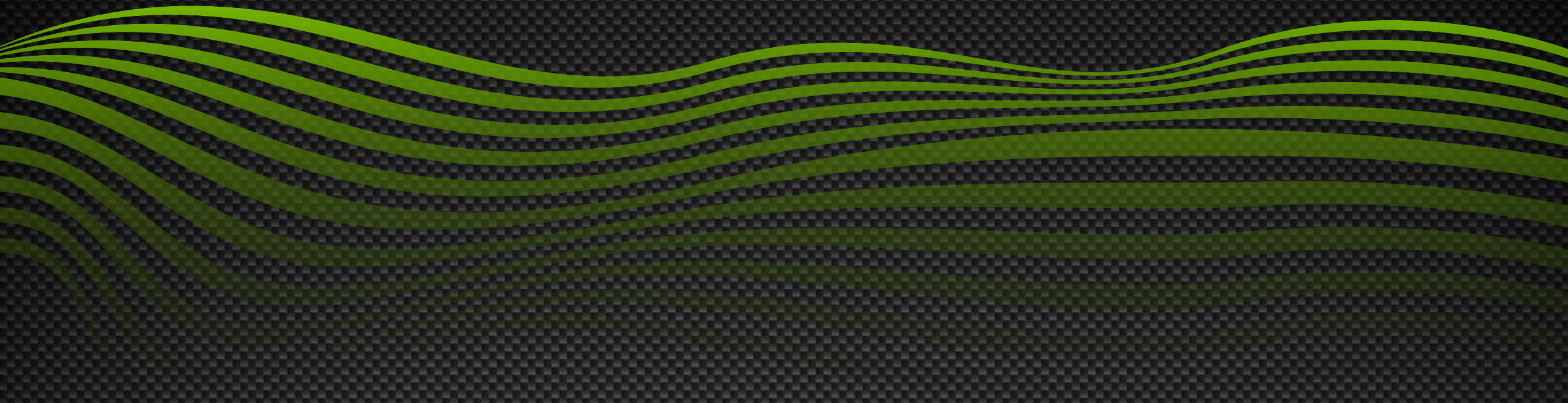


GPU understands and executes

Installing CUDA

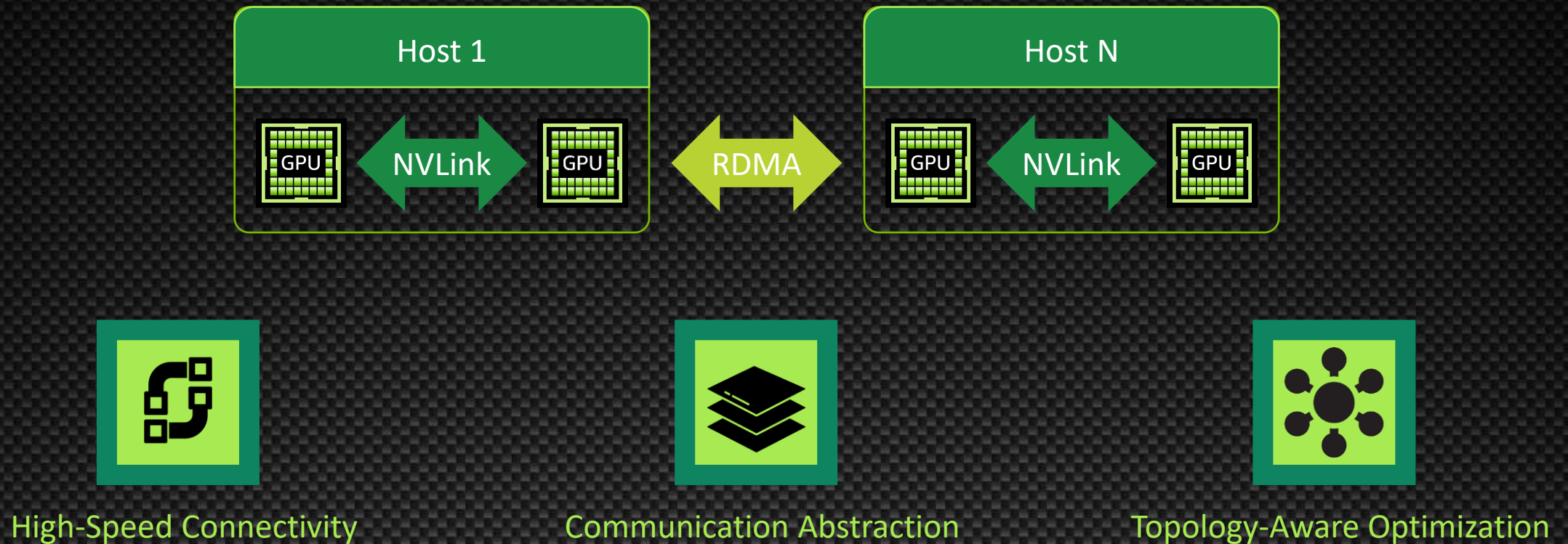


NVIDIA Collective Communications Library (NCCL)



NCCL (NVIDIA Collective Communications Library)

- A multi-GPU communication library which provides abstractions and optimizes patterns of communication between many GPUs



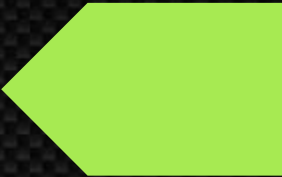
- Frameworks Like PyTorch and TensorFlow Already Include NCCL



Quick Comparison

	NVLink, NVSwitch, PCIe, RDMA	NCCL
What is it?	Hardware and Drivers	Software Library
Analogy	High Speed Express way 	Traffic Management System 
Solves What?	"How to transfer data fast?"	"How to organize many transfers efficiently"
Benefit	Low latency per transfer	Fewer transfers needed overall
Example Use	Moving 1GB from GPU A to GPU B quickly	Coordinating gradient sharing among 100 GPUs

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Layer 5: Monitoring and Management

Monitoring & Management		
nvidia-smi	DCGM	Base Command Manager

Driving Needs a Dashboard - Data Centers Need Monitoring.



Sports Car

Fuel



Resource
Availability

Speed



System
Performance

Temperature



Server or GPU
Temperature

Safety



Alerts and
incidents



Data Center

NVIDIA tools for monitoring and managing data centers

nvidia-smi

01

DCGM

02

BCM

03

nvidia-smi

Primary Purpose	• Quick GPU monitoring and basic management on single node	
Scope	• Single system/node only	
What it Monitors?	• GPU utilization (%) • Memory usage • Power consumption (watts)	• Temperature • Running processes • Clock speeds
What it Manages?	• GPU configuration settings • Power limits	• Compute modes • Process termination
Interface Type	• Command-line interface (CLI)	
Installation/Deployment	• Included with NVIDIA GPU drivers (automatic, no separate install)	
Integration	• Basic shell scripting	• Limited (output parsing)
Critical Limitation	• No historical data storage • Single-node only	• No native alerting • Manual monitoring only, no continuous monitoring

DCGM - Data Center GPU Manager



Primary Purpose	• Enterprise-scale GPU health monitoring and diagnostics	
Scope	• Multi-node GPU clusters, Continuous Monitoring, Alerting	
What it Monitors?	• GPU health metrics • Utilization patterns • Power/thermal data	• Memory bandwidth • PCIe throughput • Error rates (ECC)
What it Manages?	• GPU grouping • GPU power and clock management • Health policies	• Diagnostic tests • Alerting thresholds • Metric collection policies
Interface Type	• Command-line interface (CLI)	• Can provide metrics via an HTTP endpoint through DCGM-Exporter
Installation/Deployment	• Installed separately as a package or can be installed through <i>GPU Operator</i> (a tool)	• Requires agent on each node
Integration	• Monitoring platforms like Prometheus/Grafana	
Critical Limitation	• Monitoring focused tool, not for job scheduling • Cluster Scope, cannot natively monitor multiple independent clusters across a data center.	

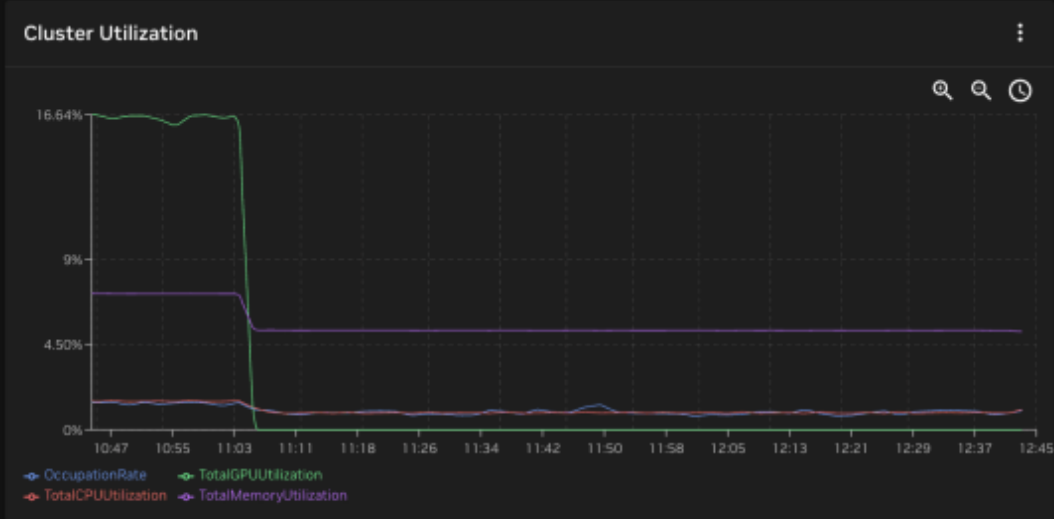
BCM - Base Command Manager

Primary Purpose	• Manage, monitor, and orchestrate GPU resources and workloads	
Scope	• Entire AI infrastructure in a data center (CPU, GPU, storage, network, workloads)*	
What it Monitors?	• All DCGM metrics • Cluster resource utilization • Infrastructure health	• Job status/progress • User quotas • Workload performance
What it Manages?	• GPU firmware, configuration, power policies • Software updates • Cluster provisioning	• Job scheduling • Resource allocation • Workload deployment
Interface Type	• CLI and Web-based UI	• REST API
Installation/Deployment	• Separate enterprise product • Multi-component deployment	• Requires dedicated management nodes
Integration	• Monitoring platforms like Prometheus/Grafana	
Critical Limitation	• Requires enterprise license (cost) • Complex setup and learning curve • Overkill for small deployments	

- Cluster
- Networking
- Provisioning
- Grouping
- Devices
- Datacenter Infrastructure
- HPC
- Cloud
- Containers
- Mission Control
- Jupyter
- Monitoring
- Identity Management

- Cluster
- Overview
- Settings
- Run Command
- License Information
- System Information
- Hardware Inventory
- Node Arch OS Information
- Version Information
- Port Forwarding
- IMEX Configuration
- Workload Ut

+ Add Widget ... Actions



Device Status

DevicesClosed	0	DevicesDown	0	DevicesTotal	0	DevicesUp	0
ManagedSwitchesClosed	0	ManagedSwitchesDown	0	ManagedSwitchesTotal	0	ManagedSwitchesUp	0
NVLinkSwitchesClosed	0	NVLinkSwitchesDown	0	NVLinkSwitchesTotal	4	NVLinkSwitchesUp	4
NodesClosed	0	NodesDown	2	NodesTotal	13	NodesUp	11
PowerShelvesClosed	0	PowerShelvesDown	2	PowerShelvesTotal	2	PowerShelvesUp	0

Resource Status

CoresDown				288
CoresTotal	CoresUp	DPUNodesDown	DPUNodesTotal	
1.872 k	1.584 k	0	2	
DPUNodesUp	GPUsDown	GPUsTotal	GPUsUp	
2	8	32	24	

Health Checks

26/28 health checks passed

Entity	Name	Value	Last Check
🖨️	interfaces	❌ Fail	14 Aug 2025 12:44:00
🖨️	cm-chroot-sw-img	❌ Fail	14 Aug 2025 12:44:00
🖨️	swraid	✅ Pass	14 Aug 2025 12:44:00
🖨️	Smart_Health/dev/nvme2@nvme	✅ Pass	14 Aug 2025 12:44:00
...	chrootprocess	✅ Pass	14 Aug 2025 12:44:00
🖨️	Smart_Health/dev/nvme3@nvme	✅ Pass	14 Aug 2025 12:44:00
🖨️	Overall_Health/dev/nvme3@nvme	✅ Pass	14 Aug 2025 12:44:00

1-7 of 28



Monitoring and Management /Configuration Capabilities

Component	Monitoring Capabilities	Management / Configuration Capabilities
CPU	✓ Monitors utilization, temperature, and health via system telemetry	✗
NVIDIA GPUs	✓ GPU utilization, memory, temperature, power draw, ECC errors, health metrics.	✓ Manages GPU firmware, configuration, power policies, and allocation to workloads.
NVIDIA Network (InfiniBand, BlueField DPUs, Spectrum Switches, SuperNICs)	✓ Monitors fabric health, throughput, latency, link errors, and device status.	✗
Storage (NFS, Lustre, NVMe)	✓ Monitors mount health, I/O throughput, latency, and available capacity.	✗
Workloads (AI, HPC)	✓ Tracks job status, resource use, GPU/CPU allocation, and performance.	✓ Manages workload submission and orchestration through Slurm, Kubernetes, and Run:AI integration.

NVIDIA tools for monitoring and management

Use nvidia-smi when:

- Checking GPU status on single system
- Quick troubleshooting
- No setup/installation needed
- Immediate results required

Use DCGM when:

- Monitoring 10+ GPU nodes
- Need historical metrics
- Require alerting/diagnostics
- Kubernetes GPU management (GPU Operator)

Use Base Command Manager when:

- Managing entire AI data center
- Need job scheduling + monitoring
- Multi-team/multi-user environment
- Enterprise-scale operation

• Memory Aids for Exam:

- nvidia-smi = **Single** System Management Interface
- DCGM = **Data** Center (many nodes) **GPU** **M**anager
- Base Command Manager = **Base** of entire AI infrastructure **Command** center
- OpenSM = **Opens** (enables) InfiniBand **S**ubnet **M**anagement

Layer 6: Applications
& Vertical Solutions

Healthcare & Hospitals	Recommendation Systems	Inference Microservices
Clara	Merlin	NVIDIA NIMs

Layer 5: Monitoring
and Management

Monitoring & Management		
nvidia-smi	DCGM	Base Command Manager

Layer 4:
Core Libraries

GPU Programming	GPU Communication
CUDA	NCCL

Layer 3: OS, Driver
and Virtualization

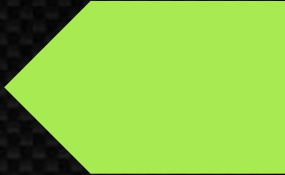
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GPU Interconnect	GPU Direct	HPC Fabric
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NVIDIA

CLARA

AI platform for healthcare and life sciences.



CAPABILITY

- Accelerates medical imaging, genomics, and smart hospital AI.

USE CASES

- Faster genome sequencing and analysis.
- AI-powered medical imaging (CT, MRI, pathology).
- Real-time AI in medical devices and hospitals.

REAL-LIFE EXAMPLE

- Accelerates medical imaging, genomics, and advanced smart hospital AI.

NVIDIA

MERLIN

Framework for building recommendation systems.



CAPABILITY

- Accelerates recommendation training and inference on GPUs.

USE CASES

- Personalized product or content recommendations.
- Session-based recommendations using recent user activity.
- Large-scale, low-latency recommendation deployment.

REAL-LIFE EXAMPLE

- Applied in large-scale retail personalization to improve product recommendations.

NVIDIA NIMs

(NVIDIA Inference Microservices)

Inference microservices for deploying AI models.



CAPABILITY

- Delivers containerized inference across LLMs, vision, and speech.

USE CASES

- Deploying large language models at scale.
- Running AI inference on any cloud or on-prem setup.
- Integrating AI into enterprise and edge applications.

REAL-LIFE EXAMPLE

- To serve pretrained NVIDIA models (e.g., Llama 3, Stable Diffusion) through standardized APIs.

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Integration

Containerization



ML Frameworks



Workload Mgmt.



Monitoring



Certified
Vendors and
Partners

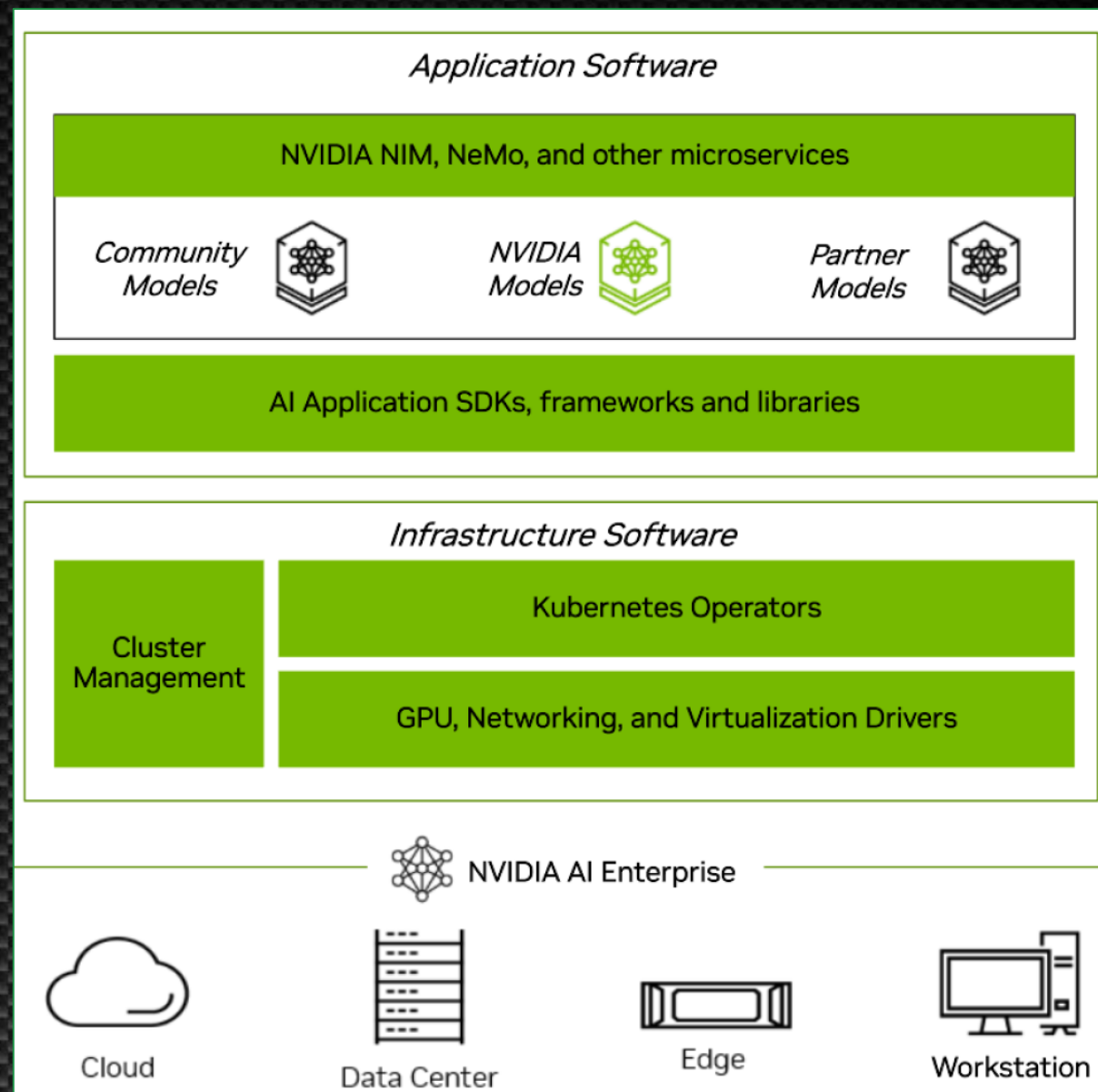
NVIDIA AI Enterprise & NVIDIA AI Factory



NVIDIA AI Enterprise:



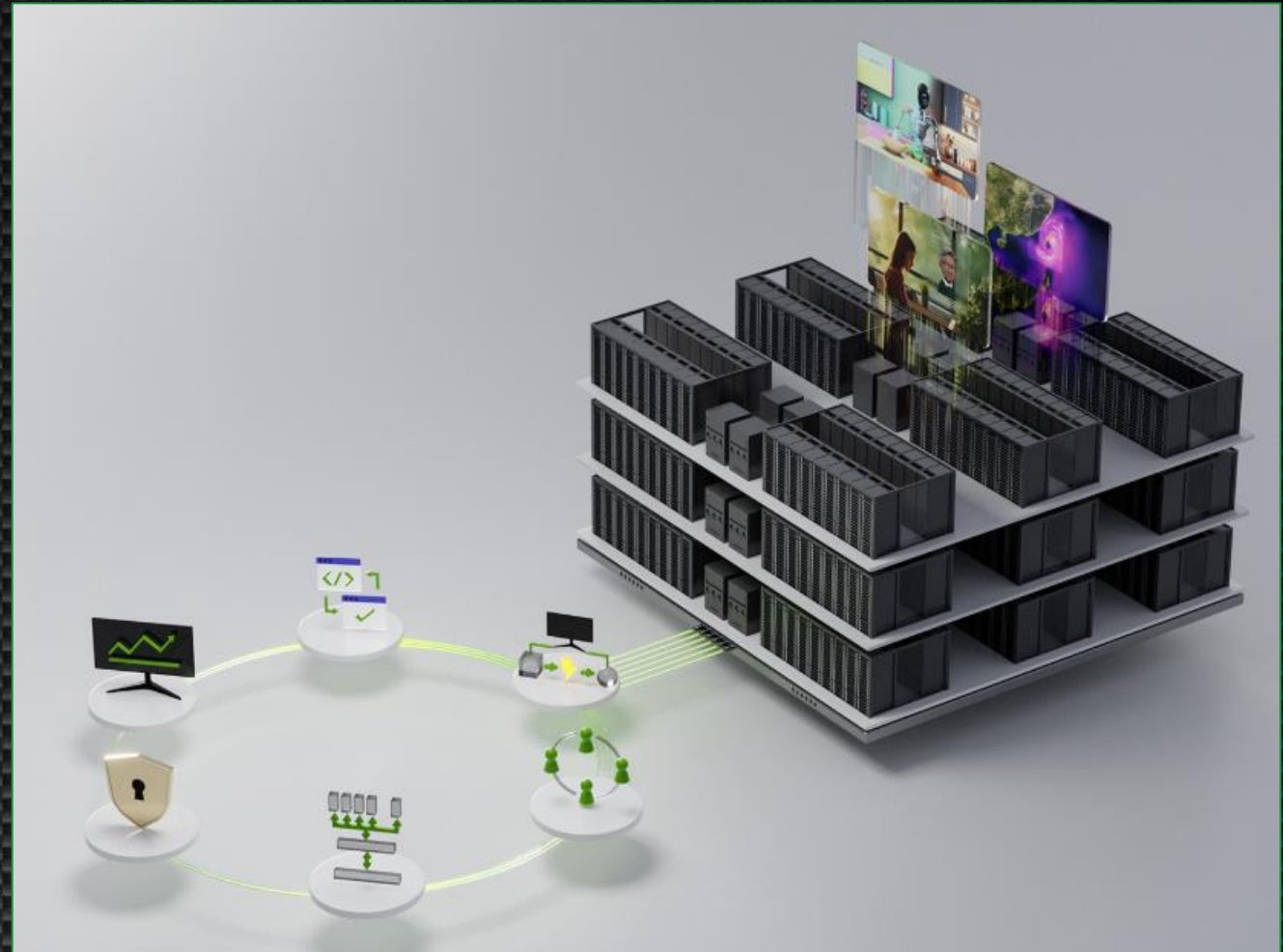
- The “operating system” for enterprise AI.
 - NVIDIA AI Enterprise is a software platform that gives companies all the tools they need to build, run, and manage AI applications on NVIDIA GPUs — safely, efficiently, and at scale.
 - It’s like a complete toolkit for businesses to use NVIDIA’s AI technology — including drivers, frameworks (like PyTorch, TensorFlow), and pre-built AI models and services — all optimized, tested, and supported for enterprise use.



NVIDIA AI Factory



- An NVIDIA AI Factory is essentially an AI-focused data center — purpose-built to produce, train, and deploy AI models at scale.
 - **Why NVIDIA calls it a Factory:**
 - It takes in raw materials (data)
 - Processes them with compute (GPUs, AI stack)
 - Outputs a product (an AI model or inference result). Just like a car factory turns steel into cars, an AI factory turns data into intelligence.
 - It covers the entire AI lifecycle: data ingestion → training → fine-tuning → inference efficiently using NVIDIA's full hardware–software ecosystem.



Module 4 - AI Workflows

Steps in AI Workflow

1



Data Processing

Clean, transform, augment and prepare raw data to make it more suitable for training a model.

NVIDIA RAPIDS

2



Model Training

Teach the model by feeding processed data so it learns patterns and behaviours.

PyTorch
TensorFlow

3



Model Optimization

Refine and compress the trained model (quantization, pruning) for better performance.

NVIDIA TensorRT

4



Inferencing / Deployment

Run the optimized AI model in production env. to make accurate predictions on new inputs.

NVIDIA Triton Inference
Server

Framework - Analogy

Imagine you're building a house:

Without a Framework



You'd need to make your own tools (hammer, saw), create your own cement, and cut your own wood. Very slow and error-prone!

VS.

With a Framework



You get a fully equipped workshop — ready-made tools, blueprints, and materials. You still build your house, but you don't waste time reinventing the basics.

Machine Learning Framework

- A machine learning framework (like PyTorch or TensorFlow) is a set of tools, libraries, and pre-written code that helps you build, train, and test machine learning models more easily — without having to code everything from scratch. Just like a workshop makes it easier to build a house.
- It provides:
 - 🧩 Building blocks (like layers, loss functions, optimizers)
 - 🚀 Hardware acceleration (uses GPU/TPU for faster training)
 - 🔧 Utilities for loading data, saving models, etc.



The NVIDIA differentiator

Your Python Code



Framework Engine
(PyTorch / TensorFlow)



cuDNN
(Optimization Layer)



CUDA



- Automatically use optimized libraries so your model trains much faster on GPUs.
- It's a collection of highly optimized, low-level GPU routines.
- Provides fast pre-built neural network operations for things like CNNs and RNNs.
- Lets the framework talk to the GPU.

Model Training vs. Model Inference

Feature	Model Training 	Model Inference 
Analogy	Learning to cook	Serving the dish
Objective	Build the model by learning parameters from large datasets through multiple iterations	Use the trained model to make predictions on new, unseen data.
Compute Requirements	Requires high compute power, often distributed across multiple GPUs. Forward pass + Backward pass + Weight updates → high compute.	Prioritizes low latency and high throughput. Compute is lighter since only the forward pass is needed.
Memory Requirements	Needs more memory for model weights, optimizer states, and gradients. Larger batch sizes also increase memory demand.	Requires less memory, as model is often optimized with quantization or compression.
Optimization Goals	Focus on convergence speed and scalability. Uses data parallelism & model parallelism.	Focus on response time, throughput, and efficiency.
Scalability	Scales horizontally (more GPUs/nodes). Expensive and time-consuming.	Scales elastically with demand (serverless or auto-scaling clusters).

Job Scheduling vs. Container Orchestration



Analogy: Airport Traffic Controller

Each plane = an AI training job
Each runway = a GPU or compute node

Controller decides:

- Which plane takes off next → job priority & queuing
- Which runway to assign → resource allocation
- Keeps all runways busy → maximum GPU utilization

Goal: Efficient use of compute and GPU for AI model training

Commonly used for: Training jobs

Implementation: Slurm schedules heavy training jobs on GPUs



Analogy: Smart City Traffic System

Each car = a containerized AI service or microservice

System controls:

- Traffic flow → load balancing
- New lanes → auto-scaling
- Breakdown recovery → auto-healing

Goal: Continuous uptime and scalability for AI inference

Commonly used for: Inference

Implementation: Kubernetes manages live inference traffic

Feature	Job Scheduling (Slurm) 	Container Orchestration (Kubernetes)  kubernetes
Primary Focus	Resource allocation & batch job management	Container lifecycle management
Workload Type	HPC, AI training, data processing	AI inference, microservices, data pipelines
Architecture Style	Static jobs, queued execution	Dynamic pods, continuous service
Execution Model	Run-to-completion batch jobs	Always-on or auto-scaled services
Scheduling Logic	Priority, queue, resource quota	Load balancing, replica scaling
GPU Integration	CUDA-aware, multi-GPU aware (Slurm GPU plugin)	GPU Operator, MIG management, DCGM metrics
Scalability	Scales to thousands of nodes for compute tasks	Scales containerized workloads across clusters
User Interface	Command-line (sbatch, srun)	API-driven (kubectl, Helm, YAML specs)
Typical Users	Researchers, data scientists, HPC admins	DevOps, MLOps, AI platform engineers
Best Suited For	Training Phase of AI lifecycle	Inference/Serving Phase of AI lifecycle

Slurm – NVIDIA Integration

- Slurm is used to schedule multi-node jobs across DGX systems.
- It uses
 - DCGM for real-time GPU health and performance monitoring.
 - NCCL to synchronize GPU data across nodes for distributed training.
 - GPUDirect RDMA for direct GPU-to-GPU memory transfers, reducing latency.
 - CUDA-aware scheduling to assign compatible jobs to GPUs.



Kubernetes – NVIDIA Integration

- Kubernetes is used to manages inference microservices on GPU-enabled clusters. It uses:
 - GPU Operator to install GPU drivers, CUDA, and DCGM automatically.
 - MIG-aware scheduling to isolate multiple GPU workloads.
 - DCGM Exporter to collect GPU metrics for Prometheus and Grafana.
 - Advanced scheduling and load balancing to assign GPU pods efficiently.
 - GPU Operator to rebalance workloads based on GPU utilization.



kubernetes

Machine Learning Operations (ML Ops)



An Analogy – ML Ops



A chef creates an amazing new dish.

Without a properly managed kitchen operation:

- The recipe isn't written down.
- Ingredients vary each time (sometimes fresh, sometimes not).
- Different cooks prepare it differently.
- No quality checks are done.
- Customers get inconsistent food, sometimes delicious, sometimes disappointing.

With a properly managed kitchen operation:

- Every recipe is documented and standardized.
- Ingredients are tracked, stored properly, and always fresh.
- Cooking steps are clear and repeatable.
- Quality control ensures each dish tastes the same every time.
- Customers always get reliable, tasty meals.

ML Ops



A data scientist creates a new ML model.

Without MLOps:

- The model isn't properly documented or versioned.
- Data used for training changes and isn't tracked.
- Different engineers deploy it in different ways.
- No monitoring after deployment.
- Users get inconsistent or poor results, sometimes accurate, sometimes not.

With MLOps:

- Every model is documented and versioned.
- Data is tracked, stored properly, and updated when needed.
- Training and deployment steps are clear, automated, and repeatable.
- Continuous monitoring ensures the model's accuracy stays reliable.
- Users always get consistent, trustworthy predictions.

Why we need ML Ops?

- Building a machine learning model is only the first step. The real challenge is:



How do we get this model into production (so real users or systems can use it)?



How do we keep it accurate when the data in the real world changes?

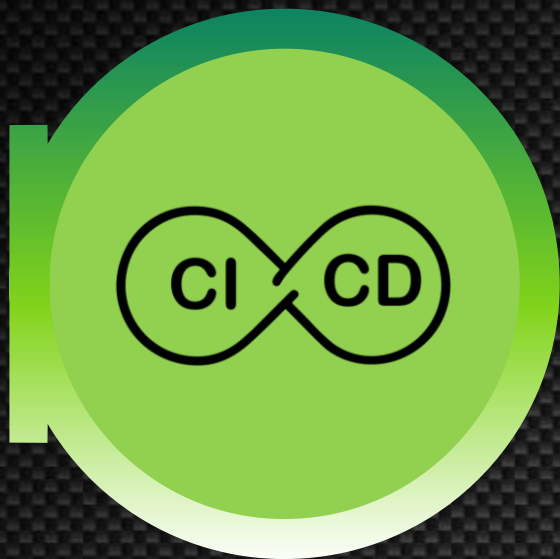


How do we update or replace it without breaking everything?

- Just like good kitchen operations make a restaurant scale reliably, ML Ops makes machine learning scale reliably.

What is ML Ops?

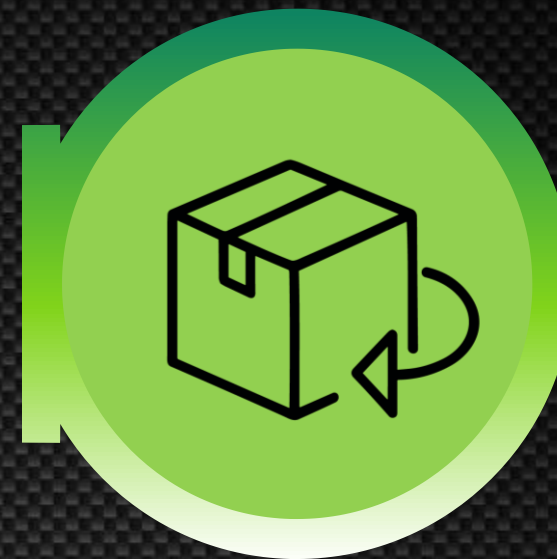
- ML Ops is tools, processes, and best practices for end-to-end machine learning system development and operations in production.



Automates the workflow with CI/CD pipelines for training, validation, and deployment.



Uses event-driven scheduling to retrain models automatically when new data arrives.



Tracks and manages model versions through a model registry for reliable experimentation.

NVIDIA Tools supporting MLOPS

MLOps Stage	NVIDIA Tools / Technologies	What They Do
1. Data Preparation & Feature Engineering	• RAPIDS, NVTabular, NeMo Data Curator	GPU-accelerated data cleaning, transformation, and feature extraction (much faster than CPU).
2. Model Training	• NVIDIA AI Enterprise, • NVIDIA Base Command Platform, • DGX Cloud, • PyTorch/TensorFlow (CUDA/cuDNN optimized)	Manage, schedule, and monitor large-scale model training efficiently across GPUs or clusters.
3. Model Optimization	• TensorRT • TAO Toolkit	Automatically optimize and compress models for fast inference on GPUs or edge devices.
4. Deployment & Inference	• NVIDIA Triton Inference Server, • NVIDIA NIM Microservices • Fleet Command	Deploy models easily in production — on-prem, in cloud, or at the edge — with autoscaling, versioning, and monitoring.
5. Monitoring & Management	• NVIDIA Base Command Manager, • Fleet Command, • NGC Registry	Centralized monitoring, model tracking, updates, and performance analytics for running AI workloads.
6. Continuous Learning / Updates	• NGC Workflows • TAO Toolkit • NeMo Framework	Automate retraining and fine-tuning of models using new data.